Supplement: Sponsorship and Race in the Housing Market

Realtor Sample

Sample selection

Sources of participants:

Realtors were sampled from the universe of publicly accessible real estate agent profiles on two platforms: Realtor.com and Homes.com. We purchased data scraped from these platforms in May 2024 that included all publicly available information on agent profiles from these two platforms, including names and contact details, office addresses, city of operation, listings history, summary sales data, market focus, and platform recommendations, among other profile attributes. We chose these platforms for several reasons. First, both platforms are among the largest real estate platforms in the United States (the 2nd and 4th largest by volume of visits in 2024).[1] Second, both are oriented more toward real estate professionals rather than property listings — unlike Zillow, the largest consumer-facing property website in the United States.[2] Each of our selected platforms maintains publicly accessible profiles for individual real estate agents (hereafter referred to as "realtors")[3] that include contact information, brokerage affiliation, transaction history, geographic focus, and more. Third, Realtor.com is the official consumer-facing platform of the National Association of Realtors (NAR), the largest real estate trade association in the United States, representing approximately 1.5 million licensed real estate professionals, and operates under a long-standing licensing agreement with the NAR. These features make the two platforms ideal sources for accessing the population of realtors in the US.

Filtering realtors:

The set of agent data from Realtor.com yielded 391,367 individual realtor profiles, and Homes.com yielded 300,666. We then engaged in a filtering process to remove realtors who (1) did not include essential data in order to send inquiries about purchasing a home in their respective coverage area, the aim of our experiment, and (2) those that were specifically inactive and therefore unlikely to be accessed in a realtor search by real prospective buyers. After removing duplicate realtors (including those with accounts on both platforms; duplicates were identified as observations sharing identical email addresses, phone numbers, or the combination of first name, last name, and city) and those without email addresses, we were left with 462,240 unique realtors in our starting dataset. Next, we applied a series of filters to this

[1] Data acquired from SimilarWeb, https://www.similarweb.com/top-websites/united-states/business-and-consumer-services/real-estate/

[2] Zillow's terms of service explicitly prohibit automated data collection, and its robots.txt configuration blocks systematic access to large portions of the site for general crawlers, meaning that efforts to acquire our data from here would have required violating their terms of service.

[3] We use the term "realtor" here colloquially to refer to real estate professionals or agents engaged in brokering home buying, selling, and renting, rather than its technical use for registered realtors with the National Association of Realtors.

combined dataset to exclude observations that would prevent us from sending inquiry emails or that would reduce the external validity of the sample. First, we dropped realtors sharing the same first and last name within a given state, to reduce the risk of inadvertent duplicate contact and thus experimental contamination (eliminating approximately 7,000 realtors). Second, we dropped realtors for whom we lacked an office address, zip code, or associated city (eliminating approximately 155,000); this information was necessary to reference the correct city in communications and to ensure we sent no more than one inquiry per office (another effort to reduce the risk of spillover effects). Third, we dropped realtors with apparent data errors in their first names, such as numeric characters or names like "realtor" (eliminating approximately 3,000). Fourth, we applied an activity filter, dropping realtors who met all three of the following low-activity criteria simultaneously: no active listings on Realtor.com; fewer than 10 recently sold properties on Realtor.com; and fewer than 20 closed sales on Homes.com. Below these thresholds, low activity strongly predicted both (1) low findability of the realtor's profile on the respective platform — and thus, that a prospective homebuyer would be unlikely to encounter them in a search — and (2) whether the realtor's email address remained active. This left 150,883 realtors across 58,240 unique offices.

Verifying email addresses:

After filtering to the set of realtors that prospective homebuyers might access when conducting a home search, we sought to verify the validity of their email addresses before reaching out. We thus selected 100,000 realtors — the maximum supported by the email verification service ZeroBounce --- with at least one realtor per office in our dataset. We then submitted all 100,000 addresses to ZeroBounce, which identified 82,600 realtor emails as valid and currently active with a high degree of certainty.

Randomly selecting the final sample.

From the set of 82,600 verified addresses, we intended to select a final set of 25,500 realtors to be included in both our pilot and final samples, with the aim to make these representative of the larger set of active realtors while ensuring we only included one realtor per office, to reduce repetition. We started by randomly selecting one realtor per office, yielding approximately 51,000 realtors. We then randomly selected 25,500 of these realtors for both our pilot and final sample. Selection probabilities were weighted by office size (i.e., the number of realtors per office), to offset the tendency of single-realtor-per-office sampling to overrepresent smaller offices relative to their true share of realtors in the market. Of these 25,500, we randomly allocated 1,500 to the pilot and 24,000 to the main study.

Check: Online Presence

To verify that our filtered dataset represented realtors that prospective buyers would plausibly encounter and contact during home search, we conducted a “reverse-search” of our realtors in March 2025. We randomly selected one zip code each in Florida, California, Texas, Illinois, and New York — the five largest realtor markets — and searched for realtors in each zip code on Realtor.com. For each zip code, we recorded the realtors listed on the first two search result pages (i.e., the first 50 realtors visible to a prospective buyer searching these zip codes). A realtor was classified as “found” if their name, state, and either brokerage or city matched an entry in our dataset. We identified 64% of the realtors appearing in these search results within our post-filtered (and pre-randomly selected) sample of 150,883. The remaining 36% that were not found could be explained by several phenomena: they may have joined the platform in the eleven months between our data collection and this check; they may have been excluded during our activity or data-completeness filters (and were searchable at the time either because their activity had increased since we collected their data, or because the platform’s algorithms surfaced paying or otherwise privileged realtors whose profiles were less complete or active).

As a second check on currency of realtors in our dataset — given that our profile data were collected eleven months prior to the start of the experiment — we manually verified the online profiles of 150 randomly selected realtors from our final sample, checking their listings on Homes.com and Realtor.com. We were unable to locate only 2 of the 150 profiles (1.33%) on either platform; for one of these, the realtor’s details were recoverable via their agency’s website. Fewer than 1% of this subsample therefore lacked any online presence. A further 7 realtors (4.67%) showed no recent activity on either platform (i.e., no recent listings or sales on Homes.com or Realtor.com). This constitutes a likely upper bound on inactivity, as some of these realtors may have continued to operate through other platforms or channels.

Taken together, these checks indicate that our sample represents a large share of the most findable realtors in the United States, and that nearly all realtors in the final sample maintained active online profiles at the time of the study. Table S1 presents characteristics of realtors in our sample alongside national benchmarks. The sample closely mirrors national figures on most dimensions. The racial and ethnic composition of sampled counties is nearly identical to 2020 Census figures — 59.8% White (national: 57.8%), 11.3% Black (national: 12.1%), and 18.7% Hispanic (national: 18.7%) — and the share of the population living in urban areas (80.5%) matches the national figure of 80.0% almost exactly. Realtors in our sample have modestly more experience than the national median reported by NAR (13.7 vs. 10 years). Realtors in our sample operate in modestly more affluent zip-codes than the national average — mean zip-code household income of $106,713 versus $81,146 nationally — though they had slightly lower than average transaction value (this might reflect sampling methods, and the National Association of Realtors uses survey data to establish benchmarks, compared to our use of Zillow data).

Table S1. Sample Characteristics: Realtors and Counties/Zips in Final Sample

Variable	N	Sample Mean (SD)	National Benchmark	Source
County and Zip--level Characteristics				
County population size	23,751	1,061,022 (1,737,372)	1,176,438	U.S. Census Bureau, 2020 Census
% White population	23,751	59.77 (20.52)	57.8%	U.S. Census Bureau, 2020 Census
% Black population	23,751	11.29 (11.16)	12.1%	U.S. Census Bureau, 2020 Census
% Hispanic population	23,751	18.69 (16.60)	18.7%	U.S. Census Bureau, 2020 Census
% Urban population	23,753	80.50% (23.70%)	80.0%	U.S. Census Bureau, 2020 Census
% Republican vote (2020)	23,731	48.60% (15.70%)	46.9%	Federal Election Commission, 2020 Presidential Election Results
% Democrat vote (2020)	23,731	49.60% (15.60%)	51.3%	Federal Election Commission, 2020 Presidential Election Results
Slavery incidence in 1860	21,708	12.40% (17.40%)	12.6%	
Racial resentment index	23,612	4.202 (0.793)	4.136	Autor et al. (2020) / CCES county-level estimates
Average income (zip-level, $000s)	23,507	106.713 (109.270)	$81,146	IRS, 2020
Average property value (zip-level)	23,587	462,091 (372,127)	$420,515	Zillow, 2020
Realtor Characteristics				
Realtor sale count	23,769	29.54 (66.56)		—
Realtor average sale value ($)	23,769	407,897 (687,455)	$435,000	National Association of Realtors, 2025 Member Profile (2024 data)
Realtor experience (years)	23,769	13.69 (7.90)	10	National Association of Realtors, 2025 Member Profile (2024 data)
Office size	23,772	8.17 (16.78)		—
Works at large brokerage	23,772	0.249 (0.433)		—

Random Assignment & Balance Checks:

Once our final sample of 24,000 realtors was established, realtors were randomly assigned to one of two buyer race experimental conditions with equal probability: (1) White buyer condition, or (2) Black buyer condition. They were also randomly assigned, with equal probability, to one of three sponsor conditions: (i) No sponsor condition, (ii) Black sponsor condition, or (iii) White sponsor condition. This 2x3 randomized design produced 6 equal-sized experimental condition cells. In order to stimulus sample, such that any effects we observe would not be due to specific elements of our stimuli, we included a range of different stimulus elements which were independently randomly assigned to participants (please see Supplement X on Stimuli for more details). This included, (1) buyer names, and (2) sponsor names (for the Black sponsor and White sponsor conditions), (3) email domain and location details of the inquiring buyer/sponsor, (4) email template, (5) email timing.

First, we randomly assigned each participant to a specific buyer name (one of the 6 different Black names for those in the Black buyer condition, and one of 6 specific White names for those in the White buyer condition). We then randomly assigned them to one of the 3 individual email accounts associated with each specific buyer name (each on a different domain, drawn from the four buyer domains: each stimulus name from our set of 12 names was used in 3 separate email/Google Workspace accounts for buyers, each on a different Google Workspace domain: see Tables S19 and S20 for all accounts and the associated domains, and the relevant section above for details about the creation of each of these domains). For participants assigned to the White sponsor condition, we also randomly assigned them to one of the 6 White names, and those in the Black sponsor condition to one of the 6 Black names. For those assigned a buyer of the same race as their assigned sponsor (e.g., a participant assigned to the Black buyer, Black sponsor condition), we randomly assigned one of the 5 *remaining* names within that race (so that buyer and sponsor never had the same name). We then randomly assigned participants to one of the 3 individual sponsor email accounts associated with that sponsor name (for each sponsor name, we also had 3 separate individual email accounts across a separate set of 4 domains). For example, a participant assigned to the Black buyer condition and Black sponsor condition might be assigned "Tyrone Banks" as a buyer. They would therefore not be assigned a sponsor by the name Tyrone Banks, but instead one of the remaining 5 Black names in our set, for example, "Kareem Williams". We would then assign them to an individual sponsor email account, for example, Kareem.Williams@BeaconGroveRealty.com. The participant would then receive an email from this email address from the Beacon Grove Realty domain, and the inquiry email would be on behalf of the randomly assigned buyer, Tyrone Banks.

We then randomly assigned each participant to one of the 6 email templates per each of sponsor and buyer emails, and randomly assigned the order that the email was sent (emails were simply ranked, and the exact send time was dictated by the scheduling system, which sent emails according to the rhythm described in the Email Domains and Sending Protocol section).

Below are a set of tables that test the balance of the sample across our 6 specific treatment cells. First, we test for balance in observable characteristics or realtors, including their personal activity and details of the geographies they operate in. Next, we test for balance in the assignment of the remaining stimulus elements, such as email template, buyer and sponsor names, and the timing of emails. Across conditions, we achieved balance in assignment for all realtor characteristics and the randomization of specific stimuli elements.

Table S2. Covariate Balance Across Treatment Conditions

			Study Condition						
Variable	**N**	**Full Sample**	**1**	**2**	**3**	**4**	**5**	**6**	**F-stat**
County / Zip Characteristics									
County population size	23,751	1,061,022.222 (1,737,371.704)	1,029,590.094 (1,668,434.183)	1,053,885.184 (1,738,077.917)	1,073,284.548 (1,704,879.866)	1,060,212.160 (1,753,158.711)	1,088,803.247 (1,767,048.253)	1,060,342.959 (1,790,326.597)	p=0.766, F=0.51
% White population	23,751	59.770 (20.522)	59.497 (20.308)	59.893 (20.498)	59.703 (20.484)	59.594 (20.553)	59.826 (20.625)	60.106 (20.671)	p=0.811, F=0.45
% Black population	23,751	11.285 (11.159)	11.612 (11.620)	11.234 (11.103)	11.322 (10.860)	11.388 (11.178)	11.186 (11.001)	10.968 (11.173)	p=0.195, F=1.47
% Hispanic population	23,751	18.691 (16.603)	18.611 (16.400)	18.731 (16.595)	18.707 (16.416)	18.849 (16.852)	18.658 (16.549)	18.592 (16.811)	p=0.987, F=0.13

% Urban population	23,753	0.805 (0.237)	0.802 (0.240)	0.804 (0.235)	0.811 (0.236)	0.803 (0.242)	0.805 (0.240)	0.806 (0.232)	p=0.647, F=0.67
% Republican vote	23,731	0.486 (0.157)	0.486 (0.157)	0.488 (0.158)	0.484 (0.157)	0.484 (0.157)	0.484 (0.156)	0.488 (0.156)	p=0.616, F=0.71
% Democrat vote	23,731	0.496 (0.156)	0.496 (0.156)	0.493 (0.157)	0.498 (0.156)	0.498 (0.156)	0.498 (0.155)	0.494 (0.155)	p=0.556, F=0.79
Slavery incidence in 1860	21,708	0.124 (0.174)	0.127 (0.176)	0.127 (0.176)	0.124 (0.175)	0.124 (0.175)	0.120 (0.172)	0.121 (0.172)	p=0.313, F=1.19
Racial resentment index (county)	23,612	4.202 (0.793)	4.201 (0.820)	4.225 (0.781)	4.194 (0.773)	4.181 (0.798)	4.205 (0.797)	4.204 (0.791)	p=0.266, F=1.29
Average income (zip)	23,507	106.713 (109.270)	108.919 (117.140)	103.645 (101.627)	108.605 (111.403)	105.050 (99.246)	106.528 (108.399)	107.524 (116.481)	p=0.221, F=1.4
Average property value (zip)	23,587	462,091.479 (372,126.570)	466,863.584 (389,315.163)	457,582.129 (354,150.048)	469,430.761 (387,295.473)	456,817.804 (347,610.253)	459,137.716 (370,041.723)	462,698.007 (382,274.299)	p=0.578, F=0.76
Realtor Characteristics									
Realtor sale count	23,769	29.535 (66.564)	29.771 (41.794)	31.091 (138.983)	28.712 (33.864)	29.260 (38.793)	28.969 (34.035)	29.408 (41.260)	p=0.671, F=0.64
Realtor average sale value	23,769	407,896.665 (687,455.248)	409,476.831 (429,513.720)	422,900.131 (1,380,223.948)	410,913.178 (509,701.553)	396,795.241 (381,847.053)	399,008.154 (425,910.004)	408,265.315 (397,582.533)	p=0.596, F=0.74
Realtor experience (years)	23,769	13.689 (7.900)	13.803 (8.139)	13.777 (7.910)	13.608 (7.698)	13.551 (7.776)	13.826 (8.121)	13.566 (7.746)	p=0.405, F=1.02
Office size	23,772	8.165 (16.775)	7.944 (16.078)	7.781 (13.611)	8.176 (15.836)	8.308 (19.406)	8.177 (14.352)	8.604 (20.279)	p=0.329, F=1.15
Works at large brokerage	23,772	0.249 (0.433)	0.248 (0.432)	0.249 (0.432)	0.240 (0.427)	0.257 (0.437)	0.251 (0.434)	0.253 (0.435)	p=0.61, F=0.72

Joint F-test (covariates → condition)	p=0.492, F=0.97	p=0.236, F=1.23	p=0.86, F=0.63	p=0.256, F=1.2	p=0.692, F=0.8	p=0.435, F=1.02

Notes: Mean (SD) reported for each variable. N reflects non-missing observations in the full sample. The rightmost column (F-stat) reports F-statistics and p-values from one-way ANOVA tests of whether each covariate's mean differs across the six study conditions. The bottom row reports joint F-statistics and p-values from OLS regressions of a binary condition-assignment indicator on all covariates simultaneously, one regression per condition. Non-significant p-values throughout indicate that covariates do not jointly predict condition assignment, and vice versa. Study conditions: (1) Black candidate, no sponsor; (2) White candidate, no sponsor; (3) Black candidate, Black sponsor; (4) White candidate, Black sponsor; (5) Black candidate, White sponsor; (6) White candidate, White sponsor.

Table S3. Balance: Buyer Name Assignment Across Conditions

	Study condition						
Buyer name	**1**	**2**	**3**	**4**	**5**	**6**	**Total**
Darius Jefferson	674 (17.0%)	–	708 (17.9%)	–	644 (16.3%)	–	2026 (8.5%)
Darnell Washington	628 (15.9%)	–	645 (16.3%)	–	693 (17.5%)	–	1966 (8.3%)
Denzel Booker	659 (16.6%)	–	655 (16.5%)	–	626 (15.8%)	–	1940 (8.2%)
Jamal Jackson	665 (16.8%)	–	693 (17.5%)	–	682 (17.2%)	–	2040 (8.6%)
Kareem Williams	655 (16.5%)	–	665 (16.8%)	–	651 (16.4%)	–	1971 (8.3%)
Tyrone Banks	678 (17.1%)	–	598 (15.1%)	–	666 (16.8%)	–	1942 (8.2%)
Cody Olson	–	640 (16.1%)	–	681 (17.2%)	–	675 (17.0%)	1996 (8.4%)
Dustin Snyder	–	655 (16.5%)	–	641 (16.2%)	–	641 (16.2%)	1937 (8.1%)
Edwin Schmidt	–	675 (17.0%)	–	650 (16.4%)	–	693 (17.5%)	2018 (8.5%)
Scott Wagner	–	665 (16.8%)	–	677 (17.1%)	–	665 (16.8%)	2007 (8.4%)
Seth Ryan	–	702 (17.7%)	–	680 (17.2%)	–	634 (16.0%)	2016 (8.5%)
Steven Jensen	–	627 (15.8%)	–	629 (15.9%)	–	657 (16.6%)	1913 (8.0%)

Note: Each cell shows count (column %). Black buyer names appear in conditions 1, 3, 5; White buyer names in conditions 2, 4, 6; '–' denotes zero by design. Chi-square test (Black names, conditions 1/3/5): p = 0.173. Chi-square test (White names, conditions 2/4/6): p = 0.662.

Table S4. Balance: Sponsor Name Assignment Across Conditions

	Study condition				
Sponsor name	**3**	**4**	**5**	**6**	**Total**
Darius Jefferson	626 (15.8%)	647 (16.3%)	–	–	1273 (8.0%)
Darnell Washington	630 (15.9%)	652 (16.5%)	–	–	1282 (8.1%)
Denzel Booker	675 (17.0%)	667 (16.9%)	–	–	1342 (8.5%)
Jamal Jackson	660 (16.6%)	679 (17.2%)	–	–	1339 (8.4%)
Kareem Williams	652 (16.4%)	684 (17.3%)	–	–	1336 (8.4%)
Tyrone Banks	721 (18.2%)	629 (15.9%)	–	–	1350 (8.5%)
Cody Olson	–	–	680 (17.2%)	677 (17.1%)	1357 (8.6%)
Dustin Snyder	–	–	697 (17.6%)	680 (17.2%)	1377 (8.7%)
Edwin Schmidt	–	–	644 (16.3%)	676 (17.0%)	1320 (8.3%)
Scott Wagner	–	–	635 (16.0%)	623 (15.7%)	1258 (7.9%)
Seth Ryan	–	–	641 (16.2%)	690 (17.4%)	1331 (8.4%)
Steven Jensen	–	–	665 (16.8%)	619 (15.6%)	1284 (8.1%)

Note: Each cell shows count (column %). Black sponsor names appear in conditions 3, 4; White sponsor names in conditions 5, 6; '–' denotes zero by design. Chi-square test (Black sponsors, conditions 3/4): p = 0.152. Chi-square test (White sponsors, conditions 5/6): p = 0.472.

Table S5. Balance: Email Template Assignment Across Conditions

Study condition

Template	1	2	3	4	5	6	Total
1	611 (15.4%)	688 (17.4%)	656 (16.5%)	651 (16.4%)	685 (17.3%)	676 (17.0%)	3967 (16.7%)
2	651 (16.4%)	631 (15.9%)	670 (16.9%)	673 (17.0%)	686 (17.3%)	645 (16.3%)	3956 (16.6%)
3	660 (16.7%)	650 (16.4%)	658 (16.6%)	656 (16.6%)	674 (17.0%)	661 (16.7%)	3959 (16.7%)
4	671 (16.9%)	693 (17.5%)	644 (16.2%)	662 (16.7%)	635 (16.0%)	658 (16.6%)	3963 (16.7%)
5	667 (16.8%)	651 (16.4%)	674 (17.0%)	663 (16.8%)	652 (16.5%)	660 (16.6%)	3967 (16.7%)
6	699 (17.7%)	651 (16.4%)	662 (16.7%)	653 (16.5%)	630 (15.9%)	665 (16.8%)	3960 (16.7%)

Note: Each cell shows count (column %). Chi-square test across all six conditions: p = 0.859.

Table S6. Balance: Send Timing Across Conditions

	Study condition							
Variable	**1**	**2**	**3**	**4**	**5**	**6**	**Overall**	**p**
Send rank (0–23,999)	11999.4 (7031.4)	12009.4 (6970.1)	12214.7 (6939.3)	11902.7 (6908.6)	11909.1 (6888)	11924.3 (6849.7)	11993.3 (6931.5)	0.334
Hour of day (0–23)	14.2 (4.1)	14.1 (4.1)	14.1 (4.1)	14 (4.1)	14.2 (4.2)	14 (4.1)	14.1 (4.1)	0.145
Day from experiment start	6.9 (3.5)	6.9 (3.4)	7 (3.4)	6.8 (3.4)	6.8 (3.4)	6.8 (3.4)	6.9 (3.4)	0.372

Note: Mean (SD) per condition. p from one-way ANOVA testing equality of means across conditions. Send rank is the ordinal position in the full pre-bounce sample (merged_df), 0-indexed; computed in 1_cleaning_base.R before the bounce filter. Hour of day extracted from send timestamp (0–23). Day from experiment start counts calendar days since Apr 24, 2025.

Audit Experiment Stimuli

Name Stimuli

Overall approach

We manipulated both buyer and sponsor race in our email stimuli using first and last names. We selected names that clearly signal Black or White race in the United States: names that are (i) common among people who identify as Black or White in the United States, respectively, and (ii) racially diagnostic — that is, names that are distinctively associated with one racial group relative to others. We combined name lists used to signal Black and White race in previous audit studies with administrative data on name incidence and racial diagnosticity in the United States, and then conducted a series of stimulus tests to select a final set of names reliably perceived as Black or White. The subsections below detail each stage of this process.

First Names

We took first names from Gaddis (2017) and Tzioumis (2018), which include a set of first names that are diagnostic of White and Black race in the United States and have been used in several prior audit studies (Gaddis, 2017, 2019, Crabtree et al., 2023; Hanson & Hawley, 2023).[4] These datasets include the number and proportion of people from each race category for each name. We linked these names to a larger dataset of racial representation of names from Rosenman (2023), which includes racial identification of 38 million voters in 6 Southern States, providing additional data on racial distinctiveness. Finally, we merged these data with all names registered with the Social Security Administration (SSA) between 1983 and 1997 (the most plausible cohort for a first-time home buyer in 2025; National Association of Realtors, 2025) to obtain the frequency of each name within this cohort in the United States population.

We then applied two filters to this merged list. First, we excluded names outside the top 1,000 most common names in this cohort, ensuring that selected names apply to a meaningful share of the population. Second, we excluded names below our thresholds for racial diagnosticity: for White names, we required that at least 75% of bearers identify as White; for Black names, the corresponding threshold was 70%.[5]

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[4] This includes names from birth cohorts between 1994 and 2012, and from a set of mortgage applicants in 2010. We ensure that these names are more common in "older cohorts" by pairing with SSA data on name frequencies for the US population.

[5] We deprioritize Tzioumis (2018) in the racial distinctiveness measures. Gaddis (2017) is a complete list of all births in NYS over 1994-2012. Rosenman et al. (2023) is a complete list of voters between 2019-2022 for 6 Southern states. These seem to have far greater coverage of Black names and the racial distinctiveness measures between these are far more correlated with each other than either is with Tzioumis (2018). There are other reasons to doubt the Tzioumis data. First, some of the most common black first names, frequently used in audits, are completely missing (e.g., "Darnell"). Another is that names very strongly correlated with black race are more white than black in this dataset (e.g., "Jamal"). Finally, these data are from mortgage lending from specific lenders: this is

To account for socio-economic status, which is also correlated with names, we used the average level of education for mothers of people born with each name, taken from Gaddis (2017), a measure commonly used to control for SES in audit studies (Bertrand & Mullainathan, 2004; Gaddis, 2017; Hanson & Hawley, 2023). Because no Black first names fell in the top quintile of socio-economic status (by mother's education), we dropped White names in this quintile, since we would be unable to match these names with a Black first name of similar socio-economic status. This procedure yielded a shortlist of 33 first names.[6] The perceived racial diagnosticity of each first name was assessed as part of a set of stimulus tests, described below. We specifically used Stimulus Test 1 to further narrow our list of names for the final stimuli.

Last Names

For last names, we followed prior audit studies and used the most common last names in the 2010 US Census (Kirgios et al., 2022, 2024; Hanson & Hawley, 2023; Crabtree et al., 2023), which includes racial identifiers for each name. As with first names, we prioritized racial diagnosticity and name frequency. For White last names, we excluded names meeting any of the following criteria: fewer than 90% of bearers identify as White; a White-to-Black bearer ratio below 20:1; or a rank outside the top 500 most common surnames. For Black last names, we excluded names meeting any of the following criteria: fewer than 50% of bearers identify as Black; a White-to-Black bearer ratio above 1:1 (i.e., more White than Black bearers); or a rank outside the top 1,000 most common surnames.[7] This produced a candidate list of 14 Black and 33 White last names, from which we selected 24 names — 12 Black and 12 White — for stimulus testing.[8] The perceived racial diagnosticity of each last name was assessed as part of a set of stimulus tests, described below, which we used to narrow the list down further and make final name selections.

Final name selection

We assessed the perceived racial diagnosticity of each first and last name in our shortlists in a set of stimulus tests conducted with Prolific participants (described below). Using the results of Stimulus test 1,

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thus a very specific form of selection, one known to be correlated with race (Hurtado, 2024): as such, this is likely not representative of race.

[6] You can find this list here.

[7] Black people in the US account for a far smaller share of the population than White people. This implies that names that are relatively common amongst Black Americans will be less common overall and also less racially distinctive, forcing us to lower these thresholds relative to White names to collect a sufficiently large pool of names.

[8] Last names are far more concentrated than first names, and thus fewer last names are diagnostic of Black race. We thus select smaller lists of last names at each stage of stimulus testing, since there were fewer viable candidate names.

we selected the 6 last names that were perceived as most racially diagnostic for each race (participants perceived each last name as the correct race more than 85% of the time and the alternate race less than 10% of the time).[9] From this same stimulus test, we selected 12 first names for each race based on the following criteria: (1) were perceived as highly racially diagnostic (above 85% for the correct race and below 10% for the alternative race, with two exceptions),[10] (2) were objectively racially diagnostic in the population, (3) were common in the population, (4) allowed us to have substantial overlap across racial groups and variance within racial groups in socio-economic status, as measured by mother's education.

To construct full names for the experiment, we paired the most racially diagnostic first names with the least diagnostic last names within each racial group, so that each full name would achieve a consistently high level of overall racial diagnosticity. Each last name was paired with two first names. We then conducted a confirmatory test of perceived race for each of these full names in Stimulus Test 2 (described below). From this test, we selected the most racially diagnostic full name for each unique last name, yielding the final set of 12 names — six Black and six White — shown in Table S7 below. [11] Each name was perceived as the correct race in more than 90% of cases, falls within the top 1,000 most common first and last names in the United States, and is highly diagnostic of race in the population.

Table S7. Final Names Used in Audit Experiment

Black	**White**
Denzel Booker	Edwin Schmidt
Tyrone Banks	Seth Ryan
Kareem Williams	Cody Olson
Jamal Jackson	Steven Jensen
Darnell Washington	Scott Wagner
Darius Jefferson	Dustin Snyder

[9] This was more accurate for white names than black names. Also, these were already screened to be both objectively racially diagnostic and fairly common in the population.
[10] Darnell, 11% White, and Jerome 12% White.
[11] With the exception that we chose Tyrone in place of DaShawn and Seth in place of Brett, since in each case these first names were outliers in SES measured by mother's education, and thus would allow less overlap in these distributions across race. Due to this same logic, we considered selecting Jabari over Darnell, but the racial recognition rate was 10% lower and the relatively far lower popularity of Jabari in the population made this a less attractive decision.

A note on SES and names.

We treat perceived SES as a potential mechanism through which racial discrimination operates, rather than as a confound, and therefore do not condition on it in our main analyses (Crabtree et al., 2023). Nonetheless, we attempted to account for objective SES in name selection using mother's education data from Gaddis (2017) as a proxy — a standard approach in this literature (Bertrand & Mullainathan, 2004; Gaddis, 2017; Hanson & Hawley, 2023), as described above.

In practice, prioritizing the primary manipulation (perceived race) and population representativeness — through a transparent and replicable selection procedure — made it impossible to achieve complete SES parity across racial groups. White names in our final set have, on average, higher SES than Black names. However, the degree of overlap is substantial: Darius, a Black name, has higher SES than four of the six White names in the set, and Edwin, a White name, has the lowest SES of all twelve names. This within- and cross-race variation in SES allows us to examine whether SES independently predicts realtor responses and to compare Black and White names at similar SES levels, providing a partial test of whether observed effects are driven by race rather than SES. Please see supplemental analysis for the effect of objective SES using this measure in the heterogeneity analysis in the main paper.

Name Stimulus Test 1

Design

We administered a survey to Prolific participants to assess the perceived race, gender, and socioeconomic status of our shortlisted first and last names. The primary purposes were to confirm that names intended as male were unambiguously perceived as such, and that names were highly racially diagnostic, ensuring a clean manipulation in the field experiment.

The name list for this test comprised 24 Black and 24 White first names and 12 Black and 12 White last names, drawn through the procedure described above. We quasi-randomly paired each first name with three different last names of the same racial group (and each last name with six first names), producing a list of 144 distinct full names: 72 White and 72 Black. Each participant was presented with 24 randomly ordered names drawn from this list. We targeted 300 participants on Prolific in order to obtain approximately 50 responses per full name (equivalent to approximately 150 responses per first name and 300 per last name). All participants were required to pass attention check questions prior to completing the survey. We obtained responses from 291 participants, slightly below our target of 300.

Questions

Each subject responded to each of the below questions for 24 randomly chosen names, after completing attention checks. We selected these questions and the wording to match those in Crabtree et al (2023).

Instructions:

*Today, you'll view a series of **names of people who live in the United States.** Our goal is to **understand what comes to mind automatically when you read these names.** Please answer honestly and as quickly as you can - your responses are never identifiable and bear no impact on your base payment.*

1. *What would you think would be the race of a person with this name?*

- *White*
- *Black or African American*
- *American Indian or Alaska Native*
- *Asian or Pacific Islander*
- *Hispanic*
- *Other*

2. *What would you think would be the gender of a person with this name?*

- *Male*
- *Female*
- *Other*

3. *What social class category do you think a person with this name is likely associated with?*

- *Lower or working class*
- *Middle class*
- *Upper class*

Results

Results are reported in Tables S8 and S9 below. Most Black first names exceeded an 80% correct-race perception threshold: 16 of the 24 Black first names were perceived as Black by more than 80% of participants. Among White first names, all but one name exceeded 80% perceived as White. Perceived gender was consistently male across both groups. Perceived SES varied modestly within each racial group, supporting subsequent selection of names with within-race SES variation. Regression coefficients confirmed these patterns closely. These results were used to select the 12 Black and 12 White first names and the six Black and six White last names carried forward to Stimulus Test 2.

Table S8 and S9: Name Stimulus Test 1 — First Name Results

Black Names

First Name	Perceived Black	Perceived White	Perceived SES (1-3)	Perceived Male
Jermaine	96.55%	2.07%	1.81	95.86%
Denzel	95.17%	2.07%	1.91	99.31%
Tyrone	95.17%	2.76%	1.59	99.31%
Dashawn	93.79%	3.45%	1.54	91.72%
Jamal	93.79%	2.76%	1.79	98.62%
Terrell	92.47%	3.42%	1.69	95.89%
Daquan	91.78%	6.85%	1.47	97.26%
Deandre	91.10%	6.85%	1.71	94.52%
Kareem	89.66%	0.69%	1.74	97.93%
Jabari	89.04%	2.74%	1.62	96.58%
Darnell	86.21%	11.03%	1.80	97.24%
Hakeem	84.25%	2.74%	1.76	99.32%
Jerome	83.56%	12.33%	1.75	98.63%
Delroy	82.88%	12.33%	1.74	97.26%
Darius	82.07%	9.66%	1.80	97.24%
Marquise	81.51%	8.90%	1.74	85.62%
Malik	78.77%	7.53%	1.76	95.21%
Kevon	74.66%	20.55%	1.71	99.32%
Jalen	73.10%	14.48%	1.80	77.93%
Cedric	67.12%	28.08%	1.90	98.63%
Reginald	65.52%	28.28%	2.22	99.31%
Quincy	64.83%	28.28%	2.06	91.72%
Amari	64.14%	10.34%	1.83	48.97%
Elijah	58.90%	30.82%	1.89	97.26%

White Names

First Name	Perceived Black	Perceived White	Perceived SES (1-3)	Perceived Male
Brett	0.00%	98.63%	2.10	96.58%
Brad	2.07%	97.24%	2.01	99.31%
Hunter	1.38%	96.55%	2.10	97.24%

Connor	2.05%	95.89%	2.18	99.32%
Scott	1.38%	95.86%	2.15	100.00%
Seth	0.69%	95.86%	2.10	99.31%
Robert	2.74%	95.21%	2.12	100.00%
Dustin	2.07%	95.17%	1.86	99.31%
Paul	1.38%	95.17%	2.20	100.00%
Steven	0.69%	95.17%	2.09	99.31%
Cody	0.69%	94.48%	1.85	97.93%
Greg	4.14%	93.79%	2.08	100.00%
Jake	4.11%	93.15%	1.88	99.32%
Todd	3.45%	92.41%	2.07	99.31%
Brian	3.42%	91.78%	1.99	100.00%
Matthew	1.37%	91.10%	2.10	99.32%
Aaron	2.74%	89.73%	2.00	100.00%
Jay	4.79%	89.73%	2.01	97.95%
Brendan	4.11%	86.99%	2.02	96.58%
Caleb	4.79%	85.62%	1.96	100.00%
Anthony	11.64%	84.25%	2.04	100.00%
Maxwell	13.01%	84.25%	2.31	100.00%
Edwin	6.90%	80.00%	2.26	100.00%
Ronny	6.90%	77.24%	1.71	94.48%

S10 & S11: Name Stimulus Test 1 — Last Name Results

Last Name	Perceived Black	Perceived White	Perceived SES (1-3)	Perceived Male
Jefferson	89.69%	4.12%	1.74	92.44%
Washington	88.66%	8.25%	1.75	95.88%
Williams	87.97%	8.25%	1.76	95.19%
Jackson	87.29%	9.62%	1.82	98.28%
Banks	87.29%	7.22%	1.78	94.85%
Booker	85.91%	7.90%	1.72	94.50%
Charles	82.82%	7.56%	1.75	95.88%
Joseph	82.13%	10.31%	1.74	99.66%
Mosley	76.98%	17.87%	1.92	95.19%
Rivers	73.88%	13.40%	1.77	85.22%
Dorsey	73.20%	16.84%	1.73	85.57%
Gaines	72.16%	18.21%	1.82	92.78%

Last Name	Perceived Black	Perceived White	Perceived SES (1-3)	Perceived Male
Carlson	7.56%	89.00%	2.11	98.97%
Wolf	5.15%	76.29%	1.96	98.28%
Hansen	4.47%	89.00%	2.03	99.66%
Fischer	4.12%	93.47%	2.02	98.63%
Meyer	3.44%	92.10%	1.98	97.94%
Ryan	3.09%	95.53%	2.14	99.66%
Hoffman	3.09%	89.69%	2.18	99.48%
Jensen	3.09%	94.85%	2.01	99.31%
Olson	3.09%	93.81%	2.05	99.31%
Wagner	2.58%	93.81%	2.06	98.20%
Snyder	2.06%	93.13%	1.99	98.97%
Schmidt	1.72%	95.53%	2.11	99.66%

Name Stimulus Test 2

Design

This survey followed the same approach and used identical question wording as Stimulus Test 1, with two key differences. First, rather than testing first and last names separately, we tested exact full-name combinations — the specific pairings we planned to use in the field experiment. Second, given the smaller set of names under evaluation, we reduced the sample to 50 Prolific participants.

The 24 full names were constructed as follows. We selected 12 Black and 12 White first names from Stimulus Test 1 based on: (1) perceived racial diagnosticity; (2) objective racial diagnosticity in the population; (3) name frequency in the SSA cohort; and (4) variation in SES within each racial group. We paired these with the 12 most perceived-diagnostic last names from Stimulus Test 1 — six Black and six White — matching the most diagnostic first names with the least diagnostic last names within each racial group, so that each full name would achieve a uniformly high level of racial diagnosticity. Each last name was paired with two first names, yielding 24 unique full names. Names were presented in random order.

Results

Results are reported in Table S12 below. All 24 full names exceeded 80% correct-race perception, and 19 of the 24 exceeded 90%. Perceived gender was male in more than 90% of cases for all names. For each unique last name, we selected the full name achieving the highest perceived racial diagnosticity, yielding the final set of 12 names shown in Table S7 above.

Table S12. Name Stimulus Test 2: Full Name Perceptions + Administrative Data

	Stimulus Test Perceptions				First Name Stats in Population					Last Name Stats in Population		
Full Name	**Perceived Black**	**Perceived White**	**Perceived SES (1-3)**	**Perceived Male**	**% Black in Population**	**% White in Population**	**Name frequency rank SSA**	**Mothers with college**	**Mother's college quintile**	**% Black in Population**	**% White in Population**	**Name frequency rank Census 2010**
DaShawn Banks	96.08%	0.00%	1.7	92.16%	91.29%	7.11%	697	14.70%	1	54.51%	39.27%	292
Denzel Booker	92.16%	1.96%	2	98.04%	82.46%	12.89%	522	34.30%	2	65.22%	28.03%	941
Tyrone Banks	94.12%	1.96%	1.82	94.12%	76.07%	18.26%	259	24.40%	1	54.51%	39.27%	292
Kareem Williams	90.20%	1.96%	1.84	98.04%	78.79%	14.76%	567	32.60%	2	47.68%	45.75%	3
Jamal Jackson	96.08%	1.96%	1.8	96.08%	65.77%	22.83%	265	31.00%	2	53.04%	39.89%	19
Darnell Washington	92.16%	3.92%	1.88	94.12%	83.19%	14.83%	380	28.30%	2	87.53%	5.17%	145
Terrell Jackson	92.16%	3.92%	1.7	94.12%	63.49%	34.90%	298	33.20%	2	53.04%	39.89%	19
Jermaine Booker	92.16%	3.92%	1.78	92.16%	88.52%	7.44%	287	27.10%	2	65.22%	28.03%	941
Jabari Washington	82.35%	3.92%	1.84	86.27%	94.63%	3.22%	869	56.80%	4	87.53%	5.17%	145
Darius Jefferson	90.20%	5.88%	1.86	96.08%	63.81%	32.15%	205	43.50%	3	74.24%	17.45%	615
DeAndre Williams	88.24%	7.84%	1.72	96.08%	87.59%	8.43%	296	31.50%	2	47.68%	45.75%	3
Jerome Jefferson	88.24%	7.84%	1.88	96.08%	42.31%	52.11%	266	37.70%	3	74.24%	17.45%	615
Paul Jensen	7.84%	80.39%	2.08	96.08%	7.14%	87.32%	51	62.60%	4	0.44%	93.97%	265
Greg Schmidt	3.92%	88.24%	1.96	96.08%	18.31%	77.64%	673	42.00%	3	0.37%	95.15%	185
Edwin Schmidt	3.92%	90.20%	2.24	98.04%	9.18%	65.09%	192	18.20%	1	0.37%	95.15%	185
Todd Snyder	3.92%	92.16%	2.08	96.08%	8.02%	89.91%	160	46.70%	3	1.46%	94.07%	165
Seth Ryan	3.92%	92.16%	1.92	96.08%	5.79%	91.39%	96	59.30%	4	3.02%	91.65%	193
Brad Wagner	3.92%	92.16%	2.24	96.08%	7.47%	87.27%	323	49.80%	4	2.60%	92.47%	173
Hunter Olson	0.00%	92.16%	1.92	92.16%	1.84%	95.73%	111	58.60%	4	0.40%	94.76%	157
Cody Olson	5.88%	92.16%	1.96	94.12%	2.74%	94.22%	35	40.50%	3	0.40%	94.76%	157
Steven Jensen	1.96%	92.16%	2.08	96.08%	6.52%	86.52%	28	40.10%	3	0.44%	93.97%	265
Scott Wagner	1.96%	94.12%	2.06	96.08%	2.21%	95.43%	53	60.20%	4	2.60%	92.47%	173
Brett Ryan	1.96%	94.12%	2.04	90.20%	1.92%	96.60%	84	66.70%	5	3.02%	91.65%	193
Dustin Snyder	0.00%	98.04%	1.72	96.08%	1.97%	94.55%	50	33.10%	2	1.46%	94.07%	165

Email Stimuli

Drafting the email text

Our email stimuli were designed to simulate a cold inquiry from a prospective first-time homebuyer to a realtor they had not previously met. We conducted two in-depth interviews with a practicing realtor with experience across Illinois, Massachusetts, and California to establish: (1) how realistic such unsolicited inquiries are in practice, and (2) what content and format such emails typically take. The realtor confirmed that both direct inquiries from out-of-state clients relocating to a new state, and referral emails sent by a client's existing realtor on their behalf under a commission-sharing arrangement, were common occurrences. This was corroborated by follow-up qualitative interviews with three additional realtors and a quantitative survey of 16 realtors at two separate events of the Illinois Association of Realtors — all 16 reported having received referrals from other realtors, and 63% had previously received such referrals from out-of-state.

Working iteratively with the first realtor, we developed email text and signatures representative of both email types: emails sent directly by the buyer (hereafter "buyer emails") and emails sent by a referring realtor on the buyer's behalf (hereafter "sponsor emails," where the sponsor is the referring realtor). To allow for stimulus sampling, we initially drafted six versions of each email type. These are provided in the All Final Stimuli section below.

Email Signatures

To verify that our stimulus sponsor signatures were typical of real realtor signatures, we conducted a pilot study in which simple solicitations were sent from our own email accounts — under our own names and contact details — to practicing realtors (details in the Pilots section below). We then analyzed the reply signatures in emails received both from this pilot and from a prior pilot, covering realtors in Boston, New York, Chicago, and San Francisco. This yielded a sample of 43 email signatures. Among these, 26% included a photo and 37% included a logo; 81% included a phone number and 70% included an email address. All signatures that followed a formal format included a phone number. The modal signature included name, position, firm name, and contact details (email address and cell phone number), as illustrated in Figure S1 below. We modelled our fictitious sponsor signatures on this format (see the All Final Stimuli section below for the final signatures).

Figure S1: Example Realtor Email Signature

Dustin Snyder, Realtor
C: (937) 853-8076
E: Dustin.Snyder@cedartrailrealty.com

Cedar Trail Realty, Milwaukee, WI
Cedartrailrealty.com

Testing the email

We validated our email stimuli through three sequential steps.

First, we conducted a small pilot audit (Pilot 1; see Pilots section below), sending a single version of the sponsor email from a fictitious realtor account to 50 practicing realtors in the Boston area. We received 23 responses, none of which indicated suspicion about the email's authenticity. This provided an initial validation that the email would not be dismissed as obviously suspicious by practicing realtors.

Second, we tested whether our emails were perceived as typical of genuine realtor solicitations among online participants, using the stimulus test described in detail in the following subsection. No stimulus email was identified as fake at a rate exceeding chance, and stimulus emails were no more likely to be identified as fake than real emails. This confirmed that participants could not systematically distinguish our stimuli from genuine solicitations.

Third, we tested our final 12 stimulus emails in a full-scale pilot (Pilot 3; see Pilots section below) with 1,500 randomly selected realtors from our actual sample, using the same stimuli as the main experiment. We received 393 responses, of which only 8 indicated any suspicion about the sincerity of the inquiry and only 1 was flagged as spam. The remaining 385 responses engaged substantively with the stimulus. No specific email template or condition was treated as differentially suspicious. We therefore proceeded to the main experiment with our stimuli unchanged.

Email stimulus test

Design

We tested whether email text stimuli for the audit experiment were perceived as "typical" or "normal" emails, or whether they stood out as unusual. Online participants conducted two short tasks: first, they attempted to identify the single fictitious email from a set of 8 emails that included 7 real emails (sent to an actual practicing realtor) and 1 fictitious audit email. Next, they rated a pair of emails - one stimulus email and one real email - on several scales regarding how typical they seemed. We elicited responses from 400 subjects on Prolific, receiving 50 responses per distinctive email. All subjects passed a set of attention check questions. Details of each section and instructions are below.

Section 1

Each participant was shown a set of eight emails, each framed as an initial inquiry to a realtor. Seven were real emails received by an actual practicing realtor (randomly selected from the set provided below), and one was a fictitious audit email (randomly selected from our final stimuli). The order of emails was randomized. Participants were incentivized to identify the single fake email, receiving a $0.25 bonus for a correct guess.

Instructions:

Below is a set of emails sent as an initial inquiry to a Realtor/Real Estate Agent. 7 of these emails are real (with identifying information changed), and were taken from the inboxes of practicing Realtors; 1 is fake. Please guess which email you think is fake. If you guess correctly, you will receive a $0.25 bonus.

[INSERT RANDOMIZED EMAIL SET]

Section 2

After completing Section 1, each participant rated one randomly assigned fictitious audit email and one randomly assigned real email on the following questions.

Instructions:

Next, you will see a few emails like the ones you just reviewed - again, they are all initial inquiries that may or may not have been actually sent to actual practicing realtors in the past few years. Please answer the following questions, about each email, as honestly as you can.

[INSERT EMAIL]

Imagine you are moving to a new city and plan to buy a house. [If Candidate or real email: How likely would you be to send this email to a realtor/real estate agent?] [If Sponsor email: You previously worked with *a realtor in your current city, and they offered to reach out to agents in the new city on your behalf.]*

1. *How likely would you be to send/approve of this email?*

- *1 (very unlikely) - 7 (very likely)*

2. *In this scenario, how typical do you find this email?*

- *1 (not at all typical) - 7 (very typical)*

3. *In this scenario, how "normal" do you find this email?*

- *1 (not at all normal) - 7 (very normal)*

4. *In this scenario, how "weird" do you find this email?*

- *1 (not at all weird) - 7 (very weird)*

Results

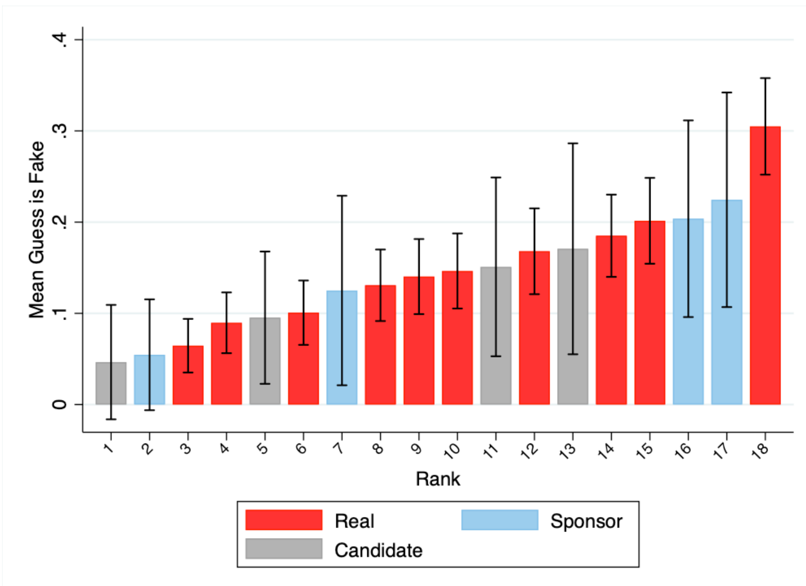
Results are reported in Table S13 and Figure S2 below. No fictitious audit email was identified as fake at a rate exceeding chance (mean identification rate: 0.13 across all stimulus emails, compared with 0.15 across all real emails), and none ranked among the three most-identified emails overall. On the typicality and likelihood-to-send measures, fictitious emails consistently outperformed real emails: the mean "likely to send" rating was 5.08 for stimulus emails versus 3.54 for real emails, and the mean normalness index was 5.06 versus 3.99. Sponsor emails and buyer emails performed comparably on both measures. These results indicate that our stimuli were indistinguishable from genuine solicitations and were perceived as at least as typical and plausible as real emails sent to practicing realtors.

Table S13. Email Stimulus Test Results

Email ID	Email Name	Guess Fake Mean	Guess Fake Rank	Likely to send Mean	Likely to send Rank	Normalness Index Mean	Normalness Index Rank
1	real1	0.17	12	3.69	12	4.06	13
2	real2	0.10	6	3.35	15	3.97	14
3	real3	0.09	4	4.18	10	4.84	8
4	real4	0.19	14	3.63	14	3.69	15
5	real5	0.30	18	2.75	18	3.19	18
6	real6	0.13	8	4.40	8	4.56	10
7	real7	0.15	10	2.87	17	3.67	16
8	real8	0.06	3	3.77	11	4.12	12
9	real9	0.20	15	3.13	16	3.50	17
10	real10	0.14	9	3.67	13	4.35	11
11	Plain 1 - Sponsor	0.05	2	5.31	4	4.84	7
12	Buyer 1 - Sponsor	0.20	16	5.46	1	5.27	1
13	Rented 1 - Sponsor	0.22	17	5.41	3	5.14	4
14	Plain 2 - Sponsor	0.13	7	5.43	2	5.15	3
15	Plain 1 - Candidate	0.17	13	5.15	5	5.21	2
16	Buyer 1 - Candidate	0.10	5	4.76	7	5.07	5
17	Rented 1 - Candidate	0.05	1	4.94	6	5.03	6

18	Plain 2 - Candidate	0.15	11	4.21	9	4.78	9
	All Real Emails	0.15	9.90	3.54	13.40	3.99	13.40
	All Stimuli Emails	0.13	9.00	5.08	4.63	5.06	4.63

Figure S2: Email Stimulus Test — Typicality and Identification Rates



<u>Table S14. Real Emails Used in Stimulus Test (Identifying Information Changed)</u>

Email ID	Deidentified Email Content
real1	Good morning John, My name is Alex, I'm looking for a pre-furnished unit around Aug/Sept and I was told I should speak to you and your team. I'm having my 30th birthday in the area, so I'd like to get a list of possible accommodation options for quite a few people. If you could help, by recommending a few places that are for rent, and the price per unit - then I can offer it to my guests (who can divide up into groups depending on the amount of people the accommodation can house). It would be for 2 nights more than likely (the weekend). I

	don't have a specific date yet - but like I said, probably mid-way between August and Sept. Thanks in advance for your assistance. Kind regards, Alex Smith
real2	Hi John, My name is Alex Smith. I have just moved to Boston, just renting for now, and will be working as an appraiser throughout Massachusetts. I would love to meet you if you are able to find some time. Best, Alex Smith
Real3	Hi There, We are looking for a house to rent over the summer holidays (July 17th - August 5th), preferably close to the beach or on the river. Do you perhaps know of anything that is available? Warmly, Alex Smith
Real4	Hi John, I'm reaching out because I'm looking at buying a property to rent out. I'd be interested either in a permanent rental or a holiday rental property. The investment has to beat returns currently achieved in bank interest.

	Are your units usually used by owners or is there a rental process available? Any further info would be appreciated. I am based in Columbus, OH. Regards, Alex Smith
Real5	Hi John I'd like to put forward an offer for Lot 75 in writing if you can. The offer is $150,000.00. I can pay $90,000.00 in the next 7 days and the remaining balance before the end of May 2025. I'll be paying cash so no bond is needed. If you would like me to come into the office, I am in Boston until October 17th. Thank you, Alex Smith
Real6	Dear John, I'm reaching out because I have had a look at your website and Realtor.com. Plymouth is one of the coastal towns I am considering for a holiday home. I am looking for a plot of land or house south of the main road, close to the river or beach. As long as it is close to the water, sea or river views are not necessary. For a house, I don't want to really pay more than $1,000,000. I might consider a really good plot at the right price and only build a home later. I live in Columbus, OH, and hope to visit Plymouth at the start of February. I am not in a hurry to buy, but if something pops up, please let me know. Also: Is the plot on 601 Island Road still available? Best, Alex Smith

Real7	Hi, Please could you send me a list, preferably with pictures, of the holiday rentals home that you have available in the Plymouth & Duxbury areas. Many thanks, Alex
Real8	Good Morning John, I will be in Plymouth from November 15–22 and I would like to look at a plot to buy. Please send me your latest map and pricing of the properties that are still available. Thank you, Alex Smith
Real9	Good afternoon John, We would like to sell this property in Boston. Ideally for $420,000 to $450,000. Can you please give me your input and a contract for marketing it? Thanks in advance. Kind regards, Alex Smith
Real10	Hello John, hope you are well Just checking if you are still involved in land sales I am interested in selling my vacant land, plot 243, Signal Hill Close

	Please could you let me know either way, or point me in the right direction of an agent colleague Thanks and best regards Alex Smith

Phone Numbers and Voicemails

Our analysis of realtor signatures indicated that including a phone number was normative — the large majority of real realtor emails contained one (81%; see Pilot 2 above). We therefore included unique phone numbers in the signatures of all 72 fictitious stimulus accounts, both to increase the realism of our stimuli and to capture any responses made by phone rather than email. To do so, we purchased a Google Voice add-on subscription for each Google Workspace domain and assigned a Google Voice license to each individual user account. Once assigned, Google provided a set of available phone numbers from which we selected those with area codes corresponding to the geographic location of each domain (i.e., Milwaukee, WI or Columbus, OH). This produced 72 unique phone numbers — one for each of the 36 sponsor accounts and one for each of the 36 buyer accounts.

For voicemails, we drafted a brief script in collaboration with a practicing realtor and recruited a professional voice actor through a freelance services platform to record a unique voicemail for each of the 72 accounts. The script was identical across accounts except for the account holder's name and, in sponsor conditions, the brokerage name.

> *Hi, this is [INSERT BUYER/SPONSOR NAME][Sponsor conditions only: from [INSERT BROKERAGE/AGENCY NAME]], sorry I couldn't get to the phone but please leave your name and I'll get back to you.*

Recordings for all 72 accounts are available at [LINK FORTHCOMING]. Because each stimulus account was linked to a unique phone number, the Google Workspace system maintained a complete log of all phone communications associated with each account, including caller contact information, time of contact, and message content. Google Voice also provides AI-generated transcriptions of voicemails, which are included in the dataset and allowed us to analyze voicemail content. An example data row, with identifying information removed, is provided in Figure S3 below.

Although we included phone numbers to maximize realism and response capture, phone communications were not our primary outcome variable. In the main pilot (Pilot 3; see above), 26.2% of realtors responded via email while approximately only 4% responded via phone — of whom 2.7 percentage points had also responded by email, meaning phone-only response rates constituted approximately 1.3%. Phone communications were not differentially distributed across treatment conditions in the pilot. Moreover, matching phone communications to participants required additional effort beyond email matching — approximately 65% were matched directly via administrative phone numbers in our dataset, with the remainder requiring manual matching as described below. While this matching was performed carefully, it added more scope for measurement error than email response. We therefore pre-registered email responses as our primary outcome variable, and pre-registered a robustness check including all phone communications within the response window as a secondary analysis.

Figure S3: Example Phone Data Row (Identifying Information Removed)

id	from_email	to_email	message_subject	sent_at	status	message_text
fa563051-f28f-4b5b-a1ca-5db7cc70204f	14142159773.[redacted]7b8-nHZnn@txt.voice.google.com	kareem.williams@summitco.org	New text message from [redacted]	4/25/25 11:00	Unrelated	Hi Kareem, this is [redacted] with [redacted] Realty. I got your email you sent last night that you were interested in looking for a house in [redacted] I sent you an email back seeing when would be a good time for us to schedule a call and talk more about your move! To respond to this text message, reply to this email or visit Google Voice.

Websites

Website Creation

To align with standard realtor communications, all sponsor emails sent to agents included a clickable link at the bottom of the email to the sponsor's fictitious associated agency website (see Figure S8). The agency website appeared alongside the agent contact information such as name and email address.

As such, four realtor agency websites were created— two websites for the agencies presented as located in Columbus, Ohio (i.e. "Beacon Grove Realty" and "Elm Valley Realty"; beacongroverealty.com and elmvalleyrealty.com, respectively), and two websites for the agencies presented as located in Milwaukee, Wisconsin (i.e. "Cedar Trail Realty" and "Maple Heights Realty"; cedartrailrealty.com and mapleheightsrealty.com, respectively). The agency websites used were the same across experimental conditions: for example, regardless of whether the condition was Black sponsor or White sponsor, the website link was always the same so long as the email itself was randomly assigned to display the associated agency name.

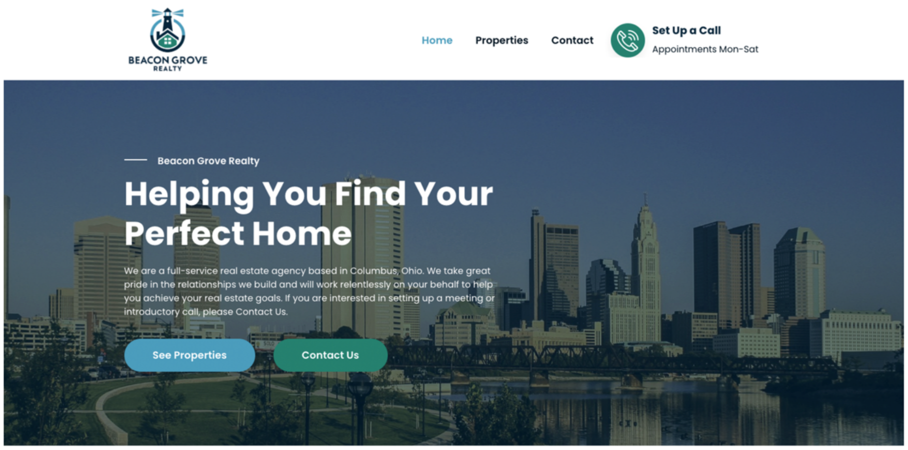
The agency websites aligned with what would be considered typical of a small, independently owned and operated real estate agency website in the states of interest. To inform typicality, we sampled agency websites in both Ohio and Wisconsin. The first five agencies from each state that were small and independently operated with working links to their websites were selected and screenshotted. This created a total of 10 agency websites that served as a template for what would be considered both realistic and minimally necessary to include on a realtor website.

Each website had three tabs: Home, Contact Us, and Properties, where text and graphics for all tabs were derived from text and graphics used in the example websites. For example, the majority of websites displayed a stock image of the city, and so the website contained a stock image of the city on the Home tab (Figure S4). Further, because half of the agency websites provided centralized contact information rather than individualized contact information for each agent, on the Contact Us tab, the website included a single central contact email for the office (e.g., beacongroverealtygroup@gmail.com). In addition to a single contact listing being sufficiently typical, the decision to include a centralized contact source also circumvented additional complications, such as the inclusion (or lack thereof) of realtors of different minority statuses influencing judgments of the sponsor and office at large. The Properties tab linked directly to the city's aggregated housing listings on Redfin, so that the websites could consistently list legitimate and active listings at the time of experiment launch.

Commented [GM1]: Highlighted in case you change around where the subject line figure appears in the supplement.

Commented [GM2]: Feel free to delete this sentence

Figure S4: Sample Home Page for Realtor Agency Website



Note: Homepage (top of page) for the Beacon Grove Realty agency website, one of the four realtor agency websites developed for the experiment (i.e., Beacon Grove Realty, Cedar Trail Realty, Maple Heights Realty, and Elm Valley Realty). Each realtor agency website looked the same with the exception of a different DALL·E 3-generated logo and company name for each fictitious agency, the homepage stock photograph for Milwaukee-based agencies contained a photograph of the Milwaukee skyline (rather than the Columbus skyline), and the city name matched the respective city in the subhead description.

The websites and domains were created by a professional website developer hired on Fiverr, the freelancer marketplace. DALL·E 3 generated each agency logo using the prompt: "Can you make a neutral logo for a real estate agency named [insert agency name]" (see Figure S5 for each agency logo).

Figure S5: Realtor Agency Logos

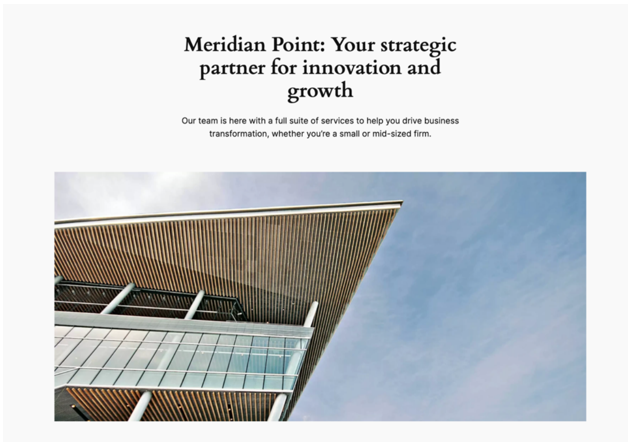
Note: Logos for the four fictitious real estate agencies used in the experiment: Beacon Grove Realty, Cedar Trail Realty, Elm Valley Realty, and Maple Heights Realty. Each logo was generated using DALL·E 3 from the prompt: "Can you make a neutral logo for a real estate agency named '[agency name]'?"

Before the experiment was run, a small group of US agents blind to the experiment hypotheses reviewed the websites: agents reported the websites felt realistic and did not invoke suspicion regarding the fictitious office's legitimacy.

Commented [GM3]: Nick, do you know the exact number? I'm assuming your mom, Ike Silver's husband, also potentially people in Ike Silver's office (though I'm not sure if that's true?), anyone else?

To parallel the level of searchable detail provided for sponsors in our experiment, we also created four websites for the buyer conditions: two fictitious Columbus, Ohio-based consulting firms (Northstream Company and Crescent Bridge; northstreamco.org and crescentbridge.org, respectively) and two fictitious Milwaukee, Wisconsin-based consulting firms (Meridian Point and Summit; meridianpoint.org and summitco.org, respectively). All four websites were also clickable via the buyer's email signature (see Figure S8), presumably representing where the buyer was employed, and described a generic consulting agency offering "a full suite of services to help you drive business transformation, whether you're a small or mid-sized firm" (see Figure S6). Like the realtor agency websites, the buyer websites contained a generic contact email (e.g., "info@meridianpoint.org").

Figure S6: Sample Home Page for Buyer's Website



Note: Homepage (top of page) for Meridian Point, one of the four buyer company websites developed for the experiment (i.e., (Meridian Point, Summit, Northstream Company, and Crescent Bridge). Each buyer website looked the same, with the exception of a different company name for each fictitious company.

<u>Website Traffic.</u>

We monitored website traffic during the full experiment duration (April 24 to May 6, 2025) and in the week that followed where email responses were still recorded (i.e., through to May 13, 2025). Site visits, number of visitors, and site visit duration for each domain are summarized in Table S15. Approximately 8% of realtors contacted viewed the sponsor or buyer's linked website, as measured by the total number of unique site visits divided by the total number of unaffected observations during the experimental period per domain (see also Table S35 for more details on unaffected vs. affected observations).

Table S15. Website Traffic Summary Statistics.

Domain	Total visitors	Average daily visitors	SD daily visitors	Total views	Average daily views	SD daily views	Average session duration (s)	SD session duration (s)	Total unaffected observations
beacongroverealty.com	461	23.05	20.55	1,242	62.10	57.46	98.10	77.63	5,732

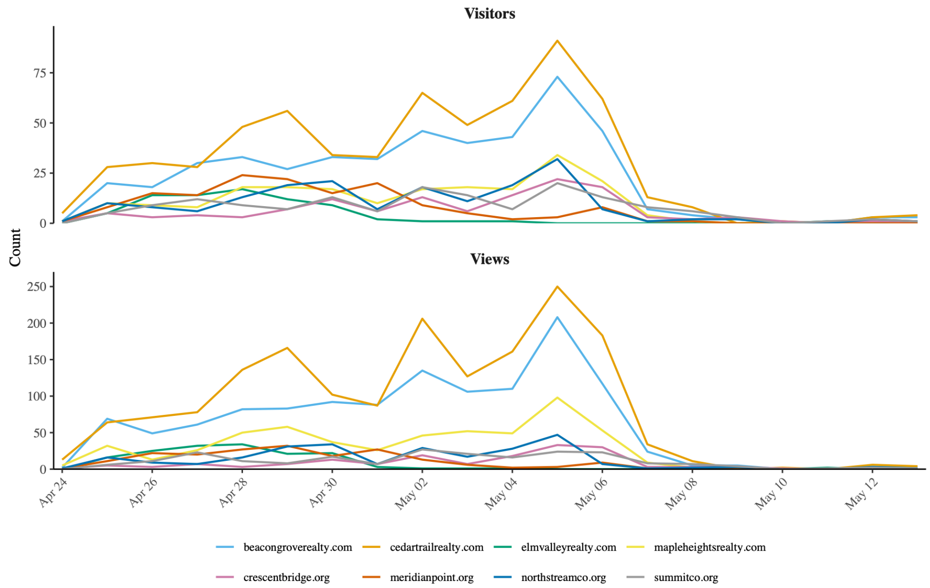
Commented [GM4]: Note I got this metric using your existing total unaffected observations table earlier in the supplement (S23)

cedartrailrealty.com	619	30.95	26.90	1,701	85.05	78.32	108.95	123.03	6,119
elmvalleyrealty.com	77	3.85	5.82	158	7.90	12.04	108.95	250.46	927
mapleheightsrealty.com	204	10.20	9.51	558	27.90	27.10	74.20	68.04	2,658
crescentbridge.org	124	6.20	6.30	164	8.20	9.64	31.15	76.00	1,562
meridianpoint.org	148	7.40	8.15	193	9.65	10.88	62.45	180.26	1,491
northstreamco.org	180	9.00	8.87	259	12.95	13.86	74.50	161.51	2,538
summitco.org	154	7.70	5.86	217	10.85	8.95	119.85	211.12	1,617
Average across all 8 domains	**245.88**	**12.29**	**11.49**	**561.5**	**28.08**	**27.28**	**84.77**	**143.5**	**2,830.5**

Note: Website traffic for each domain (four realtor agency websites and four buyer websites) are displayed above. Average session duration represents the average time-spend on the associated domain per session. Total unaffected observations represents the number of individuals sent emails during the experiment duration, Approximately 8% of realtors visited the company website per domain (i.e., total visitors divided by total unaffected observations); however, because total visitors is defined based on unique browser views (rather than unique persons), it is possible less than 8% of realtors visited the website if many realtors viewed the website from different browsers, for instance, and therefore are counted multiple times within the total visitors metric.

On average, each domain received approximately 245.88 total visitors during the entirety of the 20-day observation window, averaging 12.29 daily visits (SD = 11.49). Each website session lasted on average 84.77 seconds (SD = 143.5). Realtors tended to spend longer on the realtor (M = 97.6 seconds, SD = 16.4) than buyer sites (M = 72.0 seconds, SD = 36.8). Figure S7 summarizes daily average site traffic by domain.

Figure S7: Daily Website Visitors and Views by Domain



Note: Daily website traffic for each domain over the experiment period. Each point represents the number of website visitors (on top) and views (on bottom) per day across each domain. Site volume was low throughout the experimental period: approximately 8% of realtors contacted viewed the associated website, as measured by the total number of unique site visits divided by the total number of unaffected observations during the experimental period per domain. Four domains experienced abnormally low open rates over distinct periods: elmvalleyrealty.com, meridianpoint.org, mapleheightsrealty.com, and northstreamco.org, and emails were reassigned from Elm Valley Realty and Meridian Point following these downturns.

All Final Stimuli

Figure S8: Example Signatures

Darnell Washington

Northstream Company, Columbus, OH
C: (937) 853-8076
E: Darnell.Washington@northstreamco.org

Darnell Washington

Meridian Point, Milwaukee, WI
C: (937) 853-8076
E: Darnell.Washington@meridianpoint.org

Jamal Jackson

Realtor
Beacon Grove Realty, Columbus, OH
C: (937) 853-8076
E: Jamal.Jackson@beacongroverealty.com

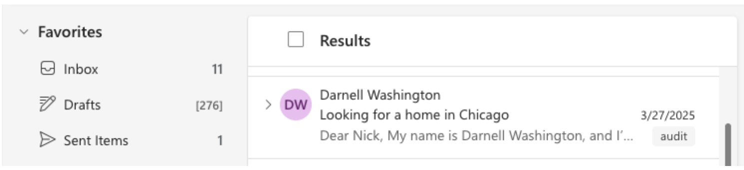
Jamal Jackson

Realtor
Cedar Trail Realty, Milwaukee, WI
C: (937) 853-8076
E: Jamal.Jackson@cedartrailrealty.com

Example Stimulus Emails

Figure S9: Examples of stimulus emails as seen in inbox

(a) Outlook



(b) Gmail

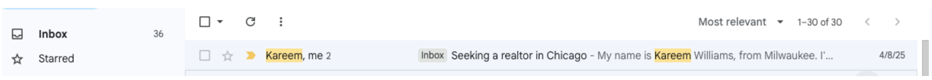
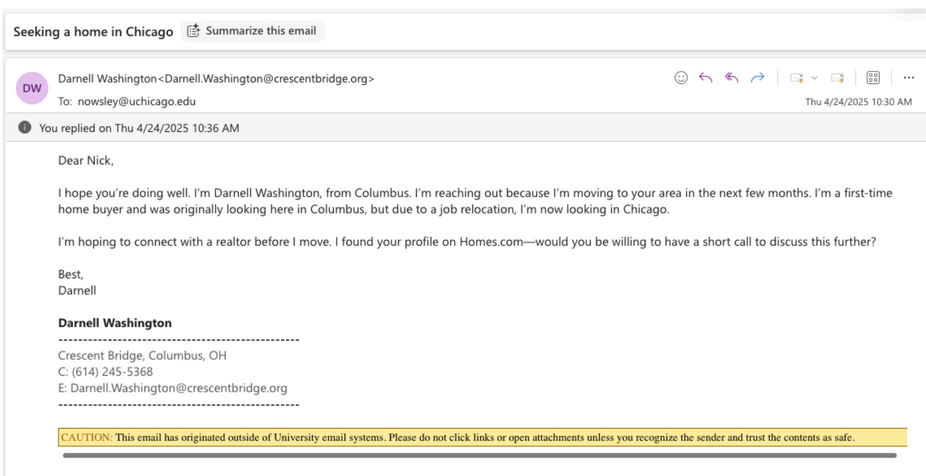


Figure S10: Example Emails as Received in Outlook (Sponsor Email; Buyer Email)



Client seeking a home in Chicago Summarize this email

SW Scott Wagner<Scott.Wagner@beacongroverealty.com>

To: Owsley, Nicholas Calbraith (Nick)

Reply Reply all Forward

Wed 3/26/2025 11:45 AM

You replied on Thu 3/27/2025 12:20 PM

Dear Nick,

I hope you're doing well. I'm Scott Wagner, and I work at Beacon Grove Realty in Columbus. I'm reaching out because a current client of mine, Darnell Washington, is moving to your area in the next few months. Darnell is a first-time home buyer who's been amazing to work with. We started looking in Columbus, but due to a job relocation, he is now looking in Chicago.

I'm hoping to connect him with a realtor in Chicago before he moves. I found your profile on Realtor.com — would you be willing to have a short call to discuss this further?

Best,
Scott

Scott Wagner

Realtor
Beacon Grove Realty, Columbus, OH
C: (614) 210-3799
E: Scott.Wagner@beacongroverealty.com

Figure S11: Example Emails as Received in Gmail (Sponsor Email; Buyer Email)

Looking to connect with a realtor in Chicago Inbox × Audit ×

Kareem Williams <Kareem.Williams@meridianpoint.org> Mon, Apr 7, 2025, 5:45 PM

to me

Dear Nick,

My name is Kareem Williams, from Milwaukee. I'm moving to the Chicago area in June, and was hoping to connect with a realtor before my move to help me look for properties there. I'm currently renting here in Milwaukee, but looking to become a first-time homeowner in Chicago.

I'm currently vetting agents and found your profile on Homes.com. Would you be willing to set up a quick call to chat about potential next steps?

Thanks,
Kareem

Kareem Williams

Meridian Point, Milwaukee, WI
C: (414) 207-9686
E: Kareem.Williams@meridianpoint.org

Referring a client looking for a home in Chicago Inbox ×

Edwin Schmidt <Edwin.Schmidt@mapleheightsrealty.com> Fri, Apr 18, 2025, 6:04 PM

to me

Dear Nick,

My name is Edwin Schmidt from Maple Heights Realty, in Milwaukee. I have a client, Scott Wagner, moving to the Chicago area in June, and I was hoping to connect him with a local realtor before his move to help him look for properties there. Scott has been amazing to work with!

I'm currently vetting agents for Scott and found your profile on Realtor.com. Would you be willing to set up a quick call to chat about potential next steps?

Thanks,
Edwin

Edwin Schmidt

Realtor
Maple Heights Realty, Milwaukee, WI
C: (414) 209-4104
E: Edwin.Schmidt@mapleheightsrealty.com

Figure S12: Example Sponsor Email — Intended Display (with Signature)

JJ   Jamal Jackson < jamal.jackson@beacongroverealty.com>

Hi <Realtor Name>,

My name is Jamal Jackson from Beacon Grove Realty, in Columbus. I have a client, Darnell Washington, moving to the Boston area in May, and I was hoping to connect him with a local realtor before his move to help him look for properties there. I helped Darnell find his current rental in Columbus, and he has been amazing to work with, but he’s looking to become a first-time homeowner in Boston.

I’m currently vetting agents for Darnell and found your profile on Realtor.com. Would you be willing to set up a quick call to chat about potential next steps?

Thanks,
Jamal

Jamal Jackson
--
Realtor
Beacon Grove Realty, Columbus, OH
C: (937) 853-8076
E: Jamal.Jackson@beacongroverealty.com
--

Figure S13: Example Buyer Email — Intended Display (with Signature)

DW

Darnell Washington < darnell.washington@northstreamco.org>

Hi <Realtor Name>,

My name is Darnell Washington, and I’m from Columbus. I’m moving to the Boston area in May, and was hoping to connect with a realtor before my move to help me look for properties there. I’m currently renting here in Columbus, but looking to become a first-time homeowner in Boston.

I’m currently vetting agents and found your profile on Realtor.com. Would you be willing to set up a quick call to chat about potential next steps?

Thanks,
Darnell

Darnell Washington

Northstream Company, Columbus, OH
C: (937) 853-8076
E: Darnell.Washington@northstreamco.org

Final Stimulus Email Templates

Table S16. Final Stimulus Email Templates

Email index	Email name	Email subject	Email body
1	Plain 1 (Sponsor)	Referring a client looking for a home in <Participant City>	Dear <Participant First Name>, My name is <Sponsor Full Name> from <Brokerage Name>, in <Sender City>. I have a client, <Candidate Full Name>, moving to the <Participant City> area in June, and I was hoping to connect him with a local realtor before his move to help him look for properties there. <Candidate First Name> has been amazing to work with! I'm currently vetting agents for <Candidate First Name> and found your profile on <Source>. Would you be willing to set up a quick call to chat about potential next steps? Thanks, <Sponsor First Name>
2	Buyer 1 (Sponsor)	Referring a client seeking a realtor in <Participant City>	Dear <Participant First Name>, My name is <Sponsor Full Name> from <Brokerage Name>, in <Sender City>. I have a client, <Candidate Full Name>, moving to the <Participant City> area in June, and I was hoping to connect him with a local realtor before his move to help him look for properties there. <Candidate First Name> is a first-time home buyer who's been amazing to work with. We started looking in <Sender City>, but due to a job relocation, he is now looking in <Participant City>. I'm currently vetting agents for <Candidate First Name> and found your profile on <Source>. Would you be willing to set up a quick call to chat about potential next steps? Thanks, <Sponsor First Name>

3	Rented 1 (Sponsor)	Connecting my client with a realtor in <Participant City>	Dear <Participant First Name>, My name is <Sponsor Full Name> from <Brokerage Name>, in <Sender City>. I have a client, <Candidate Full Name>, moving to the <Participant City> area in June, and I was hoping to connect him with a local realtor before his move to help him look for properties there. I helped <Candidate First Name> find his current rental in <Sender City>, and he has been amazing to work with, but he’s looking to become a first-time homeowner in <Participant City>. I’m currently vetting agents for <Candidate First Name> and found your profile on <Source>. Would you be willing to set up a quick call to chat about potential next steps? Thanks, <Sponsor First Name>
4	Plain 2 (Sponsor)	Client referral for a home purchase in <Participant City>	Dear <Participant First Name>, I hope you’re doing well. I’m <Sponsor Full Name>, and I work at <Brokerage Name> in <Sender City>. I’m reaching out because a current client of mine, <Candidate Full Name>, is moving to your area in the next few months. <Candidate First Name>'s been a pleasure to work with. He's looking for a place in <Participant City> and I’m hoping to connect him to a local realtor before he moves. I found your profile on <Source> — would you be willing to have a short call to discuss this further? Best, <Sponsor First Name>

<table>
<tr><td>5</td><td>Buyer 2 (Sponsor)</td><td>Client seeking a home in <Participant City></td><td>Dear <Participant First Name>,

I hope you’re doing well. I’m <Sponsor Full Name>, and I work at <Brokerage Name> in <Sender City>. I’m reaching out because a current client of mine, <Candidate Full Name>, is moving to your area in the next few months. <Candidate First Name> is a first-time home buyer who’s been amazing to work with. We started looking in <Sender City>, but due to a job relocation, he is now looking in <Participant City>.

I’m hoping to connect him with a realtor in <Participant City> before he moves. I found your profile on <Source> — would you be willing to have a short call to discuss this further?

Best,
<Sponsor First Name></td></tr>
<tr><td>6</td><td>Rented 2 (Sponsor)</td><td>Client looking for a realtor in <Participant City></td><td>Dear <Participant First Name>,

I hope you’re doing well. I’m <Sponsor Full Name>, and I work at <Brokerage Name> in <Sender City>. I’m reaching out because a current client of mine, <Candidate Full Name>, is moving to your area in the next few months. I helped <Candidate First Name> find his current rental in <Sender City>, and he has been amazing to work with, but he’s looking to become a first-time homeowner in <Participant City>.

I’m hoping to connect him with a realtor in <Participant City> before he moves. I found your profile on <Source> — would you be willing to have a short call to discuss this further?

Best,
<Sponsor First Name></td></tr>
<tr><td>7</td><td>Plain 1 (Candidate)</td><td>Looking for a home in <Participant City></td><td>Dear <Participant First Name>,

My name is <Candidate Full Name>, from <Sender City>. I’m moving to the <Participant City> area in June, and was hoping to connect with a realtor before my move to help me look for properties there.

I’m currently vetting agents and found your profile on <Source>. Would</td></tr>
</table>

			you be willing to set up a quick call to chat about potential next steps? Thanks, <Candidate First Name>
8	Buyer 1 (Candidate)	Seeking a realtor in <Participant City>	Dear <Participant First Name>, My name is <Candidate Full Name>, from <Sender City>. I'm moving to the <Participant City> area in June, and was hoping to connect with a realtor before my move to help me look for properties there. I'm a first-time home buyer and was originally looking here in <Sender City>, but due to a job relocation, I'm now looking in <Participant City>. I'm currently vetting agents and found your profile on <Source>. Would you be willing to set up a quick call to chat about potential next steps? Thanks, <Candidate First Name>
9	Rented 1 (Candidate)	Looking to connect with a realtor in <Participant City>	Dear <Participant First Name>, My name is <Candidate Full Name>, from <Sender City>. I'm moving to the <Participant City> area in June, and was hoping to connect with a realtor before my move to help me look for properties there. I'm currently renting here in <Sender City>, but looking to become a first-time homeowner in <Participant City>. I'm currently vetting agents and found your profile on <Source>. Would you be willing to set up a quick call to chat about potential next steps? Thanks, <Candidate First Name>

10	Plain 2 (Candidate)	Purchasing a home in <Participant City>	Dear <Participant First Name>, I hope you're doing well. I'm <Candidate Full Name>, from <Sender City>. I'm reaching out because I'm moving to your area in the next few months. I'm looking for a place in <Participant City> and am hoping to connect with a realtor before I move. I found your profile on <Source>—would you be willing to have a short call to discuss this further? Best, <Candidate First Name>
11	Buyer 2 (Candidate)	Seeking a home in <Participant City>	Dear <Participant First Name>, I hope you're doing well. I'm <Candidate Full Name>, from <Sender City>. I'm reaching out because I'm moving to your area in the next few months. I'm a first-time home buyer and was originally looking here in <Sender City>, but due to a job relocation, I'm now looking in <Participant City>. I'm hoping to connect with a realtor before I move. I found your profile on <Source>—would you be willing to have a short call to discuss this further? Best, <Candidate First Name>
12	Rented 2 (Candidate)	Looking for a realtor in <Participant City>	Dear <Participant First Name>, I hope you're doing well. I'm <Candidate Full Name>, from <Sender City>. I'm reaching out because I'm moving to your area in the next few months. I'm currently renting here in <Sender City>, but looking to become a first-time homeowner in <Participant City>. I'm hoping to connect with a realtor before I move. I found your profile on <Source>—would you be willing to have a short call to discuss this further? Best, <Candidate First Name>

Final Brokerage/Agency and Company Details

Table S17. Final Sponsor Brokerage Details

Index	Name	Domain	City	State
1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
3	Maple Heights Realty	mapleheightsrealty.com	Milwaukee	WI
4	Elm Valley Realty	elmvalleyrealty.com	Columbus	OH

Table S18. Final Buyer Company Details

Index	Name	Domain	City	State
1	Northstream	northstreamco.org	Columbus	OH
2	Meridian Point	meridianpoint.org	Milwaukee	WI
3	Summit	summitco.org	Milwaukee	WI
4	Crescent Bridge	crescentbridge.org	Columbus	OH

Final Stimulus Accounts: Buyers and Sponsors

Table S19. Final Stimulus Accounts: Buyers

Index	First_name	Last_name	Phone	Race	Agency id	Agency name	Agency domain	City	State
1	Denzel	Booker	(614) 210-3180	Black	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
2	Tyrone	Banks	(614) 210-3665	Black	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
3	Kareem	Williams	(614) 219-9593	Black	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
4	Jamal	Jackson	(614) 219-9044	Black	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
5	Darnell	Washington	(614) 219-9542	Black	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
6	Darius	Jefferson	(614) 219-9344	Black	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
7	Edwin	Schmidt	(614) 210-3751	White	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
8	Seth	Ryan	(614) 219-9054	White	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
9	Cody	Olson	(614) 219-9539	White	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
10	Steven	Jensen	(614) 219-9279	White	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
11	Scott	Wagner	(614) 210-3799	White	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
12	Dustin	Snyder	(614) 219-9246	White	1	Beacon Grove Realty	beacongroverealty.com	Columbus	OH
13	Denzel	Booker	(414) 207-8772	Black	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
14	Tyrone	Banks	(414) 207-8880	Black	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
15	Kareem	Williams	(414) 207-8669	Black	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
16	Jamal	Jackson	(414) 207-4473	Black	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
17	Darnell	Washington	(414) 207-9180	Black	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
18	Darius	Jefferson	(414) 207-4573	Black	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
19	Edwin	Schmidt	(414) 207-9496	White	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
20	Seth	Ryan	(414) 207-4761	White	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
21	Cody	Olson	(414) 207-4834	White	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
22	Steven	Jensen	(414) 207-4508	White	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
23	Scott	Wagner	(414) 207-9494	White	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
24	Dustin	Snyder	(414) 207-8813	White	2	Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
25	Denzel	Booker	(414) 207-4769	Black	3	Maple Heights Realty	mapleheightsrealty.com	Milwaukee	WI
26	Tyrone	Banks	(414) 207-4431	Black	3	Maple Heights Realty	mapleheightsrealty.com	Milwaukee	WI
27	Kareem	Williams	(414) 207-8333	Black	3	Maple Heights Realty	mapleheightsrealty.com	Milwaukee	WI
28	Jamal	Jackson	(614) 219-9338	Black	4	Elm Valley Realty	elmvalleyrealty.com	Columbus	OH
29	Darnell	Washington	(614) 245-5366	Black	4	Elm Valley Realty	elmvalleyrealty.com	Columbus	OH
30	Darius	Jefferson	(614) 245-5129	Black	4	Elm Valley Realty	elmvalleyrealty.com	Columbus	OH
31	Edwin	Schmidt	(414) 209-4104	White	3	Maple Heights Realty	mapleheightsrealty.com	Milwaukee	WI
32	Seth	Ryan	(414) 208-4364	White	3	Maple Heights Realty	mapleheightsrealty.com	Milwaukee	WI
33	Cody	Olson	(414) 215-9411	White	3	Maple Heights Realty	mapleheightsrealty.com	Milwaukee	WI
34	Steven	Jensen	(614) 245-5176	White	4	Elm Valley Realty	elmvalleyrealty.com	Columbus	OH
35	Scott	Wagner	(614) 245-5096	White	4	Elm Valley Realty	elmvalleyrealty.com	Columbus	OH

36	Dustin	Snyder	(614) 259-7052	White	4	Elm Valley Realty	elmvalleyrealty.com	Columbus	OH

Table S20. Final Stimulus Accounts: Sponsors

Index	First_name	Last_name	Phone	Race	Company id	Company name	Domain	City	State
1	Denzel	Booker	(614) 219-9039	Black	1	Northstream	northstreamco.org	Columbus	OH
2	Tyrone	Banks	(614) 219-9824	Black	1	Northstream	northstreamco.org	Columbus	OH
3	Kareem	Williams	(614) 219-9712	Black	1	Northstream	northstreamco.org	Columbus	OH
4	Jamal	Jackson	(614) 219-9441	Black	1	Northstream	northstreamco.org	Columbus	OH
5	Darnell	Washington	(614) 219-9273	Black	1	Northstream	northstreamco.org	Columbus	OH
6	Darius	Jefferson	(614) 210-3055	Black	1	Northstream	northstreamco.org	Columbus	OH
7	Edwin	Schmidt	(614) 210-3017	White	1	Northstream	northstreamco.org	Columbus	OH
8	Seth	Ryan	(614) 210-3841	White	1	Northstream	northstreamco.org	Columbus	OH
9	Cody	Olson	(614) 219-9128	White	1	Northstream	northstreamco.org	Columbus	OH
10	Steven	Jensen	(614) 219-9578	White	1	Northstream	northstreamco.org	Columbus	OH
11	Scott	Wagner	(614) 219-9782	White	1	Northstream	northstreamco.org	Columbus	OH
12	Dustin	Snyder	(614) 219-9793	White	1	Northstream	northstreamco.org	Columbus	OH
13	Denzel	Booker	(414) 207-9764	Black	2	Meridian Point	meridianpoint.org	Milwaukee	WI
14	Tyrone	Banks	(414) 207-8275	Black	2	Meridian Point	meridianpoint.org	Milwaukee	WI
15	Kareem	Williams	(414) 207-9686	Black	2	Meridian Point	meridianpoint.org	Milwaukee	WI
16	Jamal	Jackson	(414) 207-8419	Black	2	Meridian Point	meridianpoint.org	Milwaukee	WI
17	Darnell	Washington	(414) 207-4282	Black	2	Meridian Point	meridianpoint.org	Milwaukee	WI
18	Darius	Jefferson	(414) 207-4731	Black	2	Meridian Point	meridianpoint.org	Milwaukee	WI
19	Edwin	Schmidt	(414) 207-8633	White	2	Meridian Point	meridianpoint.org	Milwaukee	WI
20	Seth	Ryan	(414) 207-6683	White	2	Meridian Point	meridianpoint.org	Milwaukee	WI
21	Cody	Olson	(414) 207-4686	White	2	Meridian Point	meridianpoint.org	Milwaukee	WI
22	Steven	Jensen	(414) 207-4264	White	2	Meridian Point	meridianpoint.org	Milwaukee	WI
23	Scott	Wagner	(414) 207-6689	White	2	Meridian Point	meridianpoint.org	Milwaukee	WI
24	Dustin	Snyder	(414) 207-8130	White	2	Meridian Point	meridianpoint.org	Milwaukee	WI
25	Denzel	Booker	(414) 215-0568	Black	3	Summit	summitco.org	Milwaukee	WI
26	Tyrone	Banks	(414) 207-9061	Black	3	Summit	summitco.org	Milwaukee	WI
27	Kareem	Williams	(414) 215-9773	Black	3	Summit	summitco.org	Milwaukee	WI
28	Jamal	Jackson	(614) 245-5186	Black	4	Crescent Bridge	crescentbridge.org	Columbus	OH
29	Darnell	Washington	(614) 245-5368	Black	4	Crescent Bridge	crescentbridge.org	Columbus	OH
30	Darius	Jefferson	(614) 254-6692	Black	4	Crescent Bridge	crescentbridge.org	Columbus	OH
31	Edwin	Schmidt	(414) 207-9278	White	3	Summit	summitco.org	Milwaukee	WI
32	Seth	Ryan	(414) 212-5283	White	3	Summit	summitco.org	Milwaukee	WI
33	Cody	Olson	(414) 207-4859	White	3	Summit	summitco.org	Milwaukee	WI

34	Steven	Jensen	(614) 254-6692	White	4	Crescent Bridge	crescentbridge.org	Columbus	OH
35	Scott	Wagner	(614) 233-1402	White	4	Crescent Bridge	crescentbridge.org	Columbus	OH
36	Dustin	Snyder	(614) 245-5147	White	4	Crescent Bridge	crescentbridge.org	Columbus	OH

Pilots

Between December 2023 and April 2025, we conducted three separate pilots, sending email inquiries to practicing realtors. Each served a distinct purpose: Pilot 1 provided an initial test of the email stimuli and response protocols; Pilot 2 informed the design of our email signatures and helped validate our email sampling and filtering procedures; and Pilot 3 was a full-scale operational test of the complete experimental system in advance of the main study, used to inform small changes to protocol and finalize hypothesis tests and other elements of our preregistration.

Pilot 1

The goals of Pilot 1 were to: (1) establish approximate response rates to inform statistical power calculations for the main experiment; (2) trial an early version of the stimuli to inform its further development, response protocols, and email sending processes, and to gauge levels of suspicion of the stimulus; and (3) collect a sample of realtor responses to inform outcome variable coding and other elements of pre-registration.

We acquired email addresses from the publicly available profiles of 52 active practicing realtors across 24 brokerages in the Boston, MA area, including both large corporate and small independent brokerages. To sample a greater spread of realtor types and reduce the risk of contamination, we sent emails to no more than two realtors within the same office. We set up an account for a fictitious realtor, "Jerome Jackson," including an AI-generated profile photograph and company logo, and a Google Voice phone number to capture responses via call and text. On December 5th, 2023, two research assistants manually sent the email shown in Figure S14 from this fictitious realtor account, in the role of a sponsor introducing a fictitious buyer, Darnell Williams, to the recipient realtors.[12] We based these stimuli on the first round of consultations with our primary practicing realtor contact described in the previous sub-section.

We received 23 email responses from 52 inquiries (44.2%) and 4 phone calls, all from realtors who had also responded by email. Only 1 response declined to offer assistance, and none indicated suspicion of the veracity of the inquiry. These responses also yielded a sample of realtor email signatures, which we

[12] This pilot used preliminary stimulus names that were subsequently revised through the name selection process described above. The final experiment used Darnell Washington rather than Darnell Williams as the buyer name.

combined with data from Pilot 2 to inform the design of our final email signature stimuli (see Pilot 2 below).

Figure S14: Pilot 1 Stimulus Email

Hi <Realtor Name>,

My name is Jerome Jackson from Dayton Real Estate, in Dayton, OH. I have a client, Darnell Williams, moving to the Boston area in May. I'm listing his property here for sale and was hoping to connect him with a local agent before his move.

I'm currently vetting agents for Darnell, and I found your profile on Zillow. Darnell is a lovely client to work with. Would you be willing to set up a quick call to chat about potential next steps?

Thanks,

Jerome

Jerome Jackson

Realtor

DAYTON REAL ESTATE

(937) 853-8076

jerome.jackson.daytonre@gmail.com

Dayton Real Estate

S Main St, Dayton, OH 45402

IMPORTANT NOTICE: Never trust wiring instructions sent via email. Cyber criminals are hacking email accounts and sending emails with fake wiring instructions. These emails are convincing and sophisticated. Always independently confirm wiring instructions in person or via a telephone call to a trusted and verified phone number. Never wire money without double-checking that the wiring instructions are correct.

Pilot 2: Realtor Signatures

Pilot 2 had two objectives: (1) to augment the signature sample from Pilot 1 with signatures from realtors drawn from our actual sampling frame and from a broader set of cities, in order to design more representative email signatures for the main experiment; (2) to test the ZeroBounce email verification tool on a sample from our dataset, to inform our sample filtering procedure.

Three research assistants sent email inquiries from their own university accounts to 93 practicing realtors in San Francisco, New York, and Chicago, randomly selected from our full dataset. These inquiries presented the research assistants as seeking help finding a rental for the upcoming semester. We received 20 responses from the 93 emails sent.

Turning first to signatures: combining responses from Pilots 1 and 2 yielded a sample of 43 email signatures. As reported in the table below, most realtors included a formal signature (81%), all of which contained a phone number. Email addresses were included in 70% of all responses, website URLs in 56%, photos in 26%, and logos in 37%. Signatures from the two pilots were largely similar in composition. On the basis of these results, we modelled our fictitious sponsor signatures on the modal format — name, position, firm name, phone number, and email address. We thus opted not to include photographs as initially planned (and done in Pilot 1), as they were present in fewer than one in three signatures (and thus not normative), and would have introduced additional design and deliverability risks: images increase the likelihood of email service providers screening out emails or hiding content.

Table S21. Pilot 2: Signature Features

	Pilot 2	Pilot 1	All
Photo with Account	35%	48%	42%
Has a Signature	80%	83%	81%
Signature features			
Phone number	80%	83%	81%
Email	75%	65%	70%
Website url	55%	57%	56%
Photo	35%	17%	26%
Logo	35%	39%	37%
Any image	60%	39%	49%
N	20	23	43

Turning to email validity: of the 171 email addresses submitted to ZeroBounce, 93 (54%) were rated as valid and currently active with high certainty. This relatively low validity rate reflected the fact that this subsample had not yet been filtered using the activity-based criteria we subsequently developed for the main study. Indeed, several features of the dataset predicted email validity, most notably realtors' recorded activity on Realtor.com and Homes.com — the platforms from which we sourced our addresses. This finding partly informed the activity-based filtering procedure described in the Realtor Sample section above, which produced a substantially higher email validity rate in the main sample.

Pilot 3: Main Pilot

We refer to Pilot 3 as the main pilot. It was conducted on April 8–12, 2025, sixteen days before the launch of the main study, and served three broad purposes.

First, operational testing: we tested the customized automated system for sending initial inquiry emails; verified the response-capture and RA response protocols; and tested all eight email domains and 72 individual email accounts for spam complaints, deliverability issues, or other operational risks.

Second, pre-registration preparation: we used pilot responses to finalize our outcome variable definitions, conduct power calculations for the main experiment, and develop and pilot our analysis scripts and hypothesis tests.

Third, contamination and suspicion screening: we examined responses for any indication that realtors were suspicious of emails, had seen an inquiry received by a colleague, or that any public mention of the experiment had occurred. This informed our randomization approach — specifically, whether to randomize by individual participant or by state.

To these ends, we used the automated system described in the Email Domains and Sending Protocol section to send initial inquiry emails to 1,500 realtors from all 50 U.S. states, randomly selected using the same procedure described in the Realtor Sample section and thus identical in expectation to the main sample. As in the main experiment, participants were randomly assigned to one of six treatment conditions, received emails from one of the 72 individual accounts across eight domains, and were assigned a randomized email template and send order (See the relevant section above for the full randomization procedure). Team response protocols followed the same procedures as the main study, using template responses except in specified circumstances, with responses sent 36–72 hours after receiving a reply from a realtor.

We received 393 responses from 1,500 emails sent (26.2%), with a 65% open rate, 4 bounced emails, and 1 spam complaint. All emails were sent at their scheduled times, and no domains or individual accounts encountered operational issues. Cross-checks of individual inboxes confirmed that all responses were captured accurately by the automated system, and checks of 100 randomly selected outbound emails confirmed that the system correctly preserved and assigned all randomized stimulus features, including buyer and sponsor names, domain, email template, and send order. No evidence of contamination was identified. The deidentified pilot data, including all condition assignments, are available at [LINK FORTHCOMING].

Table S22: Pilot 3 (Main Pilot) — Primary Regression Results

	RQ1: No Sponsor	RQ2: Sponsor	RQ2.2: Black & White Sponsor	RQ3: Sponsor Only Subset	RQ4: BlackCandidate x Sponsor	RQ5: Full Interaction
(Intercept)	0.328***	0.282***	0.282***	0.278***	0.328***	0.328***
	(0.028)	(0.020)	(0.020)	(0.020)	(0.028)	(0.028)
black_candidate	−0.092*				−0.092*	−0.092*
	(0.040)				(0.040)	(0.040)
sponsor_dum		−0.015			−0.071*	
		(0.024)			(0.034)	
black_sponsor			−0.027	−0.023		−0.097*
			(0.028)	(0.028)		(0.040)
white_sponsor			−0.004			−0.044
			(0.028)			(0.040)
black_candidate × sponsor_dum					0.111*	
					(0.049)	
black_candidate × black_sponsor						0.141*
						(0.056)
black_candidate × white_sponsor						0.080
						(0.056)
Num.Obs.	500	1501	1501	1001	1501	1501
R2	0.010	0.000	0.001	0.001	0.004	0.005
R2 Adj.	0.008	−0.000	−0.001	−0.000	0.002	0.002
AIC	619.1	1831.9	1833.3	1210.7	1830.1	1832.3
BIC	627.5	1842.5	1849.2	1220.5	1851.4	1864.2
RMSE	0.45	0.44	0.44	0.44	0.44	0.44
Std.Errors	IID	IID	IID	IID	IID	IID

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Although this pilot was not pre-registered, its results informed and finalized our pre-registration, including outcome variable definitions and hypothesis tests. The primary results from the pilot, presented in the table above, showed that as in our main experiment, Black buyers received fewer positive responses than White buyers in the no-sponsor condition (RQ1: β = −0.092, *p* < .05), consistent with racial discrimination in realtor responsiveness. The presence of any sponsor also did not significantly affect response rates overall (RQ2: $\beta = -0.015$, p = ns). However, the interaction results and differential response by sponsor race did not show the same pattern as our main results. Given the relatively small sample, and the lack of pre-registration for these results, we chose not to interpret and explicitly compare them and treat our main experiment as the primary tests of our hypotheses.

Realtor Surveys

Realtor Conference Survey

Overview

In January 2026, we conducted a survey with realtors attending the Chicago Association of Realtors' 2026 Market Outlook Conference, held at the Radisson Blu Chicago. The goal of the survey was to understand how practicing realtors received client outreach with a focus on referrals from other realtors. We were specifically interested in how often clients reached out to realtors via email, how common realtor referrals were, including out-of-state referrals (like those used in our stimuli), the perceived costs and benefits of referrals (i.e., of being engaged by a sponsor), what criteria realtors valued in prospective buyers, and finally how sponsorship in the form of a referral might affect their perceptions of buyers. An author and a research assistant attended the conference and invited participants to complete a 5-minute survey using a QR code displayed at the event. Realtors participating in the survey stood a chance to win a $100 Amazon Gift Card. 19 realtors in attendance at the event participated in and 16 completed the survey. Below are the survey questions and results.

Survey Questions

The survey was administered in-person at the conference. Questions are presented in the order they appeared to respondents.

Study Information (IRB consent and study description displayed to participants)

1. This study focuses specifically on experiences with residential markets. Have you sold/rented residential properties?

- Yes
- No

2. How many hours per week do you work as a realtor currently?

- None, I do not work as a realtor currently
- 1–9 hours per week

- 10–50 hours per week
- 51+ hours per week

3. How many years have you worked as a realtor? [Open Text Response]

4. Through which of the following channels do new clients contact you about buying/renting a home? (Select all that apply)
   - Phone
   - Email
   - In-person
   - Events/Showings
   - Realtor.com listing
   - Zillow listing
   - Other (please specify)

5. What is the most common way that new clients contact you about buying/renting a home?
   - Phone
   - Email
   - In-person
   - Events/Showings
   - Realtor.com listing
   - Zillow listing
   - Other (please specify)

6. How often do you receive referrals from other realtors?
   - Never
   - Approximately once a year
   - Approximately once a quarter
   - Approximately once a month
   - Several times a month
   - Every week

7. How often do you receive referrals from realtors from out-of-state?
   - Never
   - Approximately once a year
   - Approximately once a quarter
   - Approximately once a month
   - Several times a month
   - Every week

8. How often do prospective (or current) clients email you with inquiries about buying a new home?
   - Never
   - Approximately once a year
   - Approximately once a quarter
   - Approximately once a month
   - Several times a month
   - Every week

9. What are the main criteria you look for when deciding to take on a new buyer? [Open Text Response]

10. If you had to pick just three primary criteria you look for when deciding to take on a new buyer, which three from the list below are your top priority? (Select up to three)

- Client seems friendly
- Client seems responsive / communicative
- Client seems financially responsible
- Client seems to have proof of funds / pre-approval
- Client seems to have a budget aligned with that of typical clients of mine
- Client seems trustworthy / reliable
- Client seems to have needs and preferences that I think I could meet
- Client shows possibility of repeat business / long-term relationship
- Other

11. What are the primary benefits of working with buyers through a referral? [Open Text Response]

12. What are the primary costs of working with buyers through a referral? [Open Text Response]

13. What are the differences between clients who contact you directly compared to those you connect with via a referral (if any)? [Open Text Response]

14. What is the most common price range of the residential properties you sell?

- Under $50,000
- $50,000–$99,999
- $100,000–$199,999
- $200,000–$299,000
- $300,000–$500,000
- $500,000–$749,000
- $750,000–$999,999
- $1,000,000–$1,499,000
- Over $1,499,000
- Prefer not to answer

15. We value your advice and comments. Do you have any additional comments or feedback for the research team? [Open Text Response]

16. Thank you for your time! If you wish to enter the Amazon raffle for a chance to win a $100 Amazon gift card, please provide an email address that we can use to contact you if you win. [Open Text Response]

Sample Descriptive Statistics

Below summarizes survey responses from attending realtors regarding their experience, target market, and current working hours. The average years of experience and average value of listings sold is very similar to the sample in the main audit experiment.

Table S23. Response overview:

Metric	N	%
Total responses	19	100.0
Completed survey	16	84.2
Sold residential property	19	100.0

Table S24. Years of experience:

Statistic	Years
Mean	12.2
Median	10.0
Min	1.0
Max	30.0

Table S25. Hours worked per week.

Hours per week	N	%
1-9 hours per week	3	16.7
10-50 hours per week	12	66.7
51+ hours per week	3	16.7

Table S26. Average listing price range

Price range	N	%
Under $50,000	1	6.2
$50,000-$99,999	0	0.0
$100,000-$199,999	0	0.0
$200,000-$299,000	4	25.0
$300,000-$500,000	8	50.0
$500,000-$749,000	2	12.5
$750,000-$999,999	0	0.0
Over $1,499,000	1	6.2
Prefer not to answer	0	0.0

Results: Buyer Outreach

Respondents were asked through which channels new clients contacted them. While phone calls were the most common mode of new client outreach, emails were highly important. More than 50% of realtors were regularly engaged by new clients through email outreach. 50% indicated that this occurred at least several times a month, while only 6.2% indicated never receiving initial client inquiries through email.

Figure S15

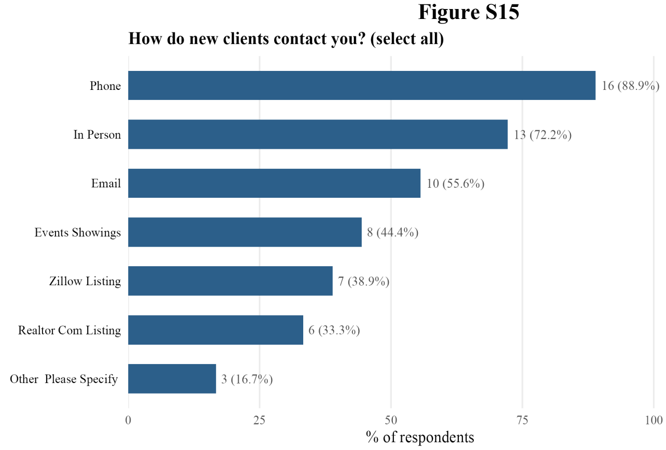
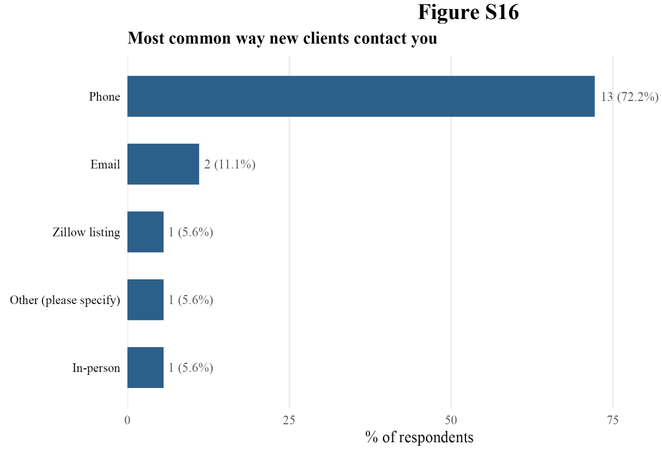


Figure S16



Respondents reported how often they received referrals from other realtors (including out-of-state). First, all realtors in the sample received referrals, and 25% of the sample received referrals at least once a month. Out-of-state referrals were less common, but still prevalent: 62.5% of realtors received at least some referrals from out-of-state.

Figure S17

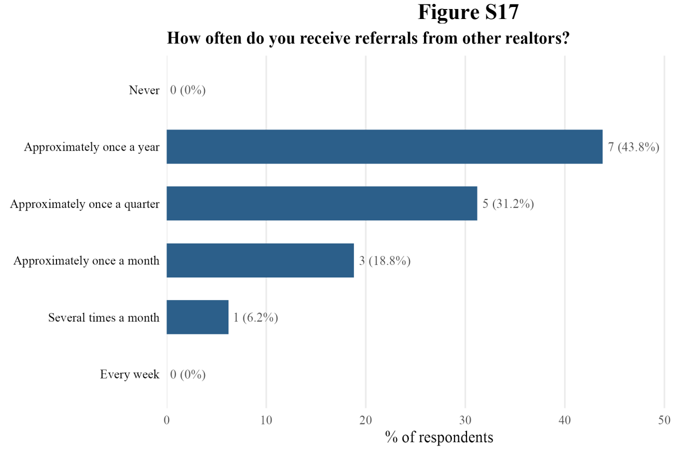


Figure S18

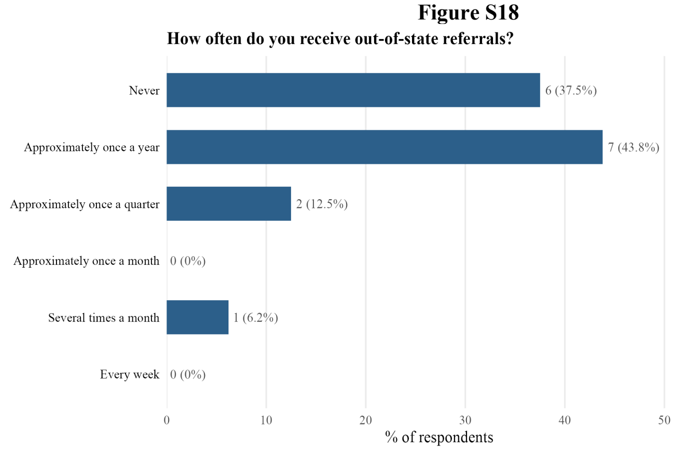
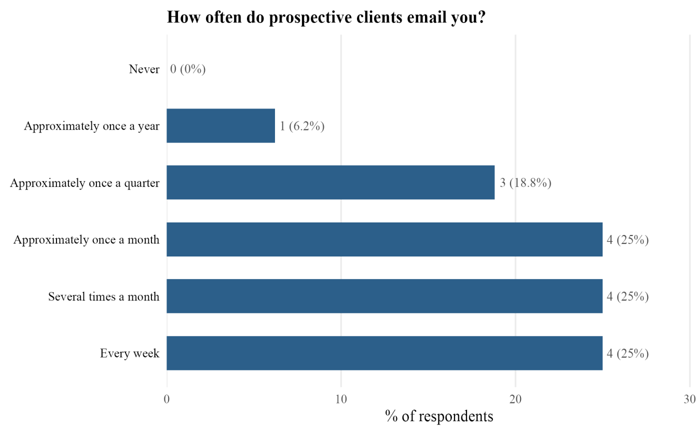


Figure S19



<u>Results: Criteria for Taking on a New Buyer</u>

Respondents selected their top 3 criteria for determining whether to take on a new buyer from a list of possible criteria, and described their main criteria in their own words (i.e., open-text). For open-text responses, we used an LLM to qualitatively classify similar responses. The authors then checked and iterated classifications and made final manual qualitative coding in cases where the LLM coding appeared inaccurate. Across both open-text and categorical responses, proof of funds / pre-approval appeared highly important factors, and were the primary criterion for taking on a new buyer as a client. Next, in the open-response data, more than half the respondents indicated (more than any other classification) buyer "readiness" or seriousness as important, which translates into buyers being prepared to purchase a home, rather than simply exploring, and thus more likely to convert interest into a purchase. This is correlated with pre-approval, which also signals readiness/seriousness, as it is a costly signal from buyers that they intend to purchase (since acquiring pre-approval takes time and effort and financial resources). Being able to meet the client's needs and client-realtor fit were also described as important, with buyer communicativeness being one of the more important categorical choices in the multiple-choice question. Budget and finances (independent of pre-approval), and client friendliness were seen as relatively less important features of a prospective buyer from the realtors' perspective.

Figure S20

Top 3 criteria for taking on a new buyer (multi-select)

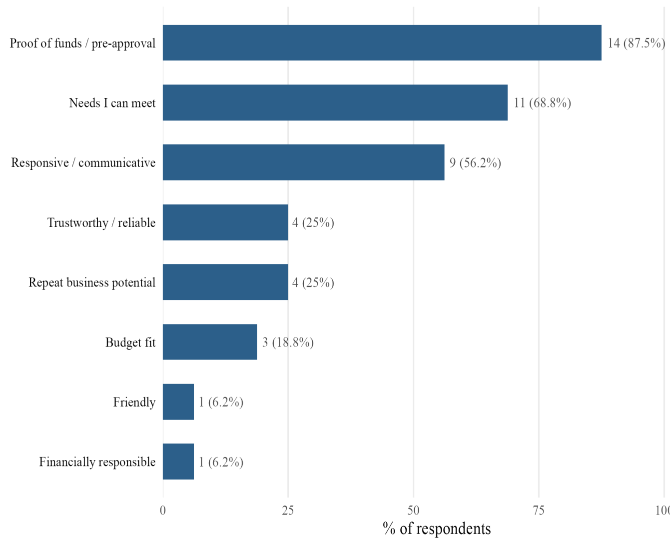
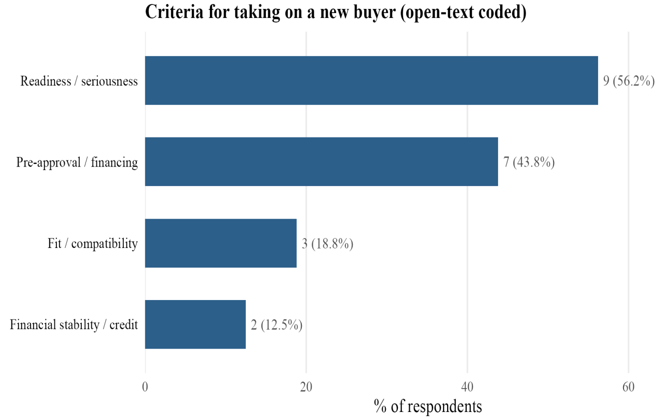


Figure S21

Criteria for taking on a new buyer (open-text coded)



<u>Results: Working with Referrals</u>

Respondents described the benefits and costs of working with clients received via referral and described whether and how clients who contact them directly differ from those they received via referral. We used an LLM to qualitatively classify similar responses from this set, using the same procedure as for the buyer criteria open-text response. The primary benefit of realtor referrals was that leads generated through this channel were “warm” (i.e., the client was more ready to purchase, and more likely to convert to a purchase for that realtor in particular) and that clients engaged through referrals were more reliable/trustworthy. This combination of factors suggests that referrals primarily provide leads that have a higher probability of generating a sale. Realtors also described clients received through referrals as frequently having a better client-relator fit and that client expectations might be better aligned with the referred realtor’s service offering. A notable minority of realtors also mentioned the future business and network opportunities of accepting referrals. Overall, most realtors described the costs of referrals as being relatively low, and that the referral fee (typically 20-30% of the final one-sided commission) and time involved in the sale were the primary costs (but it was not apparent that these time/effort costs differed from other sales).

Figure S22

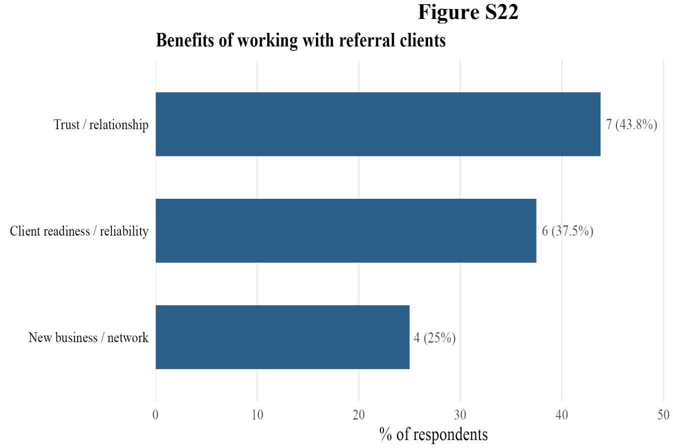


Figure S23

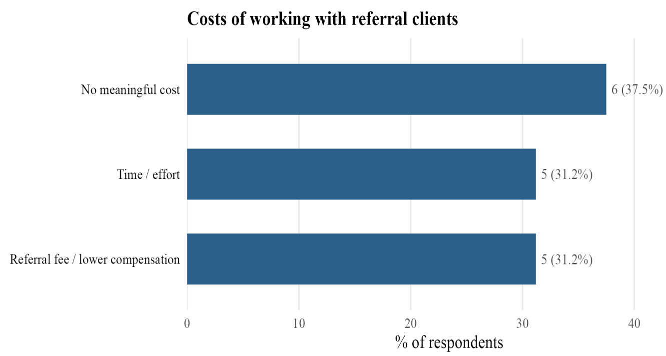
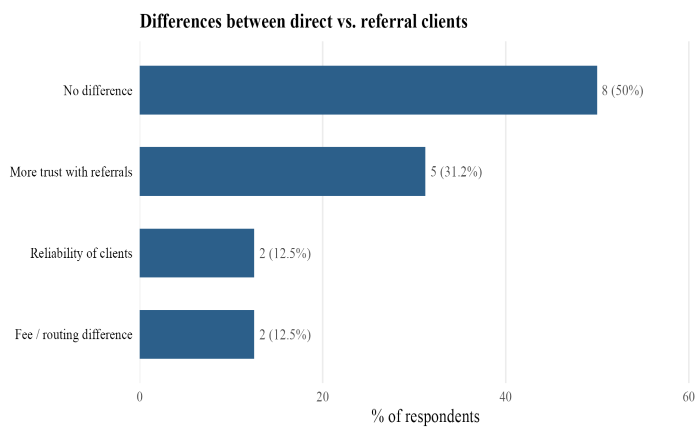


Figure S24



Realtor Email Survey

<u>Overview</u>

In June 2026, we conducted a survey with practicing realtors drawn from the same participant pool used in the main audit study. The goal of this survey was the same as for the Realtor Conference Survey (described in Supplement Section X) and survey questions were based on those from that survey, with some adjustments following feedback from that survey. Namely, we aimed to understand how practicing realtors received client outreach with a focus on referrals from other realtors. We were specifically interested in how often clients reached out to realtors via email, how common realtor referrals were, including out-of-state referrals (like those used in our stimuli), the costs and benefits of referrals (i.e., of being engaged by a sponsor) and what criteria realtors valued in prospective buyers. Additionally, with this survey we aimed to measure realtor beliefs about the relative budgets of prospective White and Black buyers. Our sample frame included the same 82,607 realtors with valid and current email addresses from which the sample for the audit experiment was drawn (please see Supplement Section X for details of this sample). We invited these realtors via email to respond to a short (5-minute) survey in exchange for a chance to win a $100 Amazon Gift card. We received 242 responses, and 167 completed survey responses. Below we provide the survey questions and analyze results.

Survey Questions

1. This study focuses specifically on experiences with residential markets. Have you sold/rented residential properties?

- Yes
- No

2. Through which of the following channels do new clients contact you about buying/renting a home? (Select all that apply)

- Phone
- Email
- In-person
- Events/Showings
- Referrals
- Realtor.com listing
- Zillow listing
- Other (please specify)

3. How often do you receive referrals from other realtors?

- Never
- Approximately once a year
- Approximately once a quarter
- Approximately once a month
- Several times a month
- Every week

4. How often do you receive referrals from realtors from out-of-state?

- Never
- Approximately once a year
- Approximately once a quarter
- Approximately once a month
- Several times a month
- Every week

5. How often do prospective (or current) clients email you with inquiries about buying a new home?

- Never
- Approximately once a year
- Approximately once a quarter
- Approximately once a month
- Several times a month
- Every week

6. What are the main criteria you look for when deciding to take on a new buyer? Please select all that apply.

- Budget aligned with typical clients
- Responsive / communicative
- Financial stability
- Proof of funds / pre-approval
- Ready to buy / serious
- Needs and preferences I can meet
- Trustworthy / reliable
- Other

7. What are the primary benefits of working with buyers through a referral? [Open Text Response]

8. What are the primary costs of working with buyers through a referral? [Open Text Response]

9. What do you think is the average price of homes purchased by Black/African-American first-time homebuyers in your state??

- Less than $100,000
- $100,000 - $199,999
- $200,000 - $299,999
- $300,000 - $399,999
- $400,000 - 499,999
- $500,000 - $599,999
- $600,000 - $749,999
- $750,000 - $999,999
- $1,000,000 - $1,499,000
- Over $1,500,000

10. What do you think is the average price of homes purchased by White/Caucasian first-time homebuyers in your state?

- Less than $100,000
- $100,000 - $199,999
- $200,000 - $299,999
- $300,000 - $399,999
- $400,000 - 499,999
- $500,000 - $599,999
- $600,000 - $749,999
- $750,000 - $999,999

- $1,000,000 - $1,499,000
- Over $1,500,000

11. How many hours per week do you work as a realtor currently?

- None, I do not work as a realtor currently
- 1-9 hours per week
- 10-50 hours per week
- 51+ hours per week

12. How many years have you worked as a realtor?

- Less than 2 years
- 2-5 years
- 6-10 years
- 11-15 years
- More than 15 years

13. What is the most common price range of the residential properties you sell?

- Under $50,000
- $50,000-$99,999
- $100,000-$199,999
- $200,000-$299,000
- $300,000-$500,000
- $500,000-$749,000
- $750,000-$999,999
- $1,000,000-$1,499,000
- Over $1,499,000
- Prefer not to answer

14. Thank you for your time! If you wish to enter the Amazon raffle for a chance to win a $100 Amazon gift card, please provide an email address that we can use to contact you if you win. [Open Text Response]

Sample Descriptive Statistics

This survey was distributed by email to 82,607 real estate agents drawn from the population of realtors on Homes.com and Realtor.com (See Supplement Section X for details of this sample). Realtors received an email invitation to complete a short survey. In total, 242 responses were received, of which 167 (69.0%) completed the full survey. Realtors in the final sample were highly experienced: 66.2% of respondents report having more than 15 years of experience, and only 0.6% had fewer than two years. The vast majority were actively practicing: only 1.9% indicated they do not currently work as a realtor, while 36.2% work more than 51 hours per week. There was also variation in the average price of realtors' listings, with the average listing price similar to those in the main audit sample. Table S31 shows the distribution of our administrative data variables for realtors participating in the survey and comparing them to the full sample pool of 82,602 from which they were drawn. Consistent with their responses, we see that realtors responding to the survey are slightly more experienced than other realtors in the sample (as of May 2024, 15.46 vs. 13.62 years, $p=0.003$). However, they did not differ on any other characteristic and the set of observable characteristics did not jointly predict participation in the survey ($F(16, 73842) = 1.59$, $p = 0.063$). As such, this set of survey respondents were highly similar to the population of realtors used for the audit study.

Table S27. Response overview:

Metric	N	%
Total responses	242	100.0
Completed survey	167	69.0
Sold residential property	239	98.8

Table S28. Hours worked per week:

Hours per week	N	%
None, I do not work as a realtor currently	3	1.9
1-9 hours per week	7	4.4
10-50 hours per week	92	57.5
51+ hours per week	58	36.2

Table S29. Years of experience:

Years as realtor	N	%
Less than 2 years	1	0.6
2-5 years	0	0.0
6-10 years	25	15.6
11-15 years	28	17.5
More than 15 years	106	66.2

Table S30. Typical listing price range:

Price range	N	%
Under $50,000	1	0.6
$50,000-$99,999	2	1.2
$100,000-$199,999	2	1.2
$200,000-$299,000	23	14.4
$300,000-$500,000	71	44.4
$500,000-$749,000	33	20.6
$750,000-$999,999	15	9.4
$1,000,000-$1,499,000	8	5.0
Over $1,499,000	2	1.2
Prefer not to answer	3	1.9

Table S31. Sample comparison: survey respondents vs. full sample.

Variable	Respondents (N = 242)	Full sample (N = 82,602)	t	p
County population size	1,100,904.89 (1,623,732.04)	1,091,528.84 (1,783,247.17)	0.09	0.928
% White population	59.90 (18.74)	59.56 (20.48)	0.28	0.780
% Black population	10.78 (9.86)	11.32 (11.09)	-0.85	0.395
% Hispanic population	18.91 (15.39)	18.80 (16.50)	0.11	0.910
% Urban population	0.82 (0.23)	0.81 (0.24)	0.69	0.491
% Republican vote	0.48 (0.15)	0.49 (0.16)	-0.37	0.711
% Democrat vote	0.50 (0.15)	0.50 (0.16)	0.35	0.724
Slavery incidence in 1860	0.12 (0.17)	0.12 (0.17)	-0.52	0.603
Racial resentment index (county)	4.11 (0.73)	4.20 (0.79)	-1.94	0.054
Average income (zip)	107.07 (91.62)	107.27 (110.84)	-0.03	0.974
Average property value (zip)	442,614.54 (266,694.12)	463,137.69 (368,869.97)	-1.20	0.232
Realtor sale count	31.86 (46.34)	29.36 (62.89)	0.84	0.400
Realtor average sale value	408,911.67 (391,213.37)	449,422.38 (983,622.87)	-1.61	0.110
Realtor experience (years)	15.46 (9.60)	13.62 (7.98)	2.99	0.003
Office size	6.06 (9.11)	5.02 (8.37)	1.79	0.075

Variable	Respondents (N = 242)	Full sample (N = 82,602)	t	p
Works at named brokerage	0.27 (0.44)	0.26 (0.44)	0.23	0.818

Note: Means and standard deviations in parentheses. t and p from two-sample t-tests comparing respondents (N = 242) to the full contacted sample (N = 82,602). Joint F-test of all covariates predicting survey participation: F(16, 73842) = 1.59, p = 0.063.

Results: Buyer Outreach

Respondents were asked through which channels new clients typically contacted them. Phone calls (94.4%) and referrals (92.7%) were near-universal, but email was also highly prevalent as a means of client outreach: 82.4% of respondents indicated that new clients contacted them via email. In-person contact (63.9%) and listing platforms such as Realtor.com (44.2%) and Zillow (42.1%) were also commonly cited. When asked directly how often prospective clients emailed them, realtors reported that email inquiries were near-universal: only 0.9% reported never receiving them, and 58.1% received them at least once a month. Together, this establishes email as a primary channel of initial client outreach, underscoring that an email-based audit study is a valid means of testing discrimination against prospective buyers in this population.

Figure S25

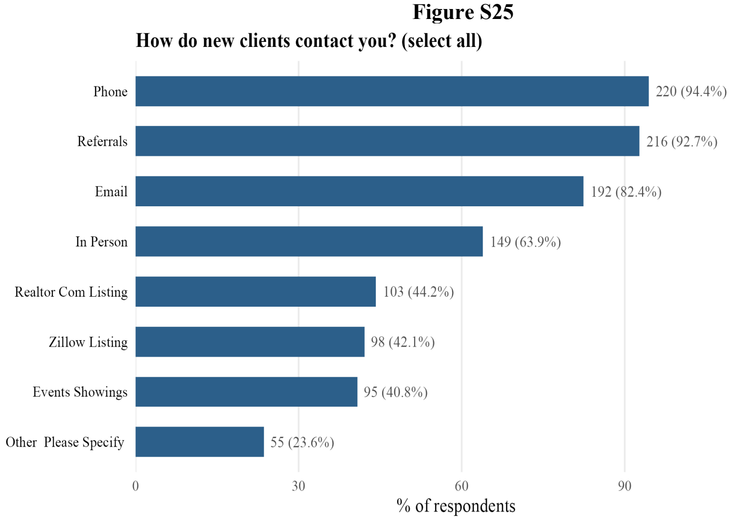
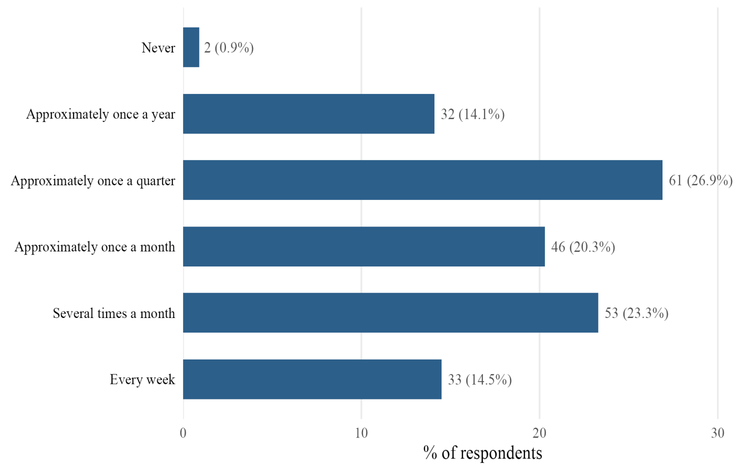


Figure S26

How often do prospective clients email you?



Respondents reported how often they received referrals from other realtors, including out-of-state referrals. Only 10.1% reported never receiving referrals, with 41.8% receiving them at least quarterly. Out-of-state referrals were somewhat less common—22.9% never receive them—but still widespread, with 77.1% receiving at least occasional out-of-state referrals and 23.5% receiving these at least quarterly.

Figure S27

How often do you receive referrals from other realtors?

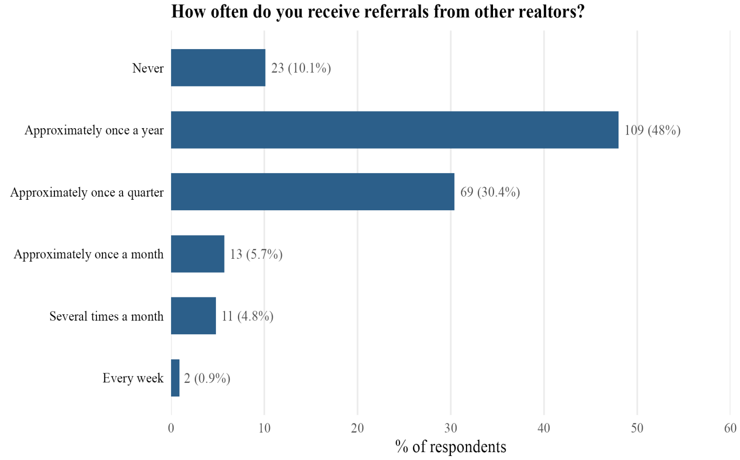
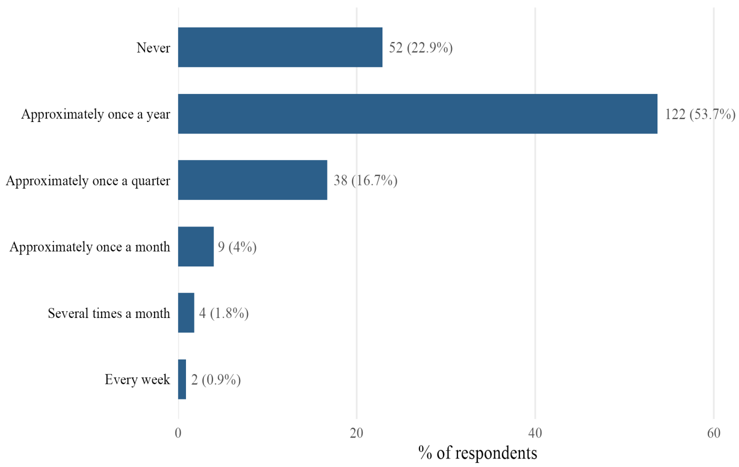


Figure S28

How often do you receive out-of-state referrals?

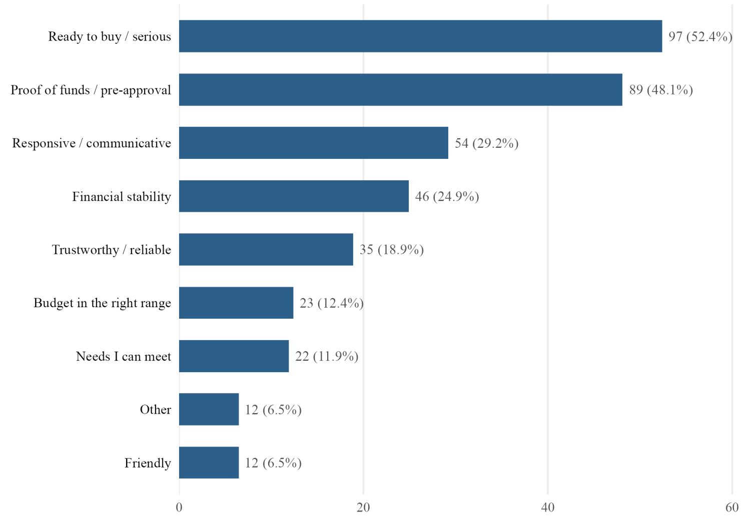


<u>Results: Criteria for Taking on a New Buyer</u>

Respondents selected their top three criteria for taking on a new buyer. The most commonly cited criterion was buyer readiness or seriousness (52.4% of respondents), closely followed by proof of funds or pre-approval (48.1%). These two reasons are arguably highly correlated (we frequently see them mentioned together in qualitative responses, and pre-approval is also a costly signal that a client has taken steps to prepare to buy a home, i.e., that they are "serious"). Client responsiveness and communicativeness ranked third at 29.2%, followed by financial stability (24.9%) and trustworthiness or reliability (18.9%). Budget was also relatively frequently mentioned (12.4%).

Figure S29

Top 3 criteria for taking on a new buyer (multi-select)



Results: Working with Referrals

Respondents were asked in their own words to describe the benefits of working with clients received via referral. Open-text responses were coded into narrow categories using an LLM. Benefit categories were then collapsed into four broader categories (Figures S30 and S31). The dominant benefit of realtor referrals, cited by 40.4% of respondents, was that referral clients were more likely to convert to a completed sale. This encompasses buyer seriousness or motivation (20.8%), the fact that the buyer has been vetted or pre-qualified by the referring agent (18.5%), and a more general sense that a referred lead is a better probabilistic bet (6.7%; this referred to comments such as “more likely to convert”, or “they would be more likely to buy with me”). The second most common theme was trust (18%), reflecting the role of the referral in establishing credibility before first contact (this was both the buyer’s and realtor’s credibility). New business and network opportunities were cited by 18.5% of respondents, with 15.2% specifically noting the referral as an additional sale that would have been unlikely to occur otherwise, and 4.5% noting that referrals were an important source of generating future business through relationships with the referring realtor. Efficiency—spending less time on marketing and client acquisition—was mentioned by 5.1% of respondents. Notably, no realtors mentioned that referrals are signals of a buyer’s potential budget or financial means, per se. Below are a set of quotes randomly selected from the categories presented above, to illustrate the wording realtors typically used for each.

Figure S30

Benefits of referral clients (narrow categories)

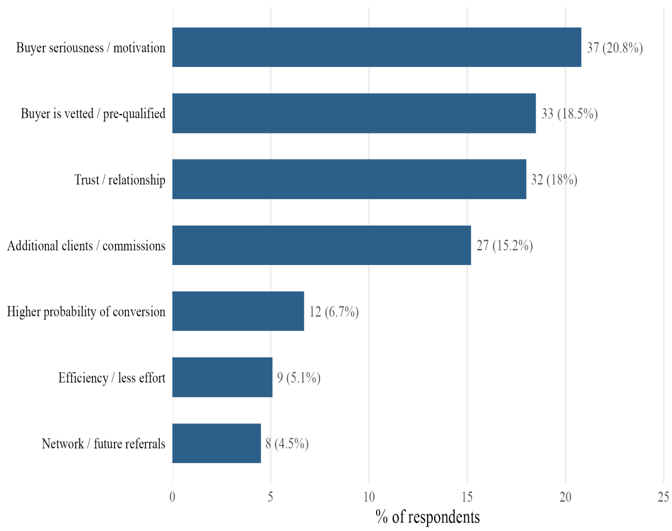
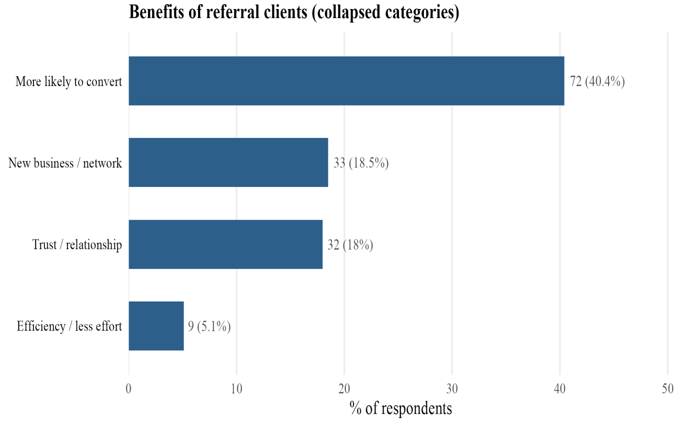


Figure S31

Benefits of referral clients (collapsed categories)



Illustrative responses by category:

Trust / relationship (18% of responses)

> *"they have already heard what kind of person I am and there is trust already there"*
>
> *"Already trust you."*

"they already trust me"

"It is a "warm" or trusted hand-off"

Buyer seriousness / motivation (20.8% of responses)

"they are warm leads"

"buyer is ready"

"They are ready to buy"

"Buyers are ready"

Buyer is vetted / pre-qualified (18.5% of responses)

"Buyer is qualified"

"Client is vetted through a third person"

"Known entity from referral"

"They already have been pre-approved and ready to purchase a property."

Additional clients / commissions (15.2% of responses)

"Of a new client"

"Client is handed to me"

"A sale of property"

"Another client and potential future referrals."

Higher probability of conversion (6.7% of responses)

"Direct contact with the buyer versus being interviewed against many agents"

"Usually they are moving to the area and need to find a place to live, as opposed to a discretionary purchase which may not happen or may take a long time"

Efficiency / less effort (5.1% of responses)

"I did not have to find them"

"Fast & efficient"

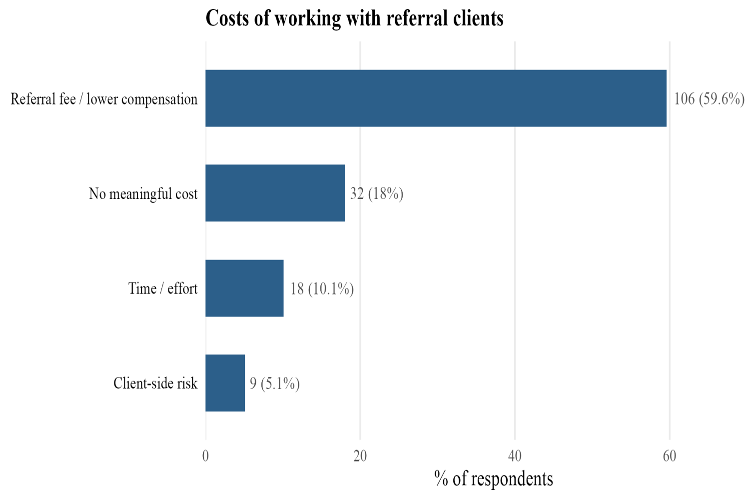
Network / future referrals (4.5% of responses)

"Another client and potential future referrals."

"Repeat business from other agents"

The dominant perceived cost of working with referral clients was the referral fee or reduced commission, cited by 59.6% of respondents—a strong majority. A notable minority (18%) indicated they perceive no meaningful cost to accepting referral clients, suggesting that for many realtors the referral fee is viewed as an acceptable price for a higher-quality or more certain lead, or for business that would otherwise not be acquired. Time and effort were mentioned by 10.1% of respondents, while client-side risk—such as buyers being a poor match—was cited by only 5.1%.

Figure S32



Illustrative responses by category:

Referral fee / lower compensation (59.6% of responses)

"Time consuming and have to payout a large % sometimes"

"The cost would be whatever the referral fee is with a Debbie, 20% 30%, etc"

"25% referral fee"

"Lower payday"

No meaningful cost (18% of responses)

"nothing"

"not too much"

"minimal,"

"Usually minimum or no cost except referral fee."

Time / effort (10.1% of responses)

"Time"

"Gas, and time."

Client-side risk (5.1% of responses)

"expectations are high"

"They can be unrealistic about the market"

Results: Budget Beliefs by Buyer Race

Respondents were asked to estimate the average home price for a first-time buyer who is Black, and separately for a first-time buyer who is White (randomly ordered). Figure S33 shows the full distribution of responses across price brackets for each group. Figure S34 presents the mean expected price with 95% confidence intervals.[13] Realtors systematically expected White first-time buyers to purchase homes at higher prices than their Black counterparts. Realtors expected White first-time buyers to pay, on average, approximately $427,914 (95% CI: $392,257–$463,571), compared with approximately $386,656 (95% CI: $353,374–$419,939) for Black first-time buyers (mean difference = $41,258, results of a paired t-test: $t(162) = 6.33$, $p < 0.001$, [95% CI: $28,379–$54,136]).[14] A Wilcoxon signed-rank test—the paired equivalent of the Mann–Whitney U test, which accounts for the possible ordinal structure of the data, finds a similar significant difference, with beliefs showing that White buyers have higher budgets ($V = 1217.5$, $p < 0.001$; this is also robust to the inclusion of the outlier: $p<0.001$).

Additionally, we find that realtors working in neighborhoods with higher property values had stronger beliefs about this White-Black buyer price gap ($r=0.273$, $p<0.001$; see Figure S35). Notably, this is the same group that discriminated more against Black homebuyers in the main audit study (this was the only moderator variable that significantly predicted higher discrimination against Black buyers).

Figure S33

[13] One respondent gave a within-person difference of −$1,850,000 (reporting that the expected price for White buyers was $1,850,000 lower than for Black buyers). No other absolute budget differences between race exceeded $500,000, and no others exceeded $200,000 in favor of Black buyers (See Figure S36 for the distribution of this difference). We thus chose to drop this observation from analyses given it is a clear outlier. However, the main statistical tests are robust to its inclusion.

[14] Including the outlier attenuates but does not eliminate the effect: $t(163) = 2.25$, $p = 0.026$, mean difference = $29,726.

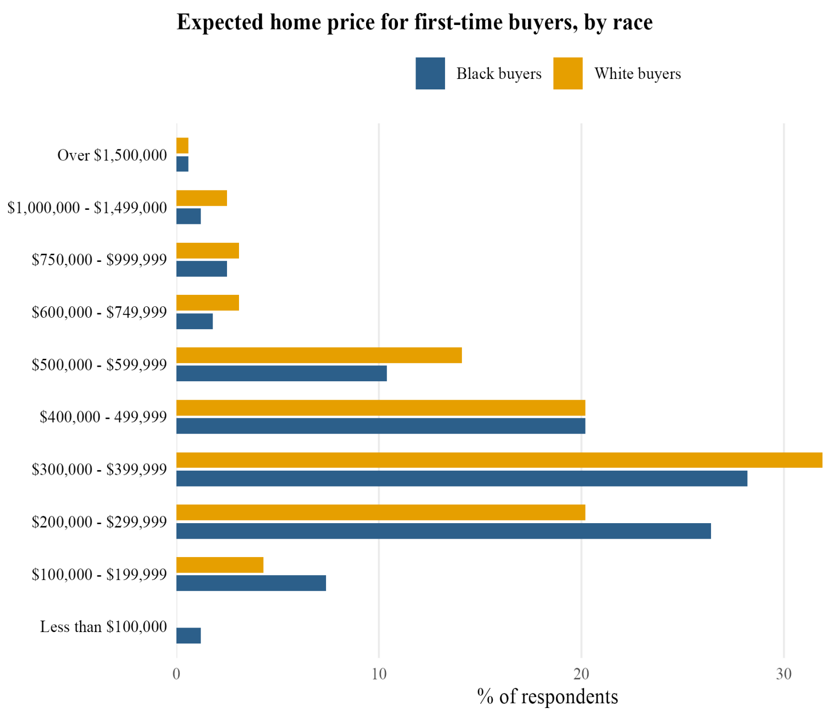

Expected home price for first-time buyers, by race
Black buyers
White buyers
Over $1,500,000
$1,000,000 - $1,499,000
$750,000 - $999,999
$600,000 - $749,999
$500,000 - $599,999
$400,000 - 499,999
$300,000 - $399,999
$200,000 - $299,999
$100,000 - $199,999
Less than $100,000
0
10
20
30
% of respondents

Figure S34

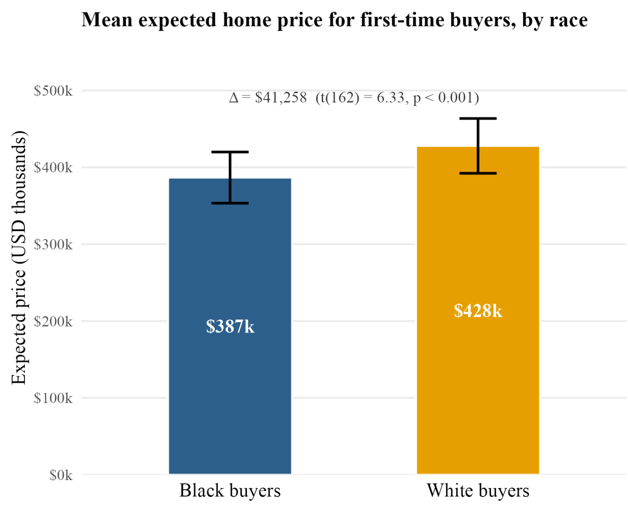


Figure S35

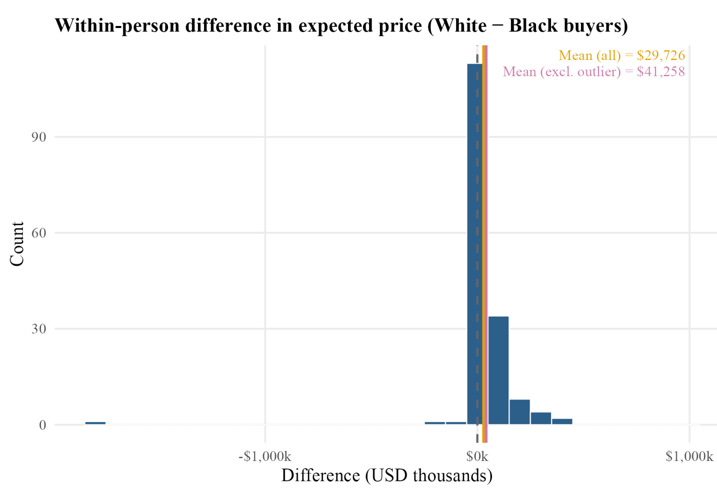
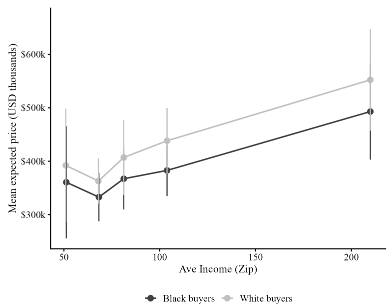
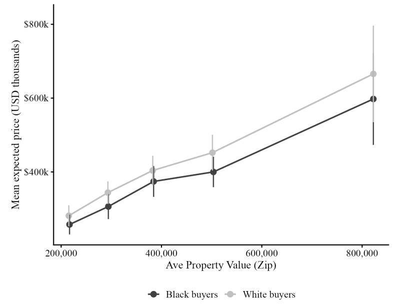


Figure S36





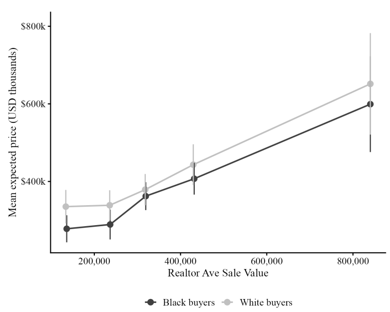


Table S32. Pearson correlations between White-minus-Black belief gap and economic correlates

Correlate	r	t	df	p
Average income (zip)	0.120	1.52	159	p = 0.130
Average property value (zip)	0.273	3.58	159	p < 0.001
Realtor average sale value	0.003	0.04	159	p = 0.971

Note: Belief gap = expected price for White first-time buyer minus expected price for Black first-time buyer (within-person). One outlier with |gap| > $1,000,000 excluded. n = 161 respondents with complete data.

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Email Domains and Sending Protocol

Email Domains and Account Structure

We sent all emails from paid Google Workspace Business accounts. These accounts offer two advantages relevant to deliverability: they support customizable email domains, allowing emails to appear as though they originate from a business rather than a personal account; and they carry stronger baseline email reputations than free accounts, reducing the likelihood of being flagged as spam. We used eight separate domains in total — four for sponsor email accounts and four for buyer email accounts, corresponding to fictitious real estate agencies and generic companies respectively (described below). Within each domain, we created multiple individual email accounts: 12 accounts in each of the first two sponsor and buyer domains, and 6 accounts in each of the remaining four. This structure allowed for stimulus sampling across domains and accounts, and ensured that no individual account or domain sent a disproportionately large volume of emails, further reducing spam risk.

Domain and Agency Names

For the four sponsor domains, we drew on a random sample of 50 independent real estate agency names from each of Wisconsin and Ohio— taken from our database of real estate agencies — and used these as input to ChatGPT to generate new agency names closely resembling the real ones. We then verified each generated name against three criteria: (1) no existing real estate agency of the same name in the United States; (2) no large company of any kind with the same name; and (3) the domain "[agencyname].com" was available for purchase.

For the four buyer domains, we asked ChatGPT to generate generic company names plausibly applicable across different sectors. We verified each against two criteria: (1) no large company of the same name in the United States; and (2) the domain "[companyname].org" was available for purchase.

This procedure produced the following eight domains, summarized in Tables S33 and S34:

Table S33. Sponsor Domains (Fictitious Real Estate Agencies)

Name	Domain	City	State
Beacon Grove Realty	beacongroverealty.com	Columbus	OH
Cedar Trail Realty	cedartrailrealty.com	Milwaukee	WI
Maple Heights Realty	mapleheightsrealty.com	Milwaukee	WI

Elm Valley Realty	elmvalleyrealty.com	Columbus	OH

Table S34. Buyer Domains (Fictitious Companies)

Name	Domain	City	State
Northstream Co	northstreamco.org	Columbus	OH
Meridian Point	meridianpoint.org	Milwaukee	WI
Summit Co	summitco.org	Milwaukee	WI
Crescent Bridge	crescentbridge.org	Columbus	OH

We acquired and set up the first two sponsor domains and first two buyer domains (Beacon Grove Realty, Cedar Trail Realty, Northstream Co, and Meridian Point) in November 2024, each with 12 individual email accounts. The remaining four domains (Maple Heights Realty, Elm Valley Realty, Summit Co, and Crescent Bridge) were added on February 11, 2025, each with 6 individual accounts. We added these additional domains to further increase stimulus sampling and to reduce the number of individual emails sent at the domain level, to reduce the chances of triggering spam filters.

Building Domain Reputation

To minimize the likelihood of our emails being flagged as spam, we undertook a structured process to build the reputation of each domain and individual account prior to the experiment. First, our project domains were integrated into Google Workspace and configured with standard cryptographic authentication protocols, specifically SPF (Sender Policy Framework), DKIM (DomainKeys Identified Mail), and DMARC (Domain-based Message Authentication, Reporting, and Conformance). This verified sender identity and prevented automated filtering systems from flagging the domains as spoofed. Second, we subscribed each individual account to two weekly newsletters from the New York Times, establishing a baseline pattern of inbound email activity. Third, we sent 10 emails per week from each individual account to a set of 25 trusted, longstanding email accounts belonging to the research team — comprising 11 institutional accounts across 6 universities and 14 personal accounts spanning the major email providers (Gmail, Yahoo, Hotmail, and Outlook) — and ensured that recipients opened and replied to each email. Sending and receiving emails with trusted, high-reputation accounts is a standard method for improving domain and account reputation and reducing spam classification. We maintained this “warm-up” protocol for approximately 20 weeks for our first four domains and 10 weeks for the remaining four. Finally, to support the high volume of emails in the experiment, outgoing message routing was migrated from standard Google Workspace inboxes to Postmark, a dedicated transactional email service. This allowed us to bypass standard inbox sending limits and ensured high inbox placement rates for the target sample, leveraging Postmark's reputation.

At the start of the experiment, all eight domains and 72 individual email accounts had neutral reputations and appeared on no block lists, as confirmed by three separate domain reputation checking services.[15]

System for Sending Emails.

To manage the sending of 24,000 emails across 8 domains and 72 individual accounts, we contracted a software consulting firm to develop a bespoke bulk mailing system integrated with Postmark and equipped with an interface for managing bulk email responses. All email templates and sponsor and buyer account signatures were loaded into this system. Prior to uploading participants to the system, we randomized: (i) the order in which participants would receive emails; (ii) treatment condition assignment; (iii) which individual buyer account each participant would be associated with; (iv) for participants assigned to a sponsor condition, which specific sponsor account they would receive; and (v) which of the 12 email templates they would receive (see Table X below for an example of the participant file with randomly assigned fields; see Supplement Section X for details of randomized assignment, and Section Y for details about stimuli). Once participants were loaded, the system sent emails automatically in batches at predetermined intervals. It recorded all inbound responses and included several safeguards to protect experimental validity: each email address could appear only once; each recipient could receive only a single inquiry email; sponsor-condition recipients could not receive buyer-domain emails and vice versa; and all fields relating to the recipient and the fictitious buyer or sponsor account were populated automatically, eliminating the risk of manual errors.

Figure S37: Automated Email Sending System Interface

index	randomization_order	first_name	last_name	email	city	study_condit	sponsor	buyer	email_templ	source
120	1	Nick	Owsley	owsley.nicholas@gmail.com	Chicago	5	36	17	4	2
109	2	Graelin	Mandel	graelinmandel@gmail.com	Chicago	1		2	7	1
116	3	Palin	Chawinwit	natpalinns@gmail.com	Chicago	4	3	31	6	1

We conducted extensive internal testing of the system's upload, send, and recording functions prior to launch, verifying that all randomized elements were correctly incorporated into outbound emails, that send order and batch timing were preserved as intended, and that all inbound responses were accurately

[15] We used the following platforms to conduct checks of domain reputation/health: https://www.talosintelligence.com/reputation_center; https://mxtoolbox.com/emailhealth/; https://www.barracudacentral.org/lookups

recorded. We also tested and built dedicated functions to handle edge cases, including forwarded inquiries, responses in separate email threads, and emoji reactions to emails.

To reduce spam risk and limit the influence of time of day on response behavior, emails were sent in batches of 20–40 (distributed across all 72 individual accounts) at intervals of 10–15 minutes, between 7AM and 9PM U.S. Eastern Time.[16] This ensured that emails arrived within or close to normal business hours and that no individual account sent more than approximately two emails per hour during peak periods. All 24,000 initial inquiry emails were sent over 13 days, between April 24 and May 6, 2025.

Of the 24,000 emails sent, 228 bounced — either because the recipient's email account was no longer active or because their mail server was temporarily unavailable — leaving a successful delivery rate of more than 99%. Of successfully delivered emails, 22 were flagged as spam by recipients. Postmark classified our campaign's spam complaint rate as "very good," indicating performance well within acceptable industry standards. This low rate also suggests that the large majority of realtors did not perceive our inquiry emails as unsolicited commercial messages.

A team of seven research assistants monitored all inbound responses throughout the study period. Each responding realtor received a single follow-up email, sent 36–72 hours after their initial reply, informing them that the buyer had found another realtor and would be proceeding with them, bringing the interaction to a close. The system provided a set of template responses covering the majority of situations; research assistants were permitted to use custom text in cases where a template response was inappropriate — for example, when a realtor indicated they were unable to help due to personal circumstances. Responses indicating a high degree of suspicion about the inquiry, or that were abusive, were flagged for analysis and did not receive a reply. In addition to reading all response emails, research assistants conducted twice-daily checks of bounce rates, spam rates, and open rates for each individual domain and sender account.

Issues with domains

These monitoring checks enabled us to quickly identify sharp declines in open rates for several domains during the experiment. We define a domain downtime period as any window during which the 95% confidence interval for open rates in a rolling 40-email bin falls below the mean open rate for the full experimental period for that domain. These periods are visually distinct: in each case, the mean open rate during the flagged window was lower than the minimum observed in any 40-email bin outside that

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[16] The accounts from Columbus, OH were nominally based in U.S. Eastern Time and those from Milwaukee, WI in U.S. Central Time, one hour behind Eastern Time.

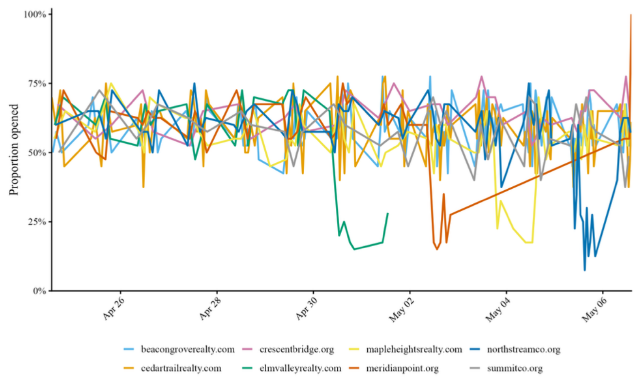
window. Figures S38 and S39 plot open rates for all eight domains, with flagged downtime windows highlighted.

Four domains were affected. Elm Valley Realty was the first, experiencing a drop in open rates from approximately 63% to 23% beginning at 9:30AM on April 30. After monitoring the domain for 12 hours, we took it offline and reassigned all participants scheduled to receive emails from that domain to other domains. Meridian Point experienced a similar decline several days later and was handled in the same way. Maple Heights Realty and Northstream Co experienced comparable drops on May 3rd and May 5th respectively. In Maple Heights' case, the decline occurred in the final hours of that day's sending window; by the time sufficient data were available to make a determination, open rates had rebounded. Given this recovery, the proximity to the end of the experiment, and the risk of introducing errors through further reassignments, we determined not to take these two domains offline.

In all reassignment cases, participants' treatment condition, email template, randomization order, and buyer and sponsor name assignments were preserved; only the sending domain changed. The downtime periods, affected observation counts, and key open-rate statistics for all eight domains are summarized in Tables S35, S36, and S37 below.

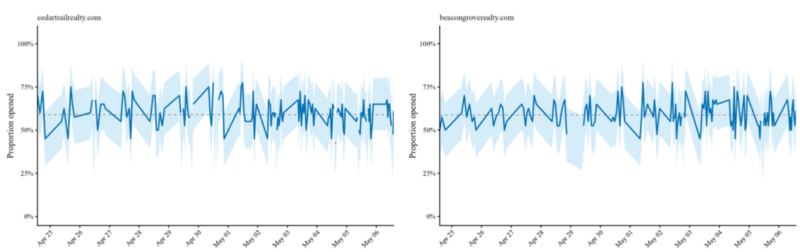
Because each domain was used exclusively for either sponsor or buyer emails, and because Black and White buyer and sponsor names were distributed equally across domains, domain failures are orthogonal to buyer race and sponsor race assignment. However, because the assignment of participants to sponsor versus no-sponsor conditions was made at the domain level, the downtime periods introduced a modest imbalance in the proportion of affected observations across these conditions. As shown in Table S37, approximately 9.9% of no-sponsor observations were affected, compared with 3.1–4.0% of sponsor observations. We address this in our analyses by including a domain-down indicator in all models, and conduct robustness checks both excluding and including all affected observations, and including domain fixed effects.

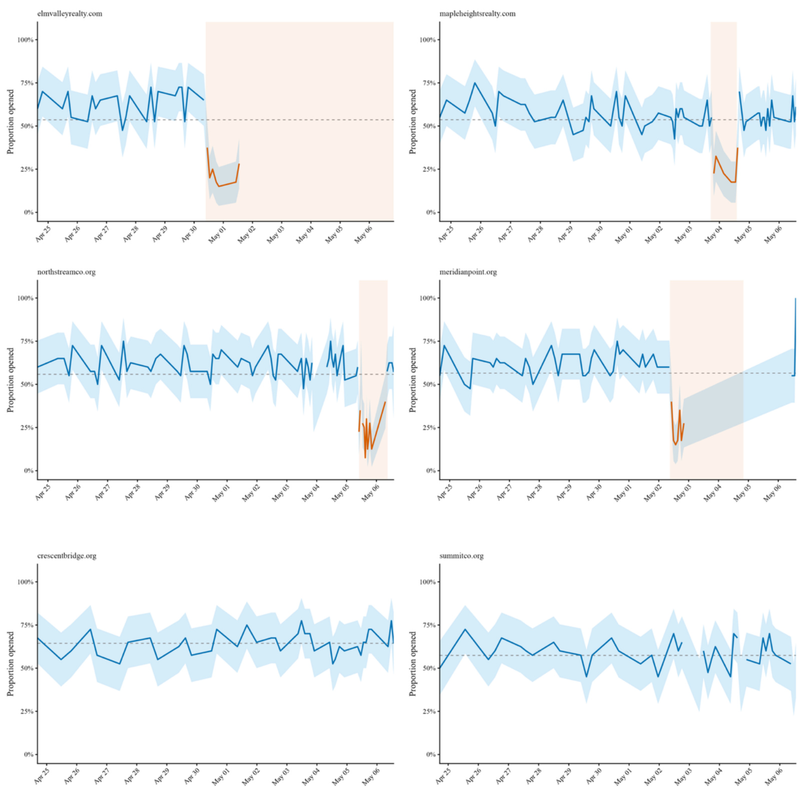
Figure S38: Email Open Rates by Sender Domain (Overview)



Note: Email open rate (proportion of sent emails marked opened) for each sender domain over the experiment period for each of the 8 experimental domains. Each point represents a sequential 40-email bin (ordered by send time within domain); the x-axis shows the median send time of each bin. Four domains experienced abnormally low open rates over distinct periods: elmvalleyrealty.com, meridianpoint.org, mapleheightsrealty.com, and northstreamco.org.

Figure S39: Email Open Rates by Sender Domain (Individual Domains)





Each plot shows the email open-rate for each individual domain used in the experiment. Each point in each plot represents a sequential 40-email bin (ordered by send time); the x-axis shows the median send time of each bin. Shaded ribbon is the 95% CI; dashed line is the domain mean open-rate for the study period. A blue line and blue 95% confidence interval indicate a period where the 95% CI for that bin either includes or is above the domain mean open-rate; red lines and red shading mark any cases of at least 3 consecutive bins where the 95% CI for that bin is below the domain mean open-rate, indicating that open-rates were surprisingly low for that domain over a sustained period.

Table S35. Domain Downtime: Summary Statistics

Domain	**Type**	**Shutdown**	**Affected**	**Downtime**	**Assigned**	**Sent**	**Unaffected obs**	**Affected obs**
beacongroverealty.com	Sponsor	No	No	–	5,219	5,732	5,732	0
cedartrailrealty.com	Sponsor	No	No	–	5,540	6,119	6,119	0
elmvalleyrealty.com	Sponsor	Yes	Yes	Apr 30 – May 6	2,548	1,209	927	282

mapleheightsrealty.com	Sponsor	No	Yes	May 3 17:00 – May 4 14:00	2,693	2,940	2,658	282
crescentbridge.org	Buyer	No	No	–	1,299	1,562	1,562	0
meridianpoint.org	Buyer	Yes	Yes	May 2 09:00 – May 4 20:00	2,741	1,694	1,491	289
northstreamco.org	Buyer	No	Yes	May 5 10:00 – May 6 09:00	2,604	3,127	2,538	503
summitco.org	Buyer	No	No	–	1,356	1,617	1,617	0
Total					24,000	24,000	22,644	1,356

Note: Assigned = original randomization count. Sent = final sending domain count, reflecting mid-experiment reassignments due to issues with the domains documented. Shutdown = domain was proactively taken offline (elmvalleyrealty.com and meridianpoint.org only). Affected = domain flagged as having a period where email open-rates dropped below the mean open-rate for the domain over a sustained period. Unaffected obs and Affected obs reflect the number of emails sent during periods where domain open-rates were flagged as unusually low vs. were not flagged, respectively. Sponsor domains are used in cases where participants were assigned to either the Black sponsor or White sponsor conditions. Buyer domains are used in No-sponsor condition emails.

Table S36. Open Rate Statistics by Domain and Period

	Pre-downtime			During downtime		
Domain	**Pre mean**	**Pre max**	**Pre min**	**During mean**	**During max**	**During min**
cedartrailrealty.com	58.9%	77.5%	37.5%	–	–	–
beacongroverealty.com	58.7%	77.5%	35.0%	–	–	–
elmvalleyrealty.com	62.9%	72.5%	47.5%	23.0%	37.5%	15.0%
mapleheightsrealty.com	57.1%	75.0%	42.5%	26.4%	55.0%	17.5%
northstreamco.org	61.4%	75.0%	37.5%	28.8%	60.0%	7.5%
meridianpoint.org	61.8%	75.0%	47.5%	24.3%	40.0%	15.0%
crescentbridge.org	64.4%	77.5%	52.5%	–	–	–
summitco.org	57.4%	72.5%	37.5%	–	–	–

Note: Each cell summarises the distribution of binned open rates across sequential 40-email bins, ordered by send time within domain. Pre-downtime: bins whose median send time falls before the domain's downtime start (full window for unaffected domains). During downtime: bins whose median send time falls within the flagged downtime window. Unaffected domains show '–' in the During columns. Affected domain rows are shown in bold.

Table S37. Domain-Down Condition Imbalance

Condition	Domain down	Domain up
Black buyer / No sponsor	395 (10.0%)	3564 (90.0%)
White buyer / No sponsor	390 (9.8%)	3574 (90.2%)
Black buyer / Black sponsor	145 (3.7%)	3819 (96.3%)
White buyer / Black sponsor	138 (3.5%)	3820 (96.5%)
Black buyer / White sponsor	122 (3.1%)	3840 (96.9%)
White buyer / White sponsor	157 (4.0%)	3808 (96.0%)
Total	1347 (5.7%)	22425 (94.3%)

Note: Each cell shows number of observations affected by domains having unusually low open-rates and those unaffected for each specific treatment cell (% of observations in brackets). Domain down refers to periods flagged as having unusually low open-rates, and Domain up refers to all other periods.

Table S38. Condition Balance: Domain-Down vs. Domain-Up Observations

Comparison	Group A	Group B	Mean A	Mean B	Difference	t	p
Buyer race	White Buyer	Black Buyer	0.058	0.056	0.002	0.642	0.5209
Sponsor race	Black Sponsor	White Sponsor	0.036	0.035	0.001	0.179	0.8576
No Sponsor vs. Black Sponsor	No Sponsor	Black Sponsor	0.099	0.036	0.063	16.032	0.0000

No Sponsor vs. White Sponsor	No Sponsor	White Sponsor	0.099	0.035	0.064	16.199	0.0000

Note: Each row reports a two-sample t-test of the proportion of emails flagged as domain-down between two treatment groups. Means are proportions (0–1). Difference = Mean A minus Mean B.

Additional Analyses

Voicemails and Texts

During the study period and response window (April 24 – May 13), we received 2,525 phone communications — comprising missed calls, voicemails, and texts — from 1,727 unique phone numbers. After removing bulk marketing and provider messages, 1,722 unique numbers remained. Table S39 summarizes the breakdown by communication type.

Table S39. Phone Communications Received by Type

Type	Frequency
Voicemail	1,428 (56.6%)
Text Message	632 (25.0%)
Missed Call	442 (17.5%)
Other	23 (0.9%)
Total	2525

We matched these communications to participants in our sample using a structured procedure. The principle was as follows: if a phone number or message content allowed us to identify the sender unambiguously and verify that the receiving stimulus account matched that realtor's randomly assigned condition, the communication was assigned to that participant; otherwise, we required at least two corroborating pieces of identifying information before making an assignment.

Specifically, we proceeded through the following steps. First, 1,223 communications were matched directly to administrative phone numbers recorded in our sample dataset. For the remaining 499, we attempted to manually matched them using two sources of information: identifying details provided in the voicemail or text itself (most voicemails and texts included the sender's name and location), and online searches of the phone number (realtor phone numbers are typically publicly listed). A match was confirmed if the realtor's name corresponded to a participant in our sample and the receiving stimulus account matched that participant's randomly assigned condition — for example, if a voicemail was left for the phone number associated with Seth Ryan at Maple Heights Realty, the realtor leaving the message needed to have been randomly assigned to that account. Where name-based matching was inconclusive, we additionally searched for any town names and agency or brokerage names mentioned in the message, and cross-referenced these against the specific receiving stimulus account. A match was confirmed if a unique participant in our sample shared the same town and agency combination and that combination was exclusively associated with the receiving account. In six cases, a match was possible on either agency or hometown (not both), or that multiple participants in our sample matched on both criteria. This created ambiguity about which realtor in our sample initiated the communication, and as such these cases were treated as unmatched. We successfully matched 341 of the 499 communications to participants in our sample using the above procedures.

Of the 158 unmatched numbers, 143 were classified as spam or marketing — comprising direct messages from Google, unsolicited commercial or scam messages, or numbers associated with known spam accounts — and 4 were from legitimate numbers with no connection to real estate, likely reflecting wrong numbers or incidental outreach. The remaining 11 could not be matched to any participant but contained content consistent with a genuine real estate inquiry; these are labeled "could not find" in the dataset. Of these, 6 were the ambiguous cases described above, and 5 we were unable to match to any specific participants in our sample. We note that following the close of the experiment, as our phone numbers became more widely indexed by marketing platforms, incoming spam volume increased. This is consistent with standard patterns of number exposure over time and had no bearing on the study period data.

In total, we successfully matched 1,564 unique phone numbers to 1,533 unique participants (31 participants contacted us from more than one phone number). Only 11 of the 1,722 numbers represented realtors we could not match; the remainder were spam or unidentified service contacts. Table S40 reports phone communications by treatment condition. Black buyers received fewer phone communications than White buyers, and Black sponsors were associated with fewer phone communications than White sponsors. Notably, Black buyers received fewer phone contacts specifically in the sponsor conditions relative to the no-sponsor condition. As shown in Table S46, including all phone communications alongside email responses in the primary hypothesis tests does not alter the main findings: the pattern of discrimination against Black buyers and Black sponsors is preserved with similar point estimates, and neither the overall sponsor effect nor the interaction between sponsor and buyer race changes substantively.

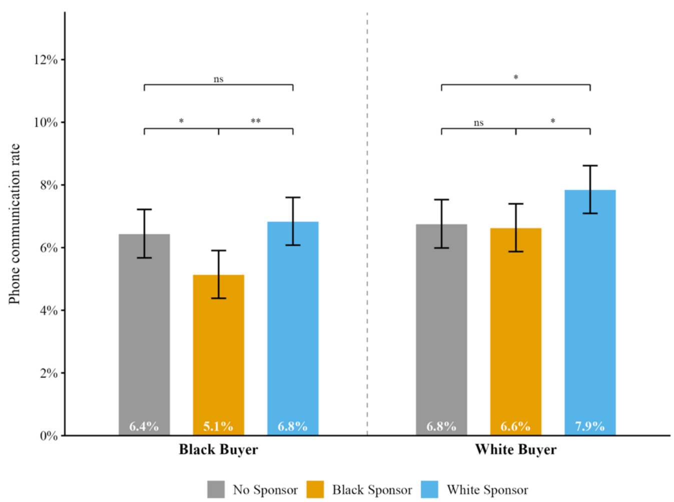
Table S40. Phone Communications by Treatment Condition

	Black Buyer			**White Buyer**		
	NS	**BS**	**WS**	**NS**	**BS**	**WS**
Total N	**3959**	**3964**	**3962**	**3964**	**3958**	**3965**

Phone communications	239 (6.0%)	198 (5.0%)	266 (6.7%)	252 (6.4%)	257 (6.5%)	305 (7.7%)
Also emailed	151 (63.2%)	145 (73.2%)	184 (69.2%)	171 (67.9%)	183 (71.2%)	223 (73.1%)
Phone only	88 (36.8%)	53 (26.8%)	82 (30.8%)	81 (32.1%)	74 (28.8%)	82 (26.9%)
Communication type						
Voicemail	142 (59.4%)	133 (67.2%)	165 (62.0%)	151 (59.9%)	172 (66.9%)	196 (64.3%)
Text message	71 (29.7%)	56 (28.3%)	86 (32.3%)	84 (33.3%)	66 (25.7%)	91 (29.8%)
Missed call	26 (10.9%)	9 (4.5%)	15 (5.6%)	17 (6.7%)	19 (7.4%)	18 (5.9%)

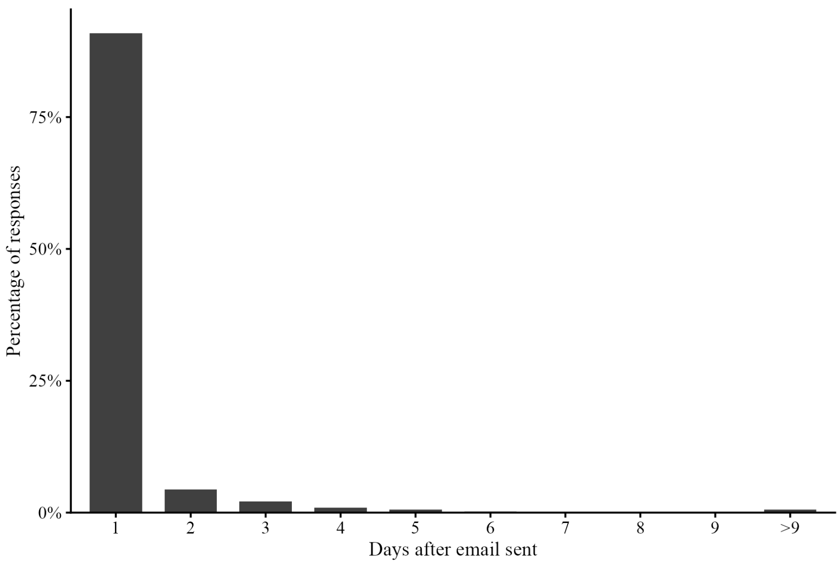
Note: NS = No Sponsor, BS = Black Sponsor, WS = White Sponsor Conditions: Phone communications = any phone contact within 7 days of email send; % is share of total N in that condition. Also emailed / Phone only: % of phone communicators in that condition that also responded via email (i.e., their response was captured already in the email response outcome variable) or those who only contacted via phone. Communication type is based on first phone contact per participant; type % is of phone communicators. Type counts may not sum to Phone communications if a participant's first contact type is missing.

Figure S40: Phone Communication Response Rate by Treatment Condition



Response Timing

Figure S41: Days until realtor response



Note: Each bar represents the proportion of all first-time email responses received from realtors on a given day after the initial inquiry was sent to that individual realtor, defined as the first 24-hour period after the inquiry email was sent, the second 24-hour period, etc. For instance, an email received by a realtor 26 hours after the initial email inquiry was sent to them would be recorded in the second bar.

Stimulus Name Socio-economic Status

We used first and last names to signal buyer and sponsor race in our experiment, selecting names that were both highly diagnostic of race and common in the United States population. The procedure for selecting these names is described in detail in the Name Selection section above. We briefly summarize the key features relevant to the SES analysis here.

Because race and socioeconomic status are highly correlated in the United States (Crabtree et al., 2022), observed differences in realtor responsiveness to Black versus White names could in principle reflect SES-based rather than race-based discrimination. To assess whether our findings are robust to this concern, we follow a set of prior audit studies (Bertrand & Mullainathan, 2004; Gaddis, 2019; Hanson & Hawley, 2023) in using mother's education—specifically, the proportion of mothers with a college degree associated with each name—as an objective proxy for the SES signal carried by each name.

In selecting names for the experiment, we attempted to match Black and White names on SES. Complete parity was not achievable given our primary selection criteria—high objective and perceived racial diagnosticity and common frequency in the population—but we achieved substantial overlap: the SES distributions of Black and White names overlap considerably, with Darius (a Black name) having higher SES than four of the six White names, and Edwin (a White name) having the lowest SES of all twelve names. This within- and cross-race variation allows us to separate the predictive power of SES from that of race.

Figures S42 and S43 show the relationship between buyer name SES and response rate in the no-sponsor and pooled sponsor conditions respectively, and Figure S44 shows the corresponding relationship for sponsor name SES. In all three figures, there is a modest positive slope, indicating that higher mother's education, our proxy for SES, is associated with higher response rates. Table S41 tests this formally by regressing a binary response indicator on buyer SES and buyer race separately, and then jointly, for both the no-sponsor and sponsor subsamples, as well as on sponsor SES and sponsor race in the sponsor subsample.

In the no-sponsor condition, name-associated SES significantly predicts response rates. However, when buyer race is included in the same model, the SES coefficient becomes only marginally significant, while buyer race remains significant. This pattern indicates that it is Black racial identity, rather than the lower average SES associated with Black names, that is the primary driver of differential responsiveness in the no-sponsor condition. In the sponsor conditions, buyer SES does not significantly predict response rates and its inclusion does not alter the point estimate for buyer race discrimination, suggesting that when a sponsor sends the email on the buyer's behalf, name-associated SES no longer affects realtor responsiveness, while the penalty faced by Black buyers persists. Similarly, sponsor SES does not significantly predict response rates, and controlling for it does not change the point estimate for sponsor race discrimination.

Finally, Table S42 replicates the main hypothesis tests for all five research questions with buyer and sponsor SES included as controls. The primary qualitative findings are unchanged across all hypothesis tests: the point estimate for buyer race discrimination in Hypothesis 1 is attenuated but remains significant, and the estimates for all remaining hypotheses are almost totally unchanged. Taken together, these results indicate that while buyer SES is a modest predictor of realtor responsiveness, the documented discrimination against Black buyers and sponsors is unlikely to reflect differences in name-associated SES. Black buyers and sponsors face a penalty due to their racial identity over and above any effect of SES.

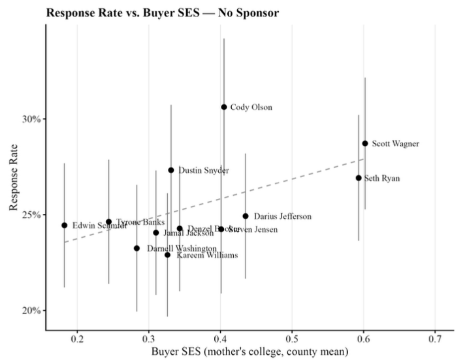
Table S41: Response Rate, Race, and Name-Based SES

	No Sponsor			**Sponsor (Buyer Race)**			**Sponsor (Sponsor Race)**		
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
Black buyer	-0.030**		-0.023*	-0.019**		-0.019*			
	(0.010)		(0.011)	(0.007)		(0.008)			
Buyer SES (mother's college)		0.110**	0.073+		0.031	-0.000			
		(0.040)	(0.043)		(0.029)	(0.031)			
Black sponsor							-0.031***		-0.032***
							(0.007)		(0.008)
Sponsor SES (mother's college)								0.041	-0.012
								(0.029)	(0.032)
Num.Obs.	7923	7923	7923	15849	15849	15849	15849	15849	15849
R2	0.013	0.013	0.013	0.005	0.004	0.005	0.005	0.004	0.005

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

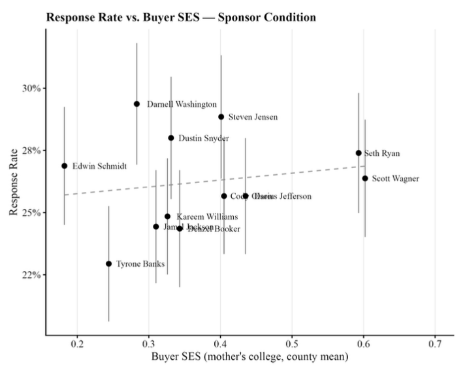
*Notes: OLS (LPM) estimates with HC1 robust standard errors. The dependent variable is whether the realtor responded via email in the pre-registered 7-day window. Models (1)–(3) use the no-sponsor subsample; models (4)–(6) use the sponsor subsample and test buyer race and buyer SES; models (7)–(9) use the sponsor subsample and test sponsor race and sponsor SES. SES is measured as the proportion of mothers with a college degree in the county associated with the name. All models control for email domain crashes (domaindown). + p<0.1, * p<0.05, ** p<0.01, *** p<0.001.*

Figure S42: Response rates and buyer SES (No sponsor)



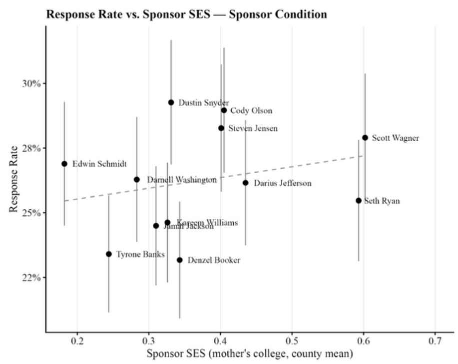
Note: Each point represents one buyer name used in the no-sponsor condition. The y-axis shows the mean response rate for that name; error bars are 95% CIs. The x-axis shows the county-level proportion of mothers with a college degree associated with the name. The dashed line shows the OLS fit across names.

Figure S43: Response rates and buyer SES (Pooled sponsor)



Note: Each point represents one buyer name used in the sponsor condition (averaged across sponsor race). The y-axis shows the mean response rate for that name; error bars are 95% CIs. The x-axis shows the county-level proportion of mothers with a college degree associated with the name. The dashed line shows the OLS fit across names.

Figure S44: Response rates and sponsor SES (Pooled sponsor)



Note: Each point represents one sponsor name used in the sponsor condition. The y-axis shows the mean response rate when that sponsor name appeared in the email; error bars are 95% CIs. The x-axis shows the county-level proportion of mothers with a college degree associated with the sponsor name. The dashed line shows the OLS fit across names.

Table S42: Hypothesis Tests controlling for Buyer and Sponsor SES

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.023*				-0.028**	-0.027**
	(0.011)				(0.010)	(0.010)
Pooled sponsor		-0.003			-0.008	
		(0.006)			(0.009)	
Black sponsor			-0.019**	-0.032***		-0.025**
			(0.007)	(0.008)		(0.010)
White sponsor			0.013+			0.009
			(0.007)			(0.010)
Black buyer × Pooled sponsor					0.011	
					(0.012)	
Black buyer × Black sponsor						0.014
						(0.014)
Black buyer × White sponsor						0.008
						(0.014)
Num.Obs.	7923	23772	23772	15849	23772	23772
R2	0.013	0.007	0.008	0.006	0.007	0.008

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

*Notes: OLS (LPM) estimates with HC1 robust standard errors. The dependent variable is whether a participant responded via email in the pre-registered 7-day window. RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates. RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All models control for email domain crashes (domaindown). All models additionally control for buyer SES (proportion of mothers with a college degree associated with the buyer name). RQ3.2 additionally controls for sponsor SES; sponsor SES is not added to full-data models (RQ2, RQ3.1, RQ4, RQ5) as no-sponsor rows have no sponsor name and would be dropped. SES coefficients are not shown. + p<0.1, * p<0.05, ** p<0.01, *** p<0.001.*

Heterogeneity Analysis

Overview and approach

The analyses in this section are exploratory. Our goal is to test whether the discrimination we observe against Black buyers and Black sponsors, and the overall effect of sponsorship on response rates, are moderated by realtor characteristics or features of the geographic areas and agencies in which realtors operate. These analyses are intended to help adjudicate between theoretical accounts of the discrimination we observe — in particular, to assess whether the pattern of results is more consistent with taste-based or statistical discrimination. Previous work has shown that such moderators can be informative: Kline, Rose & Walters (2021) found that large corporate firms discriminated significantly less than smaller ones, and Grosjean et al. (2023) found that racialized policing of Black populations intensified following right-wing political rallies, specifically in counties with higher racial resentment, histories of slavery, and Republican vote shares.

We preregistered heterogeneity analyses examining eight moderators: (i) county-level Republican vote share in the 2020 U.S. Presidential Election; (ii) a county-level racial resentment index derived from the Cooperative Congressional Election Study (CCES) and historical slavery prevalence (Grosjean et al., 2023); (iii) county percent White population; (iv) realtor race, imputed using the algorithm in Ambrose et al. (2021); (v) realtor years of experience; (vi) realtor activity, measured by the number of recent listings; (vii) realtor target market, measured by the average value of recently sold listings; and (viii) percent of the county population living in an urban area. In addition, we examine a set of exploratory non-preregistered variables that we believe can provide additional insight into potential mechanisms: (i) average property value in the realtor's zip code; (ii) average household income in the realtor's zip code; (iii) the county-level mean score on the Black race Implicit Association Test (IAT), as a measure of implicit racial bias; (iv) the number of realtors in the realtor's office; and (v) the proportion of the county population that is Black.

We test moderation for: (1) discrimination against Black buyers; (2) discrimination against Black sponsors; and (3) the overall sponsor effect (any sponsor versus no sponsor). In each case, we estimate OLS regressions with robust standard errors, regressing realtor response on the treatment indicator of interest, the moderator variable, and their interaction. In place of several binary indicators specified in the preregistration, we use the continuous versions of each moderator,[17] as these provide richer data and

[17] We use log-transformed versions of several variables to account for skewness: the average property sale value in the zip code, average household income in the zip code, the realtor's average listing price from previous sales, the number of agents in the realtor's office, and the number of previous listings sold by an agent. All had a skew greater than 4, while non-logged covariates had skews less than 1.73.

greater statistical power to detect interactions.[18] For moderation of buyer and sponsor discrimination, we also present quintile plots of response rates by buyer and sponsor race for each predictor variable (Figures S48 and S49), in order to account for possible non-linearities in moderation and to illustrate how response rates for Black and White buyers and sponsors vary across the full distribution of each moderator.

Moderation of discrimination against Black buyers and Black sponsors

Overall, we find very little evidence that discrimination against Black buyers or Black sponsors is moderated by realtor characteristics or features of the geographies and agencies in which they work. This is demonstrated in both the interaction coefficient plots (Figures S45 and S46) and the quintile plots (Figures S48 and S49). Across the 13 continuous moderator variables, Black buyers received fewer responses than White buyers in 63 of the 65 quintiles tested, and Black sponsors received fewer responses than White sponsors in every quintile. This suggests that discrimination against Black buyers and sponsors in the real estate market is pervasive and not substantially concentrated in particular types of neighborhoods or among particular types of realtors. The only quintiles in which we observe higher responses for Black buyers than for White buyers are for realtors in the lowest-income neighborhoods and those selling the lowest-value properties. This is consistent with the patterns we observe in the interaction plots: realtors operating in higher property-value zip codes discriminate significantly more against Black buyers ($\beta = -0.021$, SE = 0.009, $t = -2.14$, $p = .032$), while those selling higher-value listings discriminate significantly more against Black sponsors (regression-estimated effect of Black vs. White buyer = -3.89 pp, SE = 1.26, $t = 3.09$ $p = 0.002$). We also find suggestive evidence of more discrimination against Black buyers by realtors selling higher-value listings ($\beta = -0.005$, SE = 0.003, $t = -1.94$, $p = .052$) and those in higher-income neighborhoods ($\beta = -0.019$, SE = 0.010, $t = -1.79$, $p = .074$). Realtors in more rural areas also discriminate more against Black sponsors ($\beta = -0.071$, SE = 0.029, $t = -2.40$, $p = .016$).

Therefore, the only moderation of discrimination against Black buyers is for variables that correlate with the financial stakes of real estate transactions: neighborhood income, property values, and the average price of a realtor's prior listings. This is more consistent with an account of "statistical discrimination" – where realtors infer differences in the purchasing power or profitability of Black versus White clients, and adjust their responsiveness accordingly. Realtors selling higher value properties might perceive a prospective Black buyer as less likely to afford houses in their portfolio than a realtor targeting lower value markets. We find survey evidence supporting this: realtors from the same population from which

[18] The preregistration specified binary indicators (above vs. below the dataset median) for: (i) percent White population; (ii) realtor years of experience; (iii) realtor activity (number of recent listings); (iv) realtor target market (average value of recently sold listings); and (v) percent of the county population living in an urban area. We use the continuous versions throughout.

our audit sample was drawn perceived Black buyers to purchase lower value homes than corresponding White buyers, and this perceived difference was larger for realtors in neighborhoods with the highest property values (See supplement section X for details and results of this survey).[19]

By contrast, we find no evidence of moderation by variables associated with racial prejudice, including county-level indicators of racial resentment, implicit racial bias, slavery history, the percentage of the population that is White, or Republican vote share. For instance, discrimination is nearly identical in neighborhoods with the highest and lowest levels of racial resentment (See Figure S48). This evidence is thus inconsistent with an account in which taste-based prejudice is the primary driver of the observed discrimination against Black buyers in our study. Taken together, these patterns are more consistent with statistical discrimination — in which realtors use race as a proxy for the financial value of a prospective client — than with taste-based discrimination driven by racial animus.

Moderation of the overall sponsor effect

By contrast to the limited moderation of discrimination, the overall effect of having a sponsor — any sponsor relative to no sponsor — is substantially moderated by several variables. Sponsors are significantly more effective in urban, higher-income, higher-property-value, and more politically Democrat-voting areas, and relatively less effective in rural, lower-income, more Republican-leaning, and higher-proportion-White areas. Figure S47 presents the full set of interaction coefficients.

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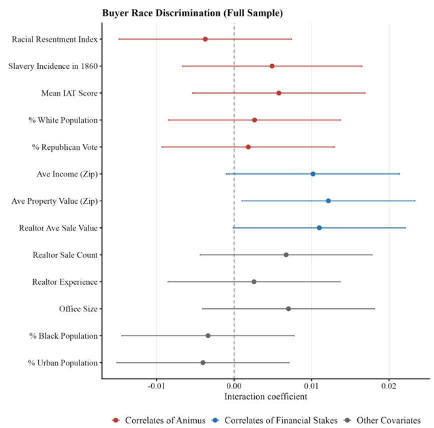
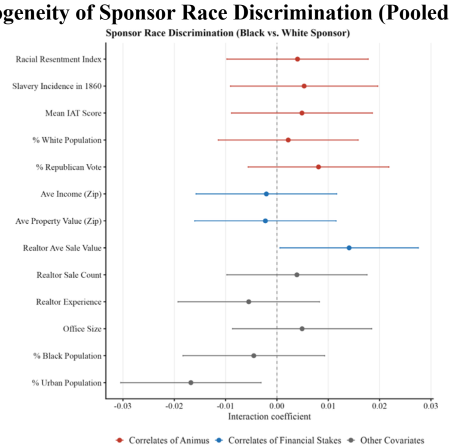
[19] Complementary survey evidence from realtors from the same sampling frame indicates that realtors believe Black prospective buyers targeting neighborhoods in their area tend to have lower budgets than comparable White buyers, providing direct support for the beliefs that may underlie this statistical discrimination channel. We discuss this survey in greater detail in Supplementary Section X.

Figure S45: Moderation of discrimination against Black buyers: Interaction coefficients



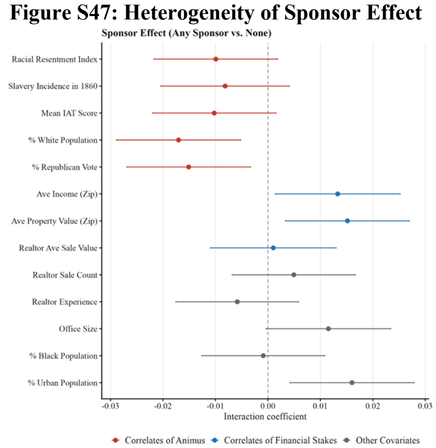
Note: Exploratory heterogeneity — Buyer Race Discrimination (Full Sample). We look at the interaction of the Black Buyer indicator and each moderator, flipped so that positive values indicate more discrimination against Black buyers. Each row shows the interaction coefficient between the treatment and a standardized version of the moderator covariate from OLS (responded ~ treatment × moderator). Error bars represent 95% confidence intervals. Covariates include: Racial Resentment Index, Slavery Incidence in 1860, Mean IAT Score, % White Population, % Republican Vote, Ave Income (Zip), Ave Property Value (Zip), Realtor Ave Sale Value, Realtor Sale Count, Realtor Experience, Office Size, % Black Population, % Urban Population. Significant at α = 0.05: Ave Property Value (Zip) (p = 0.032). Marginally significant at α = 0.10: Ave Income (Zip) (p = 0.074); Realtor Ave Sale Value (p = 0.052).

Figure S46: Heterogeneity of Sponsor Race Discrimination (Pooled sponsor condition)



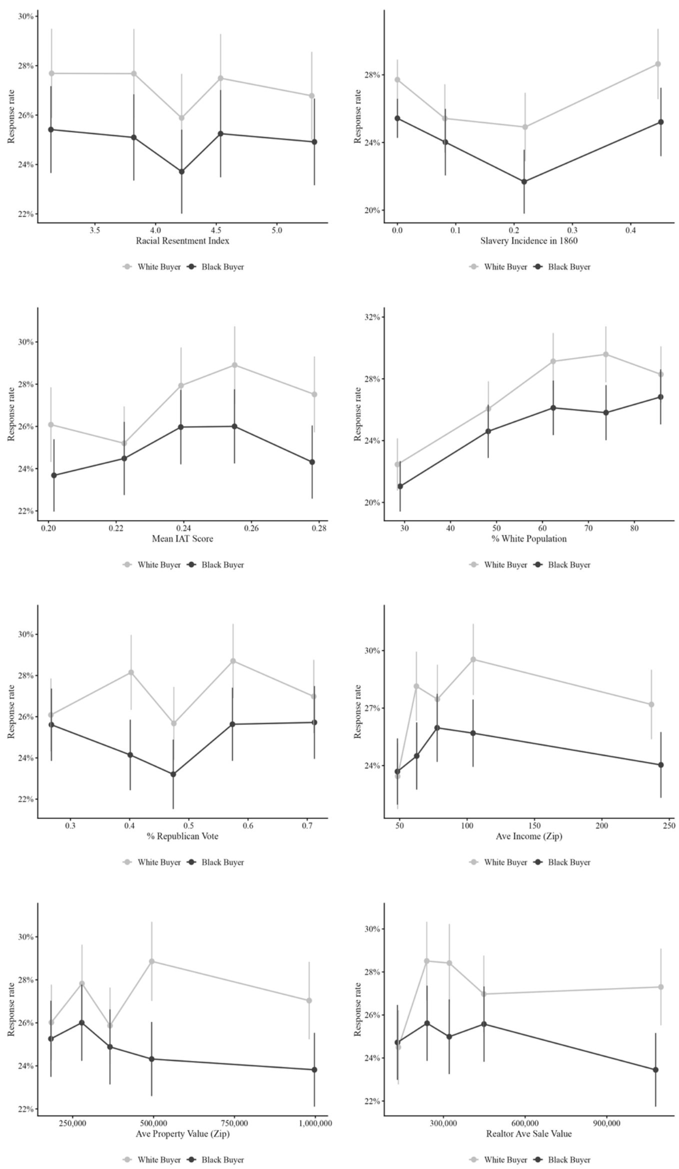
Note: Exploratory heterogeneity — Sponsor Race Discrimination (Black vs. White Sponsor) (Sponsor Conditions Only). We look at the interaction of the Black Sponsor indicator and each moderator, flipped so that positive values indicate more discrimination against Black sponsors. Each row shows the interaction coefficient between the treatment and a standardized version of the moderator covariate from OLS (responded ~ treatment × moderator + domaindown). Error bars represent 95% confidence intervals. Covariates include: Racial Resentment Index, Slavery Incidence in 1860, Mean IAT Score, % White Population, % Republican Vote, Ave Income (Zip), Ave Property Value (Zip), Realtor Ave Sale Value, Realtor Sale Count, Realtor Experience, *Office Size, % Black Population, % Urban Population. Significant at α = 0.05: Realtor Ave Sale Value (p = 0.041); % Urban Population (p = 0.016).*

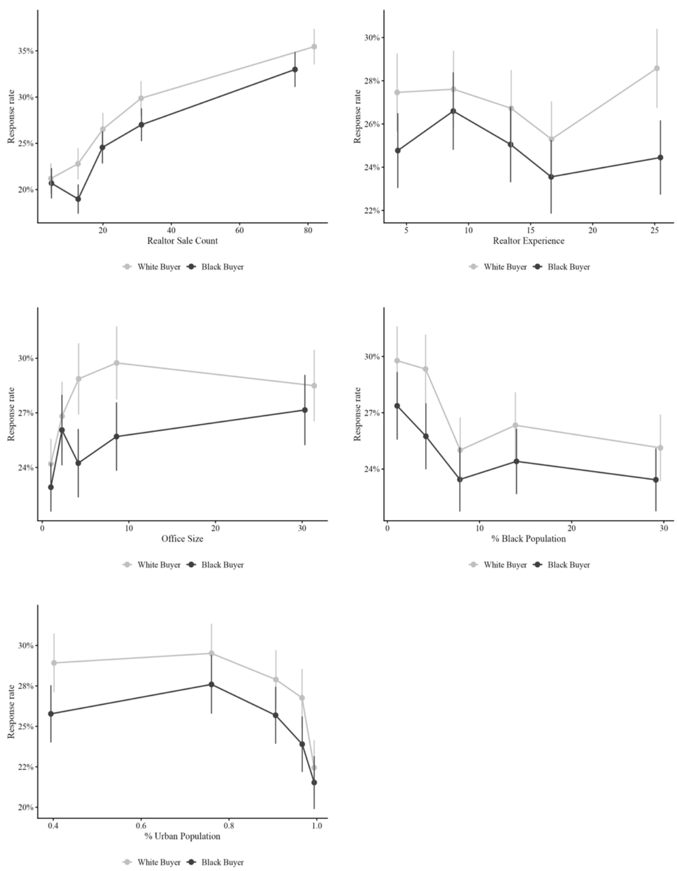
Figure S47: Heterogeneity of Sponsor Effect



Note: Exploratory heterogeneity — Sponsor Effect (Any Sponsor vs. None) (Full Sample). We look at the interaction of the Sponsor Presence indicator and each moderator, positive values indicate a larger sponsor effect at higher covariate values. Each row shows the interaction coefficient between the treatment and a standardized version of the moderator covariate from OLS (responded ~ treatment × moderator + domaindown). Error bars represent 95% confidence intervals. Covariates include: Racial Resentment Index, Slavery Incidence in 1860, Mean IAT Score, % White Population, % Republican Vote, Ave Income (Zip), Ave Property Value (Zip), Realtor Ave Sale Value, Realtor Sale Count, Realtor Experience, Office Size, % Black Population, % Urban Population. Significant at α = 0.05: % White Population (p = 0.005); % Republican Vote (p = 0.012); Ave Income (Zip) (p = 0.028); Ave Property Value (Zip) (p = 0.012); % Urban Population (p = 0.008). Marginally significant at α = 0.10: Racial Resentment Index (p = 0.099); Mean IAT Score (p = 0.088); Office Size (p = 0.057).

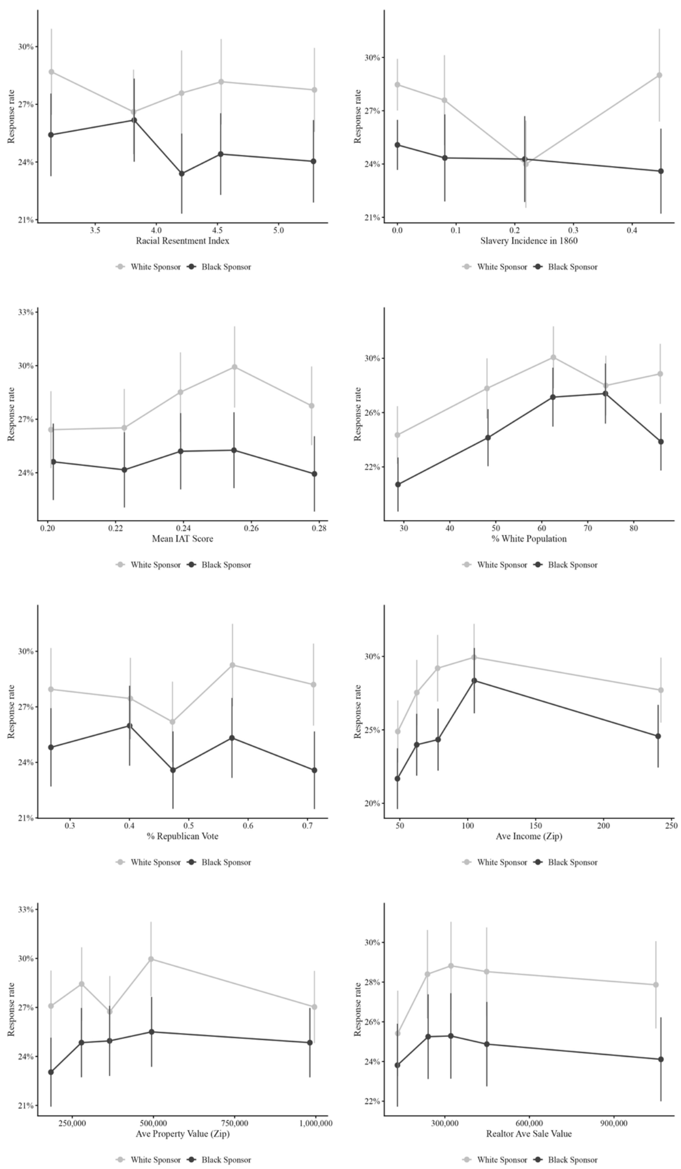
Figure S48: Moderator Quintile plots: Response rates to Black and White buyers

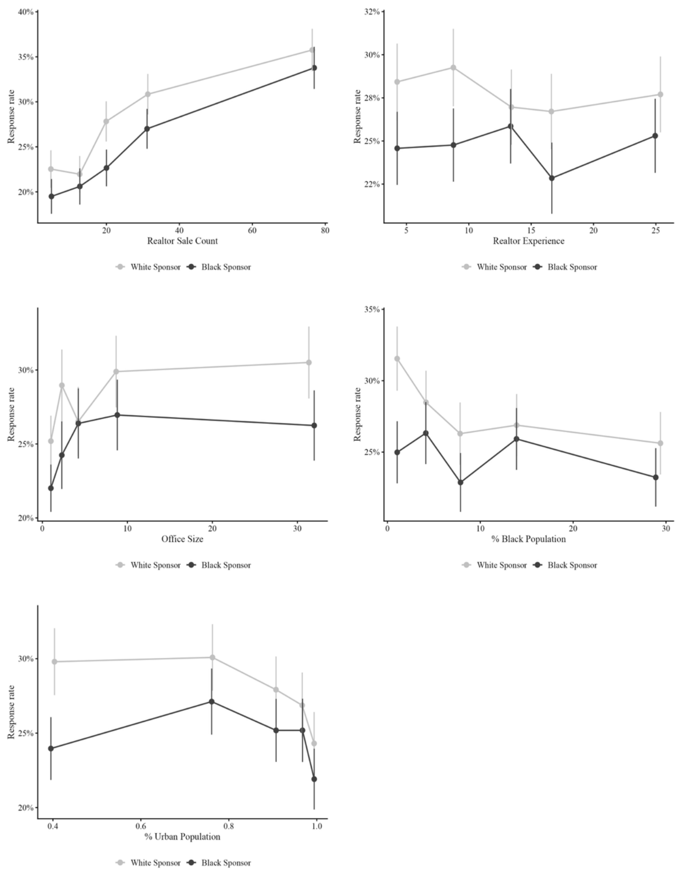




Note: Each point represents the mean email response rate within a quintile of the moderator variable (x-axis) for Black buyers (dark) and White buyers (light). Lines connect consecutive groups; error bars show 95% confidence intervals. Groups are computed on the raw (untransformed) scale. All observations are included. Two variables deviate from pure quintiles: Slavery Incidence in 1860 (pslave1860) has a large mass at zero (areas with no recorded slavery); these observations form the first group and the remaining non-zero values are split into three equal groups (4 points total). Office Size (office_size_all) similarly has a mass at its minimum value, which forms the first group; the remainder is split into four equal groups (5 points total). All other variables use standard quintiles.

Figure S49: Moderator Quintile plots: Response rates to Black and White sponsors





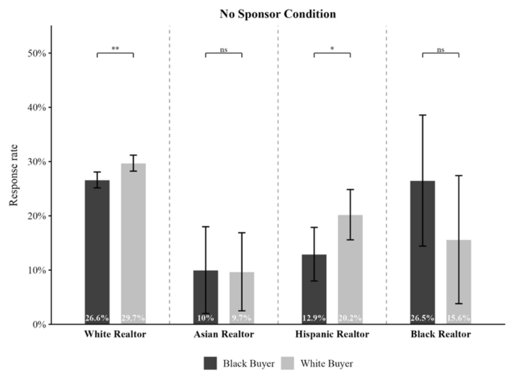
Note: Each point represents the mean email response rate within a quintile of the moderator variable (x-axis) for Black sponsors (dark) and White sponsors (light). Lines connect consecutive groups; error bars show 95% confidence intervals. Groups are computed on the raw (untransformed) scale. Analyses use the sponsor-only subsample (conditions 3–6). Two variables deviate from pure quintiles: Slavery Incidence in 1860 (pslave1860) has a large mass at zero (areas with no recorded slavery); these observations form the first group and the remaining non-zero values are split into three equal groups (4 points total). Office Size (office_size_all) similarly has a mass at its minimum value, which forms the first group; the remainder is split into four equal groups (5 points total). All other variables use standard quintiles.

Moderation by realtor race

We investigate whether the race of realtors in our participant pool in the audit experiment moderates discrimination. To test this, we first classify realtors in our sample by race using the algorithm from Ambrose et al. (2021), which uses first and last names and zip codes to predict realtor race. While this method cannot assign the correct race with perfect accuracy, the probability of correct assignment is relatively high: e.g., for Black realtors the probability of a correct classification is 85.7% and for White realtors it is 99.9%. We find that both White and Hispanic realtors discriminate against Black buyers in the absence of a sponsor (i.e., they respond less often to Black buyers than White buyers in the no sponsor condition; $p<0.03$; see Figure S50). Asian and Black realtors, meanwhile, do not discriminate against Black buyers in the absence of a sponsor ($p>0.2$). In fact, Black realtors respond to Black buyers 10.8% pp more often than to White buyers. However, Black realtors comprise approximately 1% of our sample, limiting statistical power for within- and between-group comparisons involving this group. Given this small sample size, when restricting to buyers who reach out without a sponsor, we are underpowered to detect a significant interaction between an indicator for whether a realtor is White (vs. Black) and an indicator for assignment to the Black buyer condition (even though White realtors favor White buyers over Black buyers 14.4pp more than Black realtors do; $t = 1.736$, $p = 0.083$). Across the full sample (including both the pooled sponsor and no sponsor conditions) this interaction is significant (difference in discrimination = 10.8%, $t = 2.115$, $p=0.034$). Discrimination against Black sponsors follows a similar pattern by realtor race (see Fig S15): here, only White realtors respond significantly less often to Black sponsors. Again, Black realtors show a non-significant but higher response rate to Black sponsors vs. White sponsors.

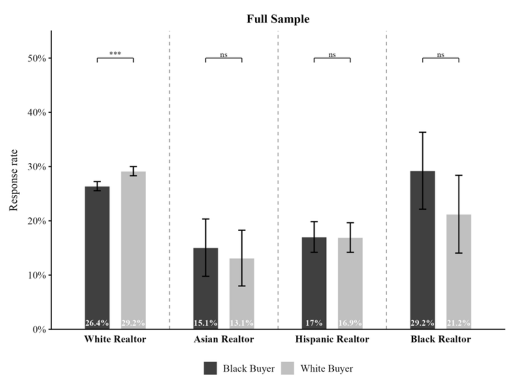
Taken together, these findings suggest that Black realtors are less likely than White realtors to discriminate against Black buyers and may in fact favor them relative to White buyers. A direct consequence is that Black buyers are more likely to end up working with Black realtors. Crucially, given we also find discrimination against Black sponsors (who are themselves realtors in this context), this suggests that racial bias in the housing market might be systemic and create a set of compounding dynamics: Black buyers face discrimination from White realtors, and are thus pushed toward Black realtors; those Black realtors are less effective as sponsors, likely due to discrimination they themselves face. Residential segregation patterns likely reinforce this cycle further (CITE). The result is that the gap in outcomes between Black and White buyers may widen further over time through this discrimination-sponsorship channel.

Figure S50: Response Rates by Realtor Race and Buyer Race (No sponsor condition)



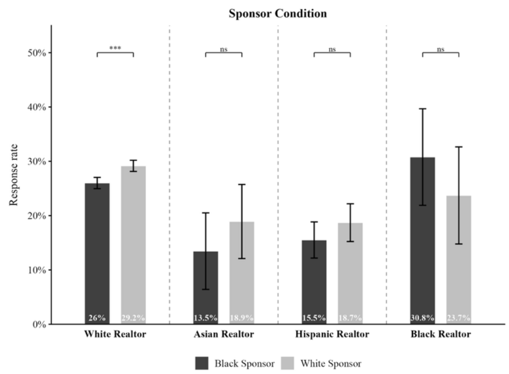
*Note: Bars show regression-adjusted response rates by realtor race (White, Asian, Hispanic, Black), with 95% CIs via the delta method. Adjusted means from OLS (responded ~ BlackBuyer/black sponsor + domaindown) at domaindown = 0. Brackets compare Black and White buyer (or sponsor) response rates within each realtor race group. No-sponsor condition; comparison is Black buyer vs. White buyer. + p<0.1, * p<0.05, ** p<0.01, *** p<0.001.*

Figure S51: Response Rates by Realtor Race and Buyer Race (Full sample)



*Note: Bars show regression-adjusted response rates by realtor race (White, Asian, Hispanic, Black), with 95% CIs via the delta method. Adjusted means from OLS (responded ~ BlackBuyer/black sponsor + domaindown) at domaindown = 0. Brackets compare Black and White buyer (or sponsor) response rates within each realtor race group. Full sample; comparison is Black buyer vs. White buyer. + p<0.1, * p<0.05, ** p<0.01, *** p<0.001.*

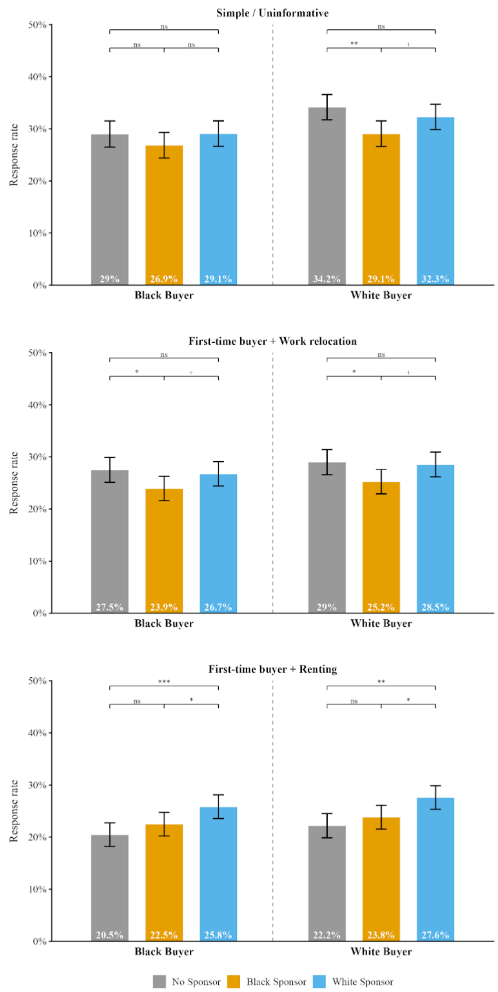
Figure S52: Response Rates by Realtor Race and Sponsor Race (Pooled sponsor conditions)



*Note: Bars show regression-adjusted response rates by realtor race (White, Asian, Hispanic, Black), with 95% CIs via the delta method. Adjusted means from OLS (responded ~ BlackBuyer/black_sponsor + domaindown) at domaindown = 0. Brackets compare Black and White buyer (or sponsor) response rates within each realtor race group. Sponsor conditions only; comparison is Black sponsor vs. White sponsor. + p<0.1, * p<0.05, ** p<0.01, *** p<0.001.*

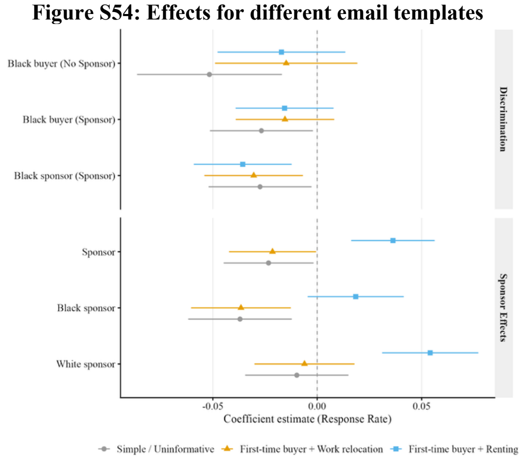
Moderation by Email Template

Figure S53: Response rates by treatment condition for different email templates



*Note: Bars show regression-adjusted response rates across all six experimental conditions, separately for three email template groups: Simple / Uninformative (templates 1 & 4), First-time buyer + Work relocation (templates 2 & 5), and First-time buyer + Renting (templates 3 & 6). Adjusted means from OLS (responded ~ factor(study_condition) + domaindown) at domaindown = 0. Delta method 95% CIs. Brackets compare sponsor conditions within each buyer race group. + p<0.1, * p<0.05, ** p<0.01, *** p<0.001.*

Figure S54: Effects for different email templates

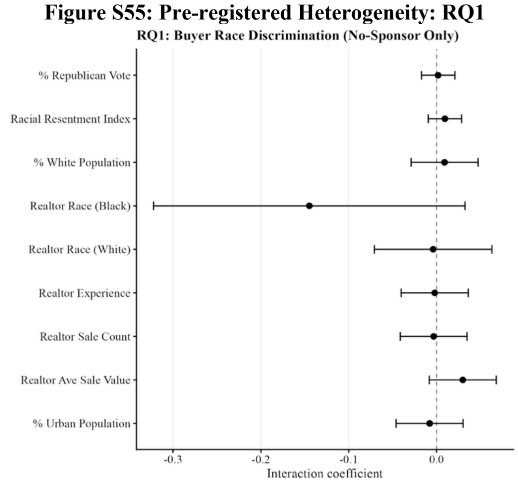


Note: Coefficient plot showing OLS estimates for response rate across three email template groups: Simple / Uninformative (grey circles), First-time buyer + Work relocation (orange triangles), and First-time buyer + Renting (blue squares), dodged vertically per row. Same two-panel structure (Discrimination / Sponsor Effects) as the main coefficient plots. Each model is run on the relevant template-filtered subsample with domaindown as a control. Error bars represent 95% confidence intervals.

Pre-registered heterogeneity analyses

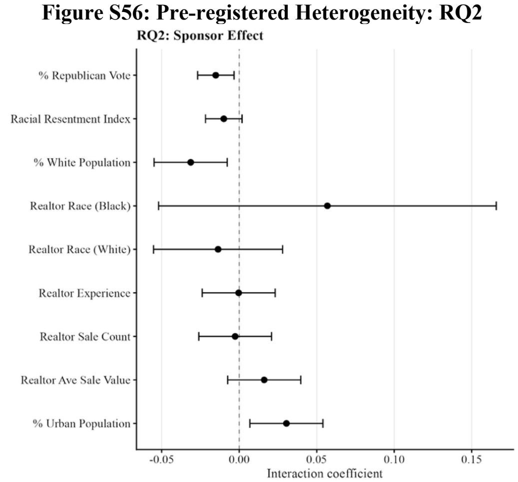
Figures S55–S61 present the full set of preregistered heterogeneity analyses for each research question (RQ1–RQ5), using the moderators and coding scheme specified in the preregistration (AsPredicted #224669). Consistent with the exploratory analyses above, we find very little evidence of meaningful moderation across the pre-registered moderators. We note, however, that these analyses have limited statistical power to detect interactions.

Figure S55: Pre-registered Heterogeneity: RQ1



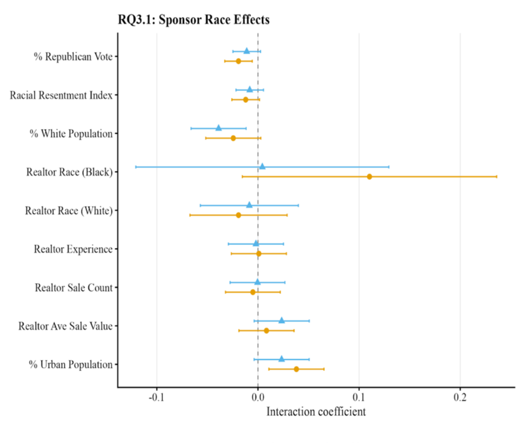
Note: Pre-registered heterogeneity analysis (AsPredicted #224669) — RQ1: Buyer Race Discrimination (No-Sponsor Only). Each row shows the interaction (or subgroup difference) between the treatment and a pre-registered moderator. Binary moderators split at the dataset median; continuous moderators (% Republican vote, Racial Resentment Index) entered at level and standardised. Realtor race entered as binary indicators. All models include domaindown as a control. Error bars represent 95% confidence intervals.

Figure S56: Pre-registered Heterogeneity: RQ2



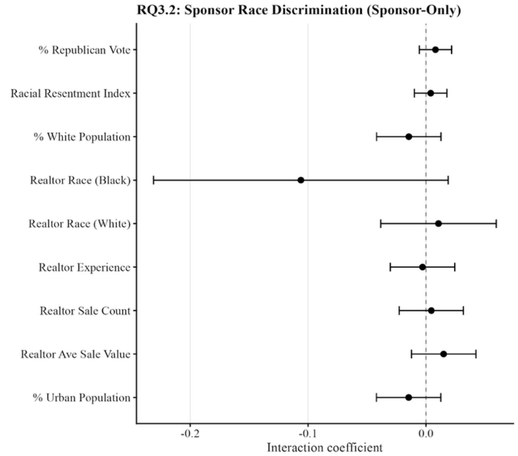
Note: Pre-registered heterogeneity analysis (AsPredicted #224669) — RQ2: Sponsor Effect. Each row shows the interaction (or subgroup difference) between the treatment and a pre-registered moderator. Binary moderators split at the dataset median; continuous moderators (% Republican vote, Racial Resentment Index) entered at level and standardised. Realtor race entered as binary indicators. All models include domaindown as a control. Error bars represent 95% confidence intervals.

Figure S57: Pre-registered Heterogeneity: RQ3.1



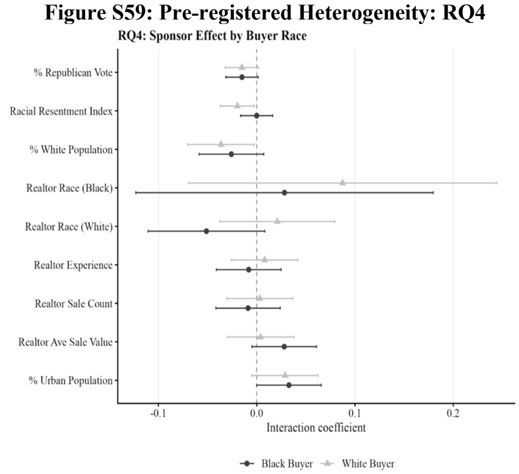
Note: Pre-registered heterogeneity analysis (AsPredicted #224669) — RQ3.1: Sponsor Race vs. No Sponsor. Each row shows the interaction (or subgroup difference) between the treatment and a pre-registered moderator. Binary moderators split at the dataset median; continuous moderators (% Republican vote, Racial Resentment Index) entered at level and standardised. Realtor race entered as binary indicators. All models include domaindown as a control. Error bars represent 95% confidence intervals.

Figure S58: Pre-registered Heterogeneity: RQ3.2



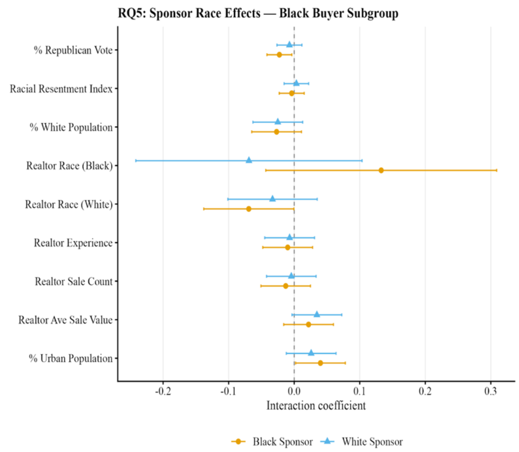
Note: Pre-registered heterogeneity analysis (AsPredicted #224669) — RQ3.2: Sponsor Race Discrimination (Black vs. White Sponsor). Each row shows the interaction (or subgroup difference) between the treatment and a pre-registered moderator. Binary moderators split at the dataset median; continuous moderators (% Republican vote, Racial Resentment Index) entered at level and standardised. Realtor race entered as binary indicators. All models include domaindown as a control. Error bars represent 95% confidence intervals.

Figure S59: Pre-registered Heterogeneity: RQ4



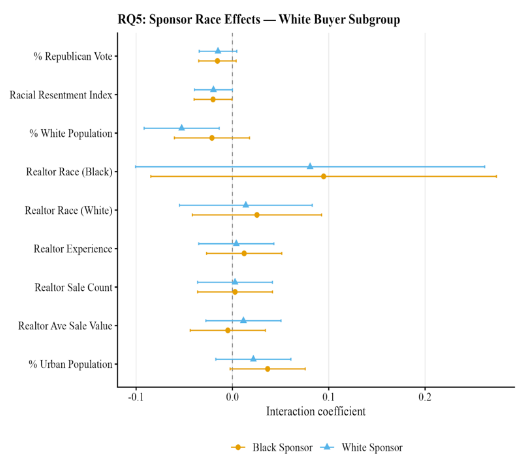
Note: Pre-registered heterogeneity analysis (AsPredicted #224669) — RQ4: Buyer Race × Sponsor Interaction. Each row shows the interaction (or subgroup difference) between the treatment and a pre-registered moderator. Binary moderators split at the dataset median; continuous moderators (% Republican vote, Racial Resentment Index) entered at level and standardised. Realtor race entered as binary indicators. All models include domaindown as a control. Error bars represent 95% confidence intervals.

Figure S60: Pre-registered heterogeneity: RQ5 - Black buyers



Note: Pre-registered heterogeneity analysis (AsPredicted #224669) — RQ5: Full Interaction — Black Buyer Subgroup. Each row shows the interaction (or subgroup difference) between the treatment and a pre-registered moderator. Binary moderators split at the dataset median; continuous moderators (% Republican vote, Racial Resentment Index) entered at level and standardised. Realtor race entered as binary indicators. All models include domaindown as a control. Error bars represent 95% confidence intervals.

Figure S61: Pre-registered heterogeneity: RQ5 - White buyers



Note: Pre-registered heterogeneity analysis (AsPredicted #224669) — RQ5: Full Interaction — White Buyer Subgroup. Each row shows the interaction (or subgroup difference) between the treatment and a pre-registered moderator. Binary moderators split at the dataset median; continuous moderators (% Republican vote, Racial Resentment Index) entered at level and standardised. Realtor race entered as binary indicators. All models include domaindown as a control. Error bars represent 95% confidence intervals.

Supplementary Tables and Figures

Response Rates

Table S43: Regression results: helpful email responses

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.029**				-0.029**	-0.029**
	(0.010)				(0.010)	(0.010)
Pooled sponsor		-0.007			-0.012	
		(0.006)			(0.009)	
Black sponsor			-0.023***	-0.032***		-0.030**
			(0.007)	(0.007)		(0.010)
White sponsor			0.009			0.006
			(0.007)			(0.010)
Black buyer × Pooled sponsor					0.009	
					(0.012)	
Black buyer × Black sponsor						0.014
						(0.014)
Black buyer × White sponsor						0.005
						(0.014)
Num.Obs.	7923	23772	23772	15849	23772	23772
R2	0.012	0.007	0.007	0.005	0.007	0.008

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS (LPM) estimates with HC1 robust standard errors. The dependent variable is whether a participant responded via email in the pre-registered response window (7 days). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily crashed.

Table S44: Regression results: email response excluding rude/suspicious responses

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.031**				-0.031**	-0.031**
	(0.010)				(0.010)	(0.010)
Pooled sponsor		-0.004			-0.010	
		(0.006)			(0.009)	
Black sponsor			-0.020**	-0.032***		-0.027**
			(0.007)	(0.007)		(0.010)
White sponsor			0.012+			0.007
			(0.007)			(0.010)
Black buyer × Pooled sponsor					0.012	
					(0.012)	
Black buyer × Black sponsor						0.015
						(0.014)
Black buyer × White sponsor						0.009
						(0.014)
Num.Obs.	7923	23772	23772	15849	23772	23772
R2	0.013	0.007	0.008	0.005	0.007	0.008

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS (LPM) estimates with HC1 robust standard errors. The dependent variable is whether a participant responded via email in the pre-registered response window (7 days). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily crashed.

Table S45: Regression results: email response with full controls

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.025**				-0.024*	-0.024*
	(0.010)				(0.009)	(0.009)
Pooled sponsor		-0.001			-0.004	
		(0.006)			(0.008)	
Black sponsor			-0.016*	-0.030***		-0.021*
			(0.007)	(0.007)		(0.010)
White sponsor			0.014*			0.012
			(0.007)			(0.010)
Black buyer × Pooled sponsor					0.007	
					(0.012)	
Black buyer × Black sponsor						0.009
						(0.013)
Black buyer × White sponsor						0.004
						(0.014)
Num.Obs.	7755	23264	23264	15509	23264	23264
R2	0.092	0.083	0.084	0.090	0.083	0.084

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS (LPM) estimates with HC1 robust standard errors. The dependent variable is whether a participant responded via email in the pre-registered response window (7 days). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily experienced interruption. All controls included were: separate fixed effects for the realtor's email provider, state of operation, brokerage, indicators for whether the realtor's profile was on Homes.com and Realtor.com, continuous controls for the realtor's history (years of experience, sale count, and average sale value), the size of the realtor's office, fixed effects for the email template used, and continuous county-based controls (percentage of county that is White, Black, and that voted for the Republican Party in the 2020 U.S. presidential election).

Table S46: Regression results: email and phone responses

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.029**				-0.029**	-0.029**
	(0.010)				(0.010)	(0.010)
Pooled sponsor		-0.007			-0.011	
		(0.006)			(0.009)	
Black sponsor			-0.024***	-0.035***		-0.029**
			(0.007)	(0.007)		(0.010)
White sponsor			0.011			0.007
			(0.007)			(0.010)
Black buyer × Pooled sponsor					0.008	
					(0.012)	
Black buyer × Black sponsor						0.009
						(0.014)
Black buyer × White sponsor						0.008
						(0.014)
Num.Obs.	7923	23772	23772	15849	23772	23772
R2	0.013	0.007	0.008	0.006	0.008	0.009

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS (LPM) estimates with HC1 robust standard errors. The dependent variable is whether a participant responded via email, phone call or text in the pre-registered response window (7 days). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails were temporarily interrupted.

Table S47: Regression results: no control for domain interruption

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.030**				-0.030**	-0.030**
	(0.010)				(0.010)	(0.010)
Pooled sponsor		0.007			0.001	
		(0.006)			(0.009)	
Black sponsor			-0.009	-0.031***		-0.016
			(0.007)	(0.007)		(0.010)
White sponsor			0.023**			0.018+
			(0.007)			(0.010)
Black buyer × Pooled sponsor					0.011	
					(0.012)	
Black buyer × Black sponsor						0.014
						(0.014)
Black buyer × White sponsor						0.009
						(0.014)
Num.Obs.	7923	23772	23772	15849	23772	23772
R2	0.001	0.000	0.001	0.001	0.001	0.002

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS (LPM) estimates with HC1 robust standard errors. The dependent variable is whether a participant responded via email in the pre-registered response window (7 days). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race.

Table S48: Regression results: observations affected by domain interruption dropped

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.030**				-0.030**	-0.030**
	(0.011)				(0.011)	(0.011)
Pooled sponsor		-0.003			-0.008	
		(0.006)			(0.009)	
Black sponsor			-0.019**	-0.032***		-0.025*
			(0.007)	(0.007)		(0.010)
White sponsor			0.013+			0.009
			(0.007)			(0.011)
Black buyer × Pooled sponsor					0.009	
					(0.013)	
Black buyer × Black sponsor						0.011
						(0.014)
Black buyer × White sponsor						0.007
						(0.015)
Num.Obs.	7138	22425	22425	15287	22425	22425
R2	0.001	0.000	0.001	0.001	0.001	0.002

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS (LPM) estimates with HC1 robust standard errors. The dependent variable is whether a participant responded via email in the pre-registered response window (7 days). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race.

Table S49: Regression results for email response: logit models

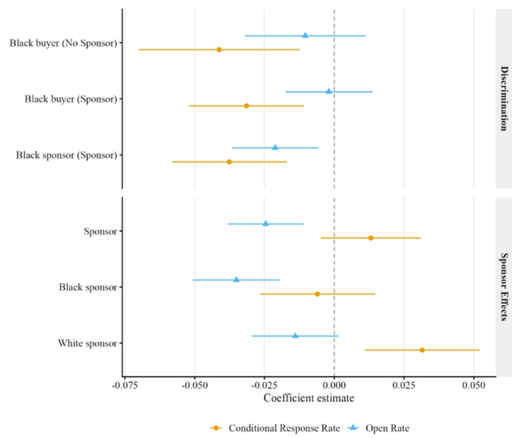
	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.160**				-0.160**	-0.160**
	(0.052)				(0.052)	(0.052)
Pooled sponsor		-0.015			-0.044	
		(0.032)			(0.044)	
Black sponsor			-0.098**	-0.163***		-0.133**
			(0.037)	(0.036)		(0.051)
White sponsor			0.064+			0.042
			(0.036)			(0.050)
Black buyer × Pooled sponsor					0.059	
					(0.063)	
Black buyer × Black sponsor						0.072
						(0.074)
Black buyer × White sponsor						0.048
						(0.072)
Num.Obs.	7923	23772	23772	15849	23772	23772
R2	0.013	0.007	0.008	0.005	0.007	0.008

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: Logit estimates with HC1 robust standard errors (coefficients reported, not marginal effects). The dependent variable is whether a participant responded via email in the pre-registered response window (7 days). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily crashed.

Open Rates and Conditional Response

Figure S62: Effects on Conditional Response and Open Rates



Note: Coefficient plot comparing OLS estimates for open rate (blue circles) and conditional response rate (orange triangles), dodged vertically per row. Panel 1 (Discrimination): Black buyer coefficients on no-sponsor and sponsor subsamples separately; Black sponsor coefficient on sponsor conditions only. Panel 2 (Sponsor Effects): Sponsor coefficient from responded ~ sponsor dum + domaindown; Black and White sponsor coefficients from responded ~ black sponsor + white sponsor + domaindown. Conditional response rate is defined among realtors who opened the email. All models include domaindown as a control. Error bars represent 95% CIs.

Table S50: Open Rate Regressions

	RQ1: Buyer Discrimination	*RQ2: Sponsor Effect*	*RQ3.1: Sponsor Race*	*RQ3.2: Sponsor Discrimination*	*RQ4: Buyer × Sponsor*	*RQ5: Full Interaction*
Black buyer	*-0.010*				*-0.010*	*-0.010*
	(0.011)				*(0.011)*	*(0.011)*
Pooled sponsor		*-0.025****			*-0.029***	
		(0.007)			*(0.009)*	
Black sponsor			*-0.035****	*-0.021***		*-0.051****
			(0.008)	*(0.008)*		*(0.011)*
White sponsor			*-0.014+*			*-0.006*
			(0.008)			*(0.011)*
Black buyer × Pooled sponsor					*0.008*	
					(0.013)	
Black buyer × Black sponsor						*0.032**
						(0.016)
Black buyer × White sponsor						*-0.015*
						(0.015)
Num.Obs.	*7923*	*23772*	*23772*	*15849*	*23772*	*23772*
R2	*0.042*	*0.025*	*0.025*	*0.017*	*0.025*	*0.026*

*+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

Notes: OLS estimates with HC1 robust standard errors. The analysis uses the full sample; observations with missing values of whether the realtor opened the email (full sample) are dropped automatically. The dependent variable is whether the realtor opened the email (full sample). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily crashed.

Table S51: Conditional Response Rate Regressions

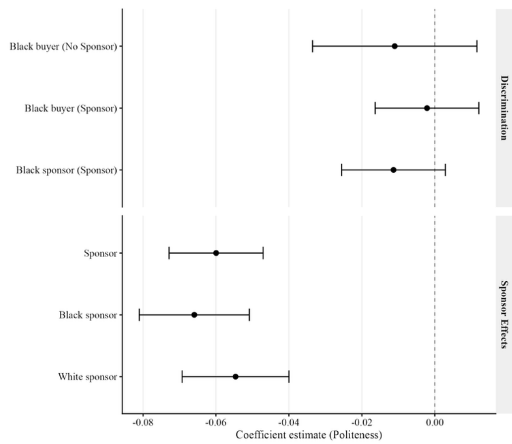
	RQ1: Buyer Discrimination	*RQ2: Sponsor Effect*	*RQ3.1: Sponsor Race*	*RQ3.2: Sponsor Discrimination*	*RQ4: Buyer × Sponsor*	*RQ5: Full Interaction*
Black buyer	-0.041**				-0.041**	-0.041**
	(0.015)				(0.015)	(0.015)
Pooled sponsor		0.013			0.008	
		(0.009)			(0.013)	
Black sponsor			-0.006	-0.038***		-0.004
			(0.010)	(0.010)		(0.015)
White sponsor			0.031**			0.020
			(0.010)			(0.015)
Black buyer × Pooled sponsor					0.010	
					(0.018)	
Black buyer × Black sponsor						-0.002
						(0.021)
Black buyer × White sponsor						0.022
						(0.021)
Num.Obs.	4617	13826	13826	9209	13826	13826
R2	0.002	0.000	0.001	0.001	0.001	0.002

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Notes: OLS estimates with HC1 robust standard errors. The analysis uses the full sample; observations with missing values of whether the realtor responded to the email, conditional on having opened it; realtors who did not open the email have a missing value for this outcome and are excluded automatically are dropped automatically. The dependent variable is whether the realtor responded to the email, conditional on having opened it; realtors who did not open the email have a missing value for this outcome and are excluded automatically. RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily was interrupted.

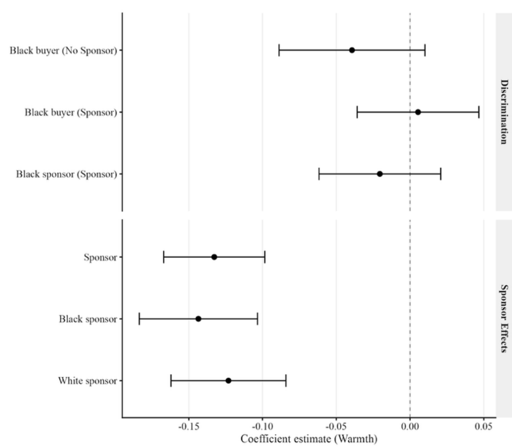
Response Quality

Figure S63: Effects on Politeness



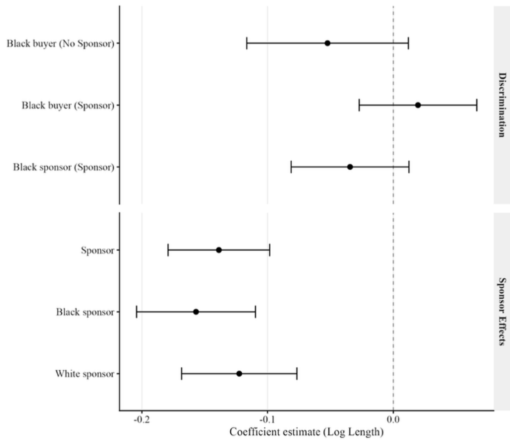
Note: Coefficient plot showing OLS estimates for Politeness, conditional on having received a response (responded = 1); non-responders are excluded. Same two-panel structure as the open rate plot above. All models include domaindown as a control. Error bars represent 95% confidence intervals.

Figure S64: Effects on Warmth



Note: Coefficient plot showing OLS estimates for Warmth, conditional on having received a response (responded == 1); non-responders are excluded. Same two-panel structure as the open rate plot above. All models include domaindown as a control. Error bars represent 95% confidence intervals.

Figure S65: Effects on Log Length



Note: Coefficient plot showing OLS estimates for Log Length, conditional on having received a response (responded == 1); non-responders are excluded. Same two-panel structure as the open rate plot above. All models include domaindown as a control. Error bars represent 95% confidence intervals.

Table S52: Politeness Regressions: Conditional on Response

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.011				-0.011	-0.011
	(0.012)				(0.012)	(0.012)
Pooled sponsor		-0.060***			-0.064***	
		(0.007)			(0.009)	
Black sponsor			-0.066***	-0.011		-0.072***
			(0.008)	(0.007)		(0.011)
White sponsor			-0.055***			-0.058***
			(0.008)			(0.010)
Black buyer × Pooled sponsor					0.009	
					(0.014)	
Black buyer × Black sponsor						0.012
						(0.016)
Black buyer × White sponsor						0.006
						(0.015)
Num.Obs.	2023	6181	6181	4158	6181	6181
R2	0.000	0.013	0.014	0.001	0.013	0.014

*+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

Notes: OLS estimates with HC1 robust standard errors. The analysis is restricted to observations where a response was received (responded = 1); non-responders are excluded. The dependent variable is the politeness of the email response, measured using the algorithm from Yeomans, Kantor, and Tingley (2018). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily crashed.

Table S53: Politeness Regressions: Unconditional (Full Sample)

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.014**				-0.014**	-0.014**
	(0.005)				(0.005)	(0.005)
Pooled sponsor		-0.016***			-0.020***	
		(0.003)			(0.004)	
Black sponsor			-0.023***	-0.013***		-0.028***
			(0.003)	(0.003)		(0.004)
White sponsor			-0.010**			-0.012**
			(0.003)			(0.005)
Black buyer × Pooled sponsor					0.007	
					(0.006)	
Black buyer × Black sponsor						0.009
						(0.006)
Black buyer × White sponsor						0.005
						(0.006)
Num.Obs.	7923	23772	23772	15849	23772	23772
R2	0.008	0.005	0.006	0.004	0.006	0.007

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS estimates with HC1 robust standard errors. The analysis uses the full sample; observations with missing values of the politeness of the email response (our dependent variable), measured using the algorithm from Yeomans, Kantor, and Tingley (2018) are dropped automatically. RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily crashed.

Table S54: Warmth Regressions: Conditional on Response

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.039				-0.039	-0.039
	(0.025)				(0.025)	(0.025)
Pooled sponsor		-0.133***			-0.154***	
		(0.016)			(0.023)	
Black sponsor			-0.144***	-0.020		-0.172***
			(0.021)	(0.021)		(0.028)
White sponsor			-0.123***			-0.139***
			(0.019)			(0.026)
Black buyer × Pooled sponsor					0.045	
					(0.033)	
Black buyer × Black sponsor						0.059
						(0.041)
Black buyer × White sponsor						0.032
						(0.037)
Num.Obs.	1878	5731	5731	3853	5731	5731
R2	0.001	0.010	0.010	0.000	0.010	0.011

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS estimates with HC1 robust standard errors. The analysis is restricted to observations where a response was received (responded = 1); non-responders are excluded. The dependent variable is the warmth of the email response, measured using the algorithm from Yeomans, Greifer, and Guenon (2021) and the definition from Cuddy, Fiske, and Glick (2008). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily crashed.

Table S55: Warmth Regressions: Unconditional (Full Sample)

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.121*				-0.121*	-0.121*
	(0.050)				(0.050)	(0.050)
Pooled sponsor		-0.046			-0.066	
		(0.031)			(0.044)	
Black sponsor			-0.123***	-0.153***		-0.149**
			(0.035)	(0.035)		(0.050)
White sponsor			0.031			0.018
			(0.036)			(0.051)
Black buyer × Pooled sponsor					0.039	
					(0.061)	
Black buyer × Black sponsor						0.052
						(0.069)
Black buyer × White sponsor						0.025
						(0.071)
Num.Obs.	7778	23322	23322	15544	23322	23322
R2	0.012	0.006	0.007	0.005	0.007	0.008

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS estimates with HC1 robust standard errors. The analysis uses the full sample; observations with missing values of the warmth of the email response, measured using the algorithm from Yeomans, Greifer, and Guenon (2021) and definition from Cuddy, Fiske, and Glick (2008) are dropped automatically. The dependent variable is the warmth of the email response, measured using the algorithm from Yeomans, Greifer, Guenon (2021) and definition from Cuddy, Fiske & Glick, (2008). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily crashed.

Table S56: Log Length of Response Regressions: Conditional on Response

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.052				-0.052	-0.052
	(0.033)				(0.033)	(0.033)
Pooled sponsor		-0.139***			-0.173***	
		(0.020)			(0.028)	
Black sponsor			-0.157***	-0.034		-0.210***
			(0.024)	(0.024)		(0.034)
White sponsor			-0.123***			-0.140***
			(0.023)			(0.032)
Black buyer × Pooled sponsor					0.072+	
					(0.041)	
Black buyer × Black sponsor						0.112*
						(0.048)
Black buyer × White sponsor						0.037
						(0.046)
Num.Obs.	2023	6181	6181	4158	6181	6181
R2	0.001	0.007	0.008	0.001	0.008	0.009

*+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

Notes: OLS estimates with HC1 robust standard errors. The analysis is restricted to observations where a response was received (responded == 1); non-responders are excluded. The dependent variable is the natural log of the number of characters in the email response. RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily crashed.

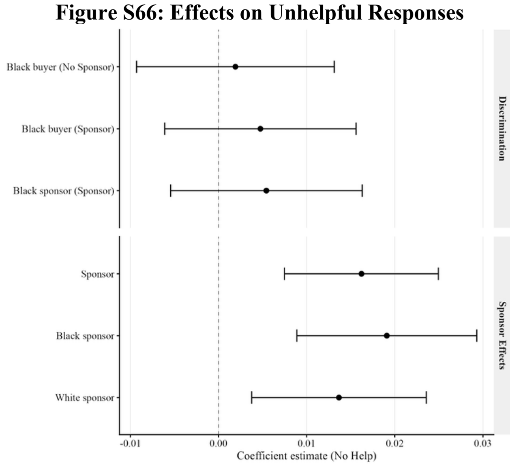
Table S57: Log Length of Response Regressions: Unconditional (Full Sample)

	RQ1: Buyer Discrimination	RQ2: Sponsor Effect	RQ3.1: Sponsor Race	RQ3.2: Sponsor Discrimination	RQ4: Buyer × Sponsor	RQ5: Full Interaction
Black buyer	-0.156**				-0.156**	-0.156**
	(0.048)				(0.048)	(0.048)
Pooled sponsor		-0.046			-0.080+	
		(0.030)			(0.043)	
Black sponsor			-0.128***	-0.164***		-0.175***
			(0.034)	(0.034)		(0.048)
White sponsor			0.036			0.016
			(0.035)			(0.050)
Black buyer × Pooled sponsor					0.067	
					(0.059)	
Black buyer × Black sponsor						0.094
						(0.067)
Black buyer × White sponsor						0.040
						(0.069)
Num.Obs.	7923	23772	23772	15849	23772	23772
R2	0.013	0.006	0.007	0.005	0.007	0.008

*+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

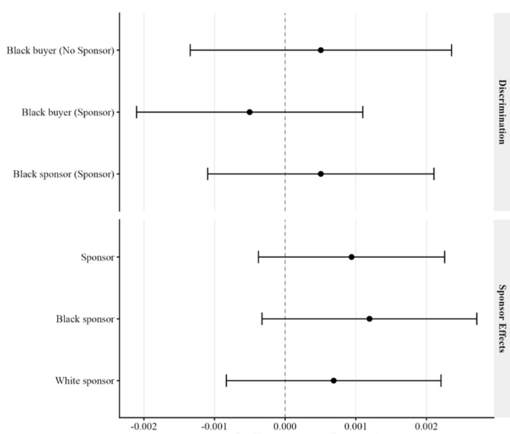
Notes: OLS estimates with HC1 robust standard errors. The analysis uses the full sample; observations with missing values of the natural log of the number of characters in the email response are dropped automatically. The dependent variable is the natural log of the number of characters in the email response. RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily crashed.

Figure S66: Effects on Unhelpful Responses



Note: Coefficient plot showing OLS estimates for "No Help" on the full sample, as coded by a team of research assistants who flagged when the response indicated an unwillingness to help. Observations with missing outcome values are dropped automatically. Same two-panel structure as the open rate plot above. All models include service interruption as a control. Error bars represent 95% confidence intervals.

Figure S67: Effects on Suspicious Responses



Note: Coefficient plot showing OLS estimates for Suspicion on the full sample, as coded by a team of research assistants when a response indicated suspicion (e.g., the realtor questioned whether the buyer/sponsor was a real person or the email was spam). Observations with missing outcome values are dropped automatically. Same two-panel structure as the open rate plot above. All models include service interruption as a control. Error bars represent 95% confidence intervals.

Table S58: No Help Regressions: Conditional on Response

	RQ1: Buyer Discrimination	*RQ2: Sponsor Effect*	*RQ3.1: Sponsor Race*	*RQ3.2: Sponsor Discrimination*	*RQ4: Buyer × Sponsor*	*RQ5: Full Interaction*
Black buyer	*-0.001*				*-0.001*	*-0.001*
	(0.005)				*(0.005)*	*(0.005)*
Pooled sponsor		*0.017****			*0.014***	
		(0.004)			*(0.005)*	
Black sponsor			*0.021****	*0.006*		*0.020***
			(0.005)	*(0.006)*		*(0.007)*
White sponsor			*0.015***			*0.010*
			(0.005)			*(0.006)*
Black buyer × Pooled sponsor					*0.006*	
					(0.008)	
Black buyer × Black sponsor						*0.001*
						(0.010)
Black buyer × White sponsor						*0.011*
						(0.009)
Num.Obs.	*2023*	*6181*	*6181*	*4158*	*6181*	*6181*
R2	*0.000*	*0.003*	*0.003*	*0.000*	*0.003*	*0.003*

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS estimates with HC1 robust standard errors. The analysis is restricted to observations where a response was received (responded = 1); non-responders are excluded. The dependent variable is whether the realtor explicitly indicated they would not help (coded 1 = not helpful). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily was interrupted.

Table S59: No Help Regressions:Unconditional (Full Sample)

	RQ1: Buyer Discrimination	*RQ2: Sponsor Effect*	*RQ3.1: Sponsor Race*	*RQ3.2: Sponsor Discrimination*	*RQ4: Buyer × Sponsor*	*RQ5: Full Interaction*
Black buyer	*0.002*				*0.002*	*0.002*
	(0.006)				*(0.006)*	*(0.006)*
Pooled sponsor		*0.016****			*0.015***	
		(0.004)			*(0.005)*	
Black sponsor			*0.019****	*0.005*		*0.020***
			(0.005)	*(0.006)*		*(0.007)*
White sponsor			*0.014***			*0.010+*
			(0.005)			*(0.006)*
Black buyer × Pooled sponsor					*0.003*	
					(0.008)	
Black buyer × Black sponsor						*-0.002*
						(0.010)
Black buyer × White sponsor						*0.007*
						(0.009)
Num.Obs.	*2026*	*6187*	*6187*	*4161*	*6187*	*6187*
R2	*0.000*	*0.002*	*0.002*	*0.000*	*0.002*	*0.003*

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS estimates with HC1 robust standard errors. The analysis uses the full sample; observations with missing values of whether the realtor explicitly indicated they would not help (coded 1 = not helpful) are dropped automatically. The dependent variable is whether the realtor explicitly indicated they would not help (coded 1 = not helpful). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily was interrupted.

Table S60: Suspicion Regressions: Conditional on Response

	RQ1: Buyer Discrimination	*RQ2: Sponsor Effect*	*RQ3.1: Sponsor Race*	*RQ3.2: Sponsor Discrimination*	*RQ4: Buyer × Sponsor*	*RQ5: Full Interaction*
Black buyer	*0.001*				*0.001*	*0.001*
	(0.003)				*(0.003)*	*(0.003)*
Pooled sponsor		*0.003+*			*0.004+*	
		(0.002)			*(0.002)*	
Black sponsor			*0.004+*	*0.002*		*0.004*
			(0.002)	*(0.003)*		*(0.003)*
White sponsor			*0.002*			*0.004*
			(0.002)			*(0.003)*
Black buyer × Pooled sponsor					*-0.002*	
					(0.004)	
Black buyer × Black sponsor						*0.000*
						(0.005)
Black buyer × White sponsor						*-0.004*
						(0.004)
Num.Obs.	*2023*	*6181*	*6181*	*4158*	*6181*	*6181*
R2	*0.001*	*0.000*	*0.001*	*0.000*	*0.000*	*0.001*

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

Notes: OLS estimates with HC1 robust standard errors. The analysis is restricted to observations where a response was received (responded = 1); non-responders are excluded. The dependent variable is whether the realtor expressed suspicion about the sender or flagged the email as spam (coded 1 = suspicious response). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily was interrupted.

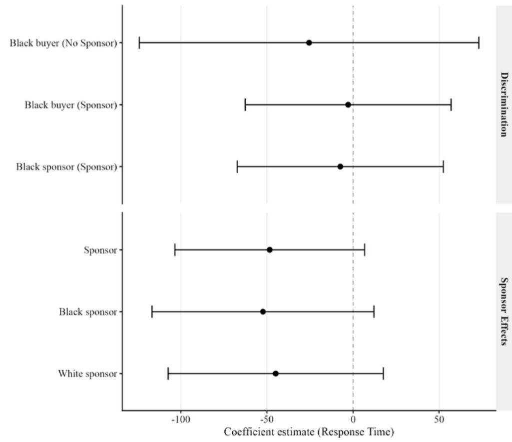
Table S61: Suspicion Regressions: Unconditional (Full Sample)

	RQ1: Buyer Discrimination	*RQ2: Sponsor Effect*	*RQ3.1: Sponsor Race*	*RQ3.2: Sponsor Discrimination*	*RQ4: Buyer × Sponsor*	*RQ5: Full Interaction*
Black buyer	*0.001*				*0.001*	*0.001*
	(0.001)				*(0.001)*	*(0.001)*
Pooled sponsor		*0.001*			*0.001+*	
		(0.001)			*(0.001)*	
Black sponsor			*0.001*	*0.001*		*0.002+*
			(0.001)	*(0.001)*		*(0.001)*
White sponsor			*0.001*			*0.001*
			(0.001)			*(0.001)*
Black buyer × Pooled sponsor					*-0.001*	
					(0.001)	
Black buyer × Black sponsor						*-0.001*
						(0.002)
Black buyer × White sponsor						*-0.001*
						(0.001)
Num.Obs.	*7923*	*23772*	*23772*	*15849*	*23772*	*23772*
R2	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*

*+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001*

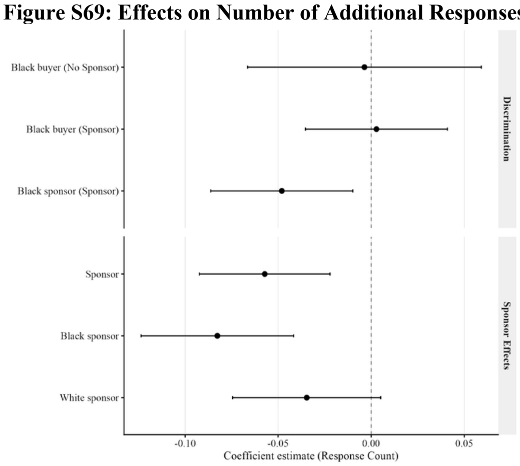
Notes: OLS estimates with HC1 robust standard errors. The analysis uses the full sample; observations with missing values of whether the realtor expressed suspicion about the sender or flagged the email as spam (coded 1 = suspicious response) are dropped automatically. The dependent variable is whether the realtor expressed suspicion about the sender or flagged the email as spam (coded 1 = suspicious response). RQ1 tests buyer race discrimination (no-sponsor condition only). RQ2 tests whether assignment to the pooled sponsor conditions increases response rates (i.e., tests for an overall sponsor effect). RQ3.1 tests Black sponsor and White sponsor effects separately vs. no-sponsor baseline. RQ3.2 tests sponsor discrimination (Black vs. White sponsor, pooled sponsor conditions only). RQ4 tests whether the sponsor effect differs by buyer race. RQ5 tests full interactions between buyer race and sponsor race. All specifications control for periods when the email domains used to send emails temporarily crashed.

Figure S68: Effects on Response Time (Minutes)



Note: Coefficient plot showing OLS estimates for Response Time (i.e., time between email send and the relator responding, in minutes), conditional on having received a response (responded = 1); non-responders are excluded. Same two-panel structure as the open rate plot above. All models include domaindown as a control. Error bars represent 95% confidence intervals.

Figure S69: Effects on Number of Additional Responses



Note: Coefficient plot showing OLS estimates for Response Count (i.e., number of email replies from the realtor), conditional on having received a response (responded == 1); non-responders are excluded. Same two-panel structure as the open rate plot above. All models include domaindown as a control. Error bars represent 95% confidence intervals.